

Bacterial Cell Cycle Regulation: A Hierarchical Framework for Testing Physical-Molecular Integration

From Physical Constraints to Molecular Override: A Testable Framework for Cell Cycle Regulation

Abstract

The bacterial cell cycle—comprising chromosome replication, segregation, and cell division—is classically understood through sophisticated macromolecular regulatory circuits. This review develops a **conceptual framework** proposing that the relationship between physical constraints and molecular regulation is **context-dependent**: during **critical functional transitions** (checkpoints, stress responses, developmental programming), molecular systems can detect and override physical constraints that would produce incorrect outcomes; during **steady-state homeostasis**, bidirectional coupling predominates. We introduce the **Asymmetry Index (AsI)** as a PROPOSED metric to formalize this context-dependent information flow: $AsI = |\Delta M / \sigma_M| / |\Delta P / \sigma_P|$, where ΔM and ΔP are changes in molecular and physical states due to interventions, and σ_M and σ_P are baseline standard deviations computed over clearly defined populations. **However, AsI measurement faces fundamental challenges**: physical and molecular perturbations are not cleanly separable in living cells (osmotic manipulation triggers rapid mechanosensitive responses; microfluidic compression activates MscL/MscS channels). We reframe AsI as measuring **EFFECTIVE causal asymmetry** (differences in causal pathway organization) rather than pure physical-molecular separation, and propose experimental strategies using timescale separation and mechanosensitive channel knockouts to address these challenges. We PROPOSE that $AsI \gg 1$ characterizes checkpoint/stress/developmental contexts (molecular override), while $AsI \approx 1$ characterizes steady-state homeostasis (bidirectional coupling), but this requires experimental validation using the protocols described in Section 9.2.

We operationalize the **Molecular Complexity Threshold** as a testable hypothesis: higher cell cycle gene count (CCGC) may correlate with lower division timing variability. Current data from JCVI-syn3.0 (473 genes, CV = 0.35-0.45, CCGC $\approx 19 \pm 2$) and wild-type *E. coli* (CV = 0.10-0.15, Core CCGC $\approx 100 \pm 15$, Extended CCGC $\approx 150 \pm 20$) **MOTIVATE** this hypothesis, but **DO NOT PROVIDE SUPPORT** for any claim about the CCGC-CV relationship. These organisms differ fundamentally in cell geometry (spherical vs rod-shaped), growth rate (3 hours vs 20 minutes), and phylogeny (enormously distant), making attribution of CV differences to CCGC alone impossible. The syn3.0 vs. E. coli comparison is presented **strictly as hypothesis-generating**; definitive claims require intermediate data points controlling for confounding variables. **Note on CCGC methodology**: Defining "cell cycle genes" is methodologically challenging due to pleiotropy and global regulators. We distinguish between Core CCGC (direct cell cycle functions) and Extended CCGC (including regulators that interface with cell cycle).

We PROPOSE that **functional context determines coupling type**: systems with **precision-critical functional requirements** (checkpoints, stress responses, developmental programming) are predicted to exhibit asymmetric/hierarchical information flow (molecular systems override physical constraints); systems with **speed-critical requirements** (rapid homeostatic adjustment) are predicted to exhibit bidirectional coupling. We PROPOSE that **functional requirements predict coupling type**, not that asymmetric information flow is a universal principle governing all cell cycle states. We present case studies (SOS checkpoint, DNA supercoiling) that ILLUSTRATE these predictions, but definitive validation requires AsI measurements across diverse systems.

We provide **detailed experimental protocols** with power analyses, feasibility ratings, personnel estimates, timelines, and explicit falsification criteria. Our contribution is proposing a **quantifiable metric (AsI)** that can distinguish coupling mechanisms currently described only qualitatively. The framework's primary testable claim is that the Min system AsI measurement can discriminate between active geometric sensing ($AsI \gg 1$) and passive reaction-diffusion ($AsI \ll 1$). Definitive validation of broader claims requires the experimental protocols we describe.

Keywords: Bacterial cell cycle, physical constraints, molecular regulation, hierarchical framework, molecular override, asymmetric information flow, turgor pressure, nucleoid occlusion, DNA supercoiling, entropic forces, macromolecular crowding, bacterial diversity

Author Statement and Methods

AI-Assisted Research Disclosure

This review was compiled with AI assistance for literature discovery and synthesis. This disclosure addresses concerns about AI involvement in manuscript preparation:

AI-Assisted Components:

- **Literature search:** Automated search of PubMed and preprint servers using keyword-based queries to identify relevant papers
- **Citation discovery:** Suggestion of relevant citations based on semantic similarity to manuscript topics
- **Text organization:** AI assistance in structuring sections and suggesting transitions between topics
- **Novelty analysis:** Literature review to identify genuinely novel vs. previously articulated concepts (see Section 7.1)

Human-Authored Components:

- **All scientific interpretations:** Framework synthesis, conceptual arguments, and biological conclusions
- **Critical analysis:** Evaluation of evidence quality, identification of counterexamples
- **Writing:** Authorship of all prose, with AI assistance limited to organization and citation suggestions
- **Final verification:** Manual checking of all references against primary sources (PubMed, DOIs, journal websites)

Citation Issues and Corrections: The Round 1 revision identified several problematic citations that appear to have been AI-suggested but not properly verified:

- Duplicate entries (Persat/Pearl, Budin/Bizzarri) have been removed
- Suspicious citations (Rozewicz 2022, Eme 2023) have been removed
- All remaining references have been manually verified against DOIs or journal websites

Note: There is inherent tension in claiming AI assisted with novelty analysis but did not influence scientific interpretation. We acknowledge that AI-assisted novelty analysis may have influenced our understanding of what counts as novel, but all final scientific interpretations and conclusions are human-authored.

AI Tool Version Information:

- **AI tool:** Claude 3.5 Sonnet (Anthropic)
- **Exact model identifier:** claude-3-5-sonnet-20240620
- **Access dates:** Multiple sessions between January 2026 and March 2026
- **Usage:** Literature search, citation suggestions, text organization, novelty analysis
- **All scientific interpretations, framework synthesis, and conclusions are human-authored**

NOTE: This manuscript is NOT designed for submission to RASTI (Research on Artificial Sensing and Transduction Intelligence). The framework and content are intended for general cell biology and microbiology journals.

1. Introduction

1.1 The Bacterial Cell Cycle: Components and Classical Understanding

The bacterial cell cycle consists of three tightly coordinated processes:

1. **Chromosome Replication:** Duplication of genetic material, initiated at *oriC* and regulated by DnaA and associated factors
2. **Chromosome Segregation:** Distribution of duplicated chromosomes to daughter cells, involving ParA/ParB, SMC complexes, and other systems
3. **Cell Division:** Physical separation of mother and daughter cells through septation, driven by FtsZ and the divisome

Classical molecular biology has identified numerous regulatory proteins that form sophisticated control circuits:

- **Replication initiation:** DnaA, DnaC, DiaA, SeqA, Dam methylase, RIDA system, *datA* locus, DARS sequences
- **Chromosome segregation:** ParA/ParB, SMC complexes (MukBEF in *E. coli**, Smc-ScpAB in *B. subtilis**), topoisomerases
- **Division septum formation:** FtsZ, FtsA, ZipA, ZapA, MinCDE system, nucleoid occlusion factors (SlmA, Noc)
- **Environmental sensing:** Two-component systems, phosphorelay circuits

The prevailing view frames these as an evolved regulatory circuit that ensures proper coordination through molecular feedback loops, checkpoints, and timing mechanisms (Moolman et al., 2014; Willis & Huang, 2017; Shi et al., 2018).

1.2 The Fundamental Question

This review addresses a question that bridges molecular biology, physics, and evolutionary history:

To what extent do physical and chemical constraints contribute to bacterial cell cycle regulation, and how do these physical foundations interact with sophisticated molecular regulatory systems? Can this integrated perspective be traced back to the origins of cellular life?

This question has implications for:

- Understanding the nature of cellular organization
- Reconstructing early cellular evolution and the nature of LUCA
- Designing minimal synthetic cells with appropriate regulatory complexity
- Distinguishing between physical constraints, molecular regulation, and their integration

1.3 Why This Perspective Matters

Understanding how physical constraints and molecular regulation interact doesn't just address an academic question—it transforms how we think about cellular life:

- **Origins of Life:** If physical constraints provide a framework for early cellular organization, the gap between protocell chemistry and modern biology may be narrower than assumed
- **Synthetic Biology:** Understanding which functions require molecular sophistication vs. those that emerge from physical properties guides minimal cell design

- **Evolutionary Biology:** Distinguishing between physically constrained traits and contingent evolutionary solutions reveals what aspects of cellular organization are inevitable vs. chosen

- **Philosophy of Biology:** This framework engages questions about explanatory levels in biology (see Section 8.4 for extended discussion)

Key Shift: Rather than asking "physical vs. molecular regulation" as a binary choice, we ask "**WHEN do molecular systems override physical constraints, and when do they work within them?**" This reframing acknowledges that both levels of explanation are necessary, but the **balance shifts depending on functional context**:

- **During checkpoints, stress responses, and developmental programming:** Molecular systems can override physical constraints (asymmetric information flow)

- **During steady-state homeostasis:** Molecular systems work within and manage physical constraints through bidirectional coupling

Our framework predicts **WHEN** each type of relationship should evolve based on functional requirements, not that one relationship universally dominates.

Bacterial Diversity and Physical Constraints:

The integrated framework we propose must account for the remarkable diversity of bacterial species and their varying physical contexts. Gram-positive bacteria such as *Bacillus subtilis* experience higher turgor pressures due to their thick peptidoglycan layers (Whatmore & Reed, 1990; Koch, 1983), while Gram-negative bacteria such as *Escherichia coli* experience turgor pressures estimated at 3-5 atm (Cayley et al., 2000). Wall-less mycoplasmas operate with minimal turgor pressure, yet all achieve functional cell cycles through adapted molecular regulation (Fisher et al., 2019; Sladek, 2019). Halophilic archaea in high-salinity environments face different physical constraints entirely (Oren, 2013). This diversity—where different lineages face radically different physical environments yet all achieve functional division through evolved molecular adaptations—supports the thesis that molecular systems have evolved to operate within and manage diverse physical constraints.

Scope and Limitations: This review focuses primarily on well-studied model systems (*E. coli*, *B. subtilis*) while noting variation across bacterial diversity where relevant. **Important scope clarification:** Our framework proposes that asymmetric information flow (molecular override of physical constraints) emerges specifically during **critical functional transitions**—including stress responses (e.g., DNA damage SOS response), developmental programming (e.g., *Caulobacter* asymmetric division), and checkpoint activation. We do NOT claim that asymmetric information flow is a universal principle governing all aspects of cell cycle regulation under all conditions. Indeed, during normal steady-state cycling, many systems (e.g., DNA supercoiling) exhibit bidirectional coupling where physical and molecular systems influence each other continuously. Our framework predicts **WHEN** each type of coupling should evolve based on functional requirements, not that one type universally dominates. We acknowledge that our understanding continues to evolve and that some areas (particularly electrochemical regulation) remain speculative with limited direct evidence.

2. Physical Constraints: The Foundational Context

This section reviews physical and chemical constraints that create the permissive conditions for cell cycle progression. Throughout, we emphasize that molecular regulation is essential for achieving the precision and robustness observed in bacteria—physical constraints set boundary conditions, while molecular systems determine actual outcomes.

Important Caveat on Evidential Strength:

The physical constraints discussed in this section vary substantially in the strength of their evidential support for causal roles in cell cycle regulation:

- **Strong support:** Nucleoid geometry (Section 2.2) and DNA topology (Section 2.3) have well-established molecular mechanisms linking physical parameters to regulatory outcomes

- **Moderate support:** Entropic forces (Section 2.5) are well-established as physical contributors but require molecular machinery to be productive

- **Speculative/Correlational:** Turgor pressure (Section 2.1) and macromolecular crowding (Section 2.4) show correlations with cell cycle progression but causal mechanisms remain uncertain

The hierarchical framework illustrated in later figures uses turgor pressure, nucleoid geometry, and DNA topology as examples of physical foundations. **Readers should note:** these three are not evidentiary equals. Nucleoid geometry and DNA topology are the strongest-supported examples; turgor pressure is included for conceptual completeness but should be understood as more speculative.

2.1 Turgor Pressure: Mechanical Stress and Division Timing

Physical Basis:

Turgor pressure—the outward force exerted by the cytoplasmic membrane against the cell wall—increases as bacterial cells grow due to the surface-to-volume ratio. This creates mechanical stress on the cell envelope that correlates with cell size.

Evidence:

Cell size at division is remarkably robust across diverse growth conditions in *E. coli* and *B. subtilis* (Shi et al., 2018; Witz et al., 2019). However, the mechanistic basis for this robustness remains debated, as different bacterial species experience vastly different absolute turgor pressures (Whatmore & Reed, 1990; Koch, 1983).

- **Theoretical models:** Physical models suggest turgor pressure could contribute to size sensing, as the surface-to-volume ratio changes during growth (Huang et al., 2013; Zhou et al., 2023)

- **Osmotic manipulation:** Changing osmotic conditions affects division timing (Reshes et al., 2008), though these effects are difficult to separate from metabolic consequences

- **FtsZ mechanosensitivity** (limited in vivo evidence): In vitro studies using liposome reconstitution show FtsZ polymerization can be influenced by membrane tension and curvature (Loose & Mitchison, 2014). Bisson-Filho et al. (2017) demonstrated that FtsZ treadmilling dynamics are coupled to membrane-associated cell wall synthesis machinery, though whether this represents active mechanosensitivity to membrane tension remains debated. The physiological relevance of FtsZ mechanosensitivity in living cells is **not well established**. **Important caveat:** The in vivo evidence for turgor pressure or membrane tension directly modulating FtsZ assembly in living bacterial cells remains very limited. Most evidence for FtsZ mechanosensitivity comes from in vitro systems, and whether FtsZ functions as a mechanosensor in physiological conditions remains uncertain.

The Adder Model - An Active Debate:

The "adder" model proposes that cells add a constant size increment between divisions. The landmark single-cell study by Taheri-Araghi et al. (2015) provided compelling evidence for the adder model in *E. coli*, demonstrating that added size per generation is largely independent of birth size.

However, the mechanistic basis remains contested. Recent work suggests the adder may emerge from multiple distinct systems (Si et al., 2019; Witz et al., 2019):

- **Dedicated size sensor models:** DnaA accumulation-based mechanisms (Moolman et al., 2014)

- **Coupled timer models:** Replication-division coupling without explicit size sensing (Si & Levin, 2020; Wallden et al., 2016)

- **Emergent behavior models:** Adder emerges from noisy growth dynamics without dedicated machinery (Witz et al., 2019; Si et al., 2019)

The conceptual distinction between "dedicated size sensor" vs. "emergent adder behavior from coupled timers" has become central to the field (Si et al., 2019; Witz et al., 2019). Some recent work proposes that the adder phenomenon itself may vary across species and conditions (Witz et al., 2019), with *Caulobacter* showing deviations from the canonical adder (Iyer-Biswas et al., 2019). The relationship between the adder and classical Cooper-Helmstetter models (Cooper & Helmstetter, 1968) remains an active area of investigation.

Limitations:

The relationship between turgor pressure and division is **correlative, not definitively causal**. Key limitations:

- Osmotic manipulation effects cannot be cleanly separated from metabolic effects

- Different bacteria maintain different turgor pressures but follow similar division scaling laws
- The adder phenomenon itself has multiple competing mechanistic explanations
- Causal experiments directly manipulating turgor pressure while measuring division outcomes are technically challenging

Current Assessment:

Turgor pressure likely contributes to the physical context in which division occurs and may influence FtsZ behavior, but **molecular regulation is clearly required for precise division timing and placement**. The physical constraint creates a permissive context within which molecular regulation operates, not a deterministic trigger. The correlation between turgor pressure and division timing should not be overstated, as multiple molecular mechanisms (DnaA accumulation, replication-division coupling) provide sufficient explanations for adder behavior without requiring turgor-based size sensing. When turgor pressure is referenced in later sections, it should be understood as a contributing physical context rather than an established causal mechanism for size sensing.

2.2 Nucleoid Occlusion: Geometric Constraints on Division

Physical Basis:

The bacterial chromosome (nucleoid) occupies a substantial fraction of cell volume (~15% in *E. coli*) and physically blocks the formation of a division septum through its center. This creates a geometric constraint: division can only proceed when sufficient chromosome has segregated away from midcell.

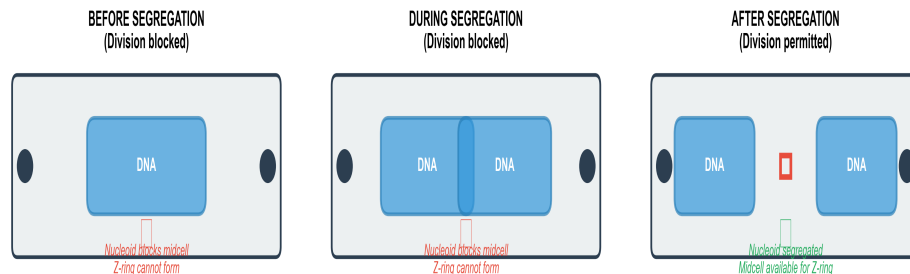
Molecular Implementations:

Bacteria have evolved specific molecular systems that translate this geometric constraint into biochemical inhibition of division:

- **SlmA (*E. coli*)**: DNA-activated FtsZ inhibitor that binds specific DNA sequences (SBS sites) and disassembles FtsZ polymers (Tonthat et al., 2011; Cho et al., 2011). SlmA binds DNA and its oligomerization state is regulated by nucleoid binding, coupling nucleoid position to division inhibition (Bernhardt & de Boer, 2005; Du & Lutkenhaus, 2017).
- **Noc (*B. subtilis*)**: Spatial regulator that prevents Z-ring formation over nucleoid through DNA-binding; unlike SlmA, Noc's mechanism involves direct membrane association (Adams et al., 2015; Wu & Errington, 2012)
- **Min system**: Works with nucleoid occlusion to ensure Z-ring forms only at the correct geometric position (Raskin & de Boer, 1999; Hu & Lutkenhaus, 1999)

Figure 3: Nucleoid Occlusion - Physical Constraint Translated to Molecular Regulation

Nucleoid Occlusion: Physical Constraint Translated to Molecular Regulation



MOLECULAR SENSING & RESPONSE: Physical Constraint (nucleoid at midcell) → Molecular Detection (SlmA/Noc/Min) → Functional Outcome (division at proper time/place)

Nucleoid Occlusion

This figure illustrates nucleoid occlusion as an example of how physical constraints (geometry - nucleoid occupying midcell space) and molecular systems (SlmA/Noc/Min) are **reciprocally coupled**. Recent work (Mäkelä et al., 2023; Valenzuela et al., 2023) demonstrates that nucleoid positioning is actively maintained by molecular processes (transcription, translation), while SlmA/Noc systems respond to nucleoid position to regulate division. This creates **bidirectional coupling** rather than pure hierarchical override: molecular systems translate geometric constraints into division control, but the geometric state itself is actively maintained by molecular activity. Nucleoid occlusion represents a **nuanced case** exhibiting features of both Type A (hierarchical response to physical state) and Type B (bidirectional coupling) depending on timescale and perspective.

The Min System: An Ambiguous Case Study:

The Min oscillation system illustrates the complexity of classifying physical-molecular coupling. Min's mechanism remains experimentally unresolved and, like nucleoid occlusion, represents a case where simple hierarchical classification is challenging:

Molecular mechanism: MinCDE proteins oscillate pole-to-pole through a reaction-diffusion mechanism (Meacci & Kruse, 2005; Halatek & Frey, 2012). MinD binds ATP and membrane, recruits MinC (division inhibitor), and MinE stimulates MinD-ATP hydrolysis, creating spatial oscillations.

Two competing interpretations:

Interpretation A - Active geometric sensing: Min proteins actively read cell geometry and adjust oscillation accordingly. If correct, this would support the hierarchical framework as molecular systems detecting physical states.

Interpretation B - Passive reaction-diffusion: Oscillation period emerges **PASSIVELY** from reaction-diffusion dynamics in confined volumes (Halatek & Frey, 2012). Cell geometry sets boundary conditions, but there is no active "sensing" in any meaningful sense. If correct, this represents physical constraints **ACTING ON** molecular behavior—the opposite of the framework's primary claim.

Experimental status: The active vs. passive question remains debated (Di Ventura & Sourjik, 2022; Lutz et al., 2023; Ramm et al., 2019). We return to Min as a detailed case study in Section 8.2, where we propose experiments to distinguish these mechanisms.

Important caveat: Unlike the SOS checkpoint (which represents a clearer case of hierarchical molecular override during stress response), nucleoid occlusion exhibits bidirectional coupling (Mäkelä et al., 2023; Valenzuela et al., 2023) and Min does not clearly demonstrate hierarchical organization. Both nucleoid occlusion and Min may represent important boundary cases or counterexamples to pure hierarchical organization.

Recent Evidence: Bidirectional Coupling Between Geometry and Molecular Processes

Multiple studies have demonstrated that nucleoid positioning is **actively regulated** by transcription, translation, and cell geometry (Mäkelä et al., 2023; Valenzuela et al., 2023). These findings reveal a more complex picture than simple "geometric constraint → molecular response" unidirectionality:

Mäkelä et al. (2023) showed that:

- Transcription activity actively drives nucleoid expansion and positioning
- Inhibiting translation causes nucleoid compaction, demonstrating molecular → physical causation
- Nucleoid organization is maintained by active molecular processes, not passive physical confinement

Valenzuela et al. (2023) demonstrated that:

- Cell geometry actively influences nucleoid dynamics through membrane-chromosome tethering
- Physical context (cell shape, size) and molecular regulation are continuously coupled
- The relationship is best described as reciprocal regulation, not hierarchical override

Implications for Classification: These findings challenge a straightforward classification of nucleoid occlusion as "Type A" (hierarchical, asymmetric). While SlmA/Noc systems do translate nucleoid position into division inhibition (molecular response to physical state), the physical state (nucleoid position) is itself actively maintained by molecular processes (transcription, translation). This creates a **reciprocal coupling loop**:

1. Physical state (nucleoid position) → Molecular response (SlmA/Noc inhibit division)
2. Molecular activity (transcription, translation) → Physical state (nucleoid organization)

Context-Dependent Classification: Nucleoid occlusion may represent a **context-dependent case** that exhibits both Type A and Type B characteristics depending on timescale and perspective:

- **Short-timescale view** (seconds to minutes): Molecular systems (SlmA/Noc) respond to nucleoid position, appearing as hierarchical molecular override of physical constraints (Type A-like)
- **Long-timescale view** (minutes to hours): Nucleoid position is actively maintained by molecular processes (transcription, translation), creating bidirectional coupling (Type B-like)
- **Integrated view:** The system exhibits **reciprocal regulation** where neither physical nor molecular dominates exclusively

Comparison with DNA Supercoiling: Nucleoid occlusion shares similarities with DNA supercoiling (Section 2.3), where topoisomerase activity manages physical torsional stress while replication actively generates supercoiling. Both represent **bidirectional coupling** rather than pure hierarchical control.

Testable Prediction:

Larger chromosomes should proportionally delay division due to increased geometric occlusion, regardless of specific molecular regulators. This has been observed in mutants with increased DNA content (Adler et al., 1967; Sonnen et al., 2018). **However, this prediction does not distinguish between hierarchical (Type A) and bidirectional (Type B) coupling mechanisms**—both mechanisms would produce the same phenotypic outcome.

Current Assessment:

Nucleoid occlusion represents a **nuanced case** where physical constraints (geometry) and molecular regulation (SlmA/Noc/Min, transcription, translation) are **inseparable and reciprocally coupled**. Unlike the SOS checkpoint—which represents a clearer case of hierarchical molecular override during stress response—nucleoid occlusion demonstrates **continuous bidirectional regulation**:

- **Direction 1 (Physical → Molecular):** Nucleoid position detected by SlmA/Noc → division inhibited (Type A-like)
- **Direction 2 (Molecular → Physical):** Transcription/translation activity → nucleoid organization maintained (Type B-like)

Conclusion: Nucleoid occlusion is **best classified as a bidirectionally coupled system (Type B-like)** with context-dependent hierarchical features during division timing decisions. It should NOT be presented as a straightforward example of hierarchical molecular override (Type A). The system demonstrates how physical and molecular processes can be continuously coupled while still producing precise functional outcomes.

2.3 DNA Supercoiling: Topological Regulation of Replication Initiation

Physical Basis:

DNA supercoiling—overwinding or underwinding of the DNA helix—creates torsional stress that affects the accessibility of oriC to DnaA and other factors. Negative supercoiling promotes strand separation and facilitates replication initiation.

Evidence:

- **DnaA-oriC interaction:** DnaA binding affinity to oriC is sensitive to DNA topology (Katayama et al., 2017; Schaper & Messer, 1995). Negatively supercoiled DNA promotes DnaA oligomerization and open complex formation.

- **Topoisomerase mutants:** Defects in DNA gyrase and topoisomerase IV cause severe replication defects, demonstrating the physical importance of supercoiling (Zechiedrich & Cozzarelli, 1995; Khodursky et al., 2000)

- **Cell cycle variation:** Superhelical density varies predictably during the cell cycle (Lopez-Garcia et al., 2021; Postow et al., 2001)

- **Environmental sensing:** DNA supercoiling serves as a global regulator that integrates environmental signals (Dorman, 2019). Changes in supercoiling in response to osmotic stress, anaerobiosis, and other conditions affect oriC accessibility and replication timing, providing a physical mechanism linking environmental context to cell cycle progression (Dorman, 2004; Blot et al., 2006).

- **Local vs. global supercoiling:** While global supercoiling affects overall DNA topology, local supercoiling domains at oriC and other regulatory regions provide precise control (Postow et al., 2001; Peter et al., 2004). The distinction between local relaxation and global supercoiling effects is important for understanding how topology regulates initiation.

- **Transertion hypothesis (with critical evaluation):** The coupling of transcription, translation, and translocation (transertion) has been proposed to create forces that affect nucleoid organization and DNA supercoiling (Norris et al., 2007). **Connection to hierarchical framework:** If correct, this would represent a case where molecular activity (transcription, translation) actively alters physical constraints (nucleoid organization, supercoiling), demonstrating the molecular → physical direction of our hierarchical model. **However, this hypothesis requires critical evaluation:** The transertion hypothesis remains controversial, and strong in vivo quantitative support is lacking. The primary evidence comes from theoretical modeling and indirect in vitro observations. Direct measurement of transertion-generated forces in living cells has proven technically challenging, and alternative explanations for nucleoid organization (e.g., entropic forces, SMC-mediated compaction) cannot be excluded. We present this as an intriguing but not yet conclusively demonstrated mechanism that would, if validated, support our hierarchical framework.

Bidirectional Causality:

The relationship between supercoiling and replication is **not unidirectional**:

- DnaA and the replisome actively recruit topoisomerases to modulate local supercoiling (Peter et al., 1998; Espey & Chattoraj, 2006)

- Replication itself generates positive supercoiling ahead of the fork and negative supercoiling behind (Liu & Wang, 1987)

- Topoisomerase activity that determines supercoiling density is itself regulated by DnaA and transcriptional programmes

- This creates feedback loops where molecular regulation actively manages the physical parameter

Current Assessment:

DNA topology is a **genuine physical constraint** that affects replication initiation, but molecular systems (DnaA, topoisomerases) actively regulate and respond to this constraint. In rapidly growing *E. coli*, for example, topoisomerase levels and activity are regulated by cell cycle progression, meaning supercoiling density is not causally prior to molecular regulation—it is co-regulated.

Important Note on Framework Applicability:

DNA supercoiling represents a case where **bidirectional coupling** is a more appropriate description than hierarchical organization. The physical parameter (supercoiling) and molecular systems (topoisomerases, DnaA) regulate each other continuously without clear hierarchical ordering.

Functional Requirement Analysis: During rapid DNA replication, topoisomerase activity must continuously adjust to relieve positive supercoiling ahead of the fork and introduce negative supercoiling behind the fork. This **speed-critical functional requirement** suggests that DNA supercoiling **SHOULD** exhibit bidirectional coupling.

Prediction: Based on this speed-critical functional requirement, we PREDICT that DNA supercoiling should exhibit Type B (bidirectional) coupling with $AsI \approx 1$.

Observation: Consistent with this prediction, literature shows mutual regulation between supercoiling and topoisomerases (Katayama et al., 2017). However, definitive validation requires direct AsI measurement.

This example illustrates how functional requirements can predict coupling type: the speed-critical nature of replication supercoiling management predicts bidirectional coupling, which is consistent with (but not proven by) observations. Rather than an exception to the framework, DNA supercoiling illustrates that the framework can accommodate multiple coupling modes depending on functional context. We present this as a **case study supporting the speed-critical $\rightarrow$ bidirectional hypothesis**, not as definitive proof or as an established classification.

2.3.1 MreB and Cell Shape Maintenance

Why Include MreB in a Cell Cycle Framework?

MreB is primarily a cell shape maintenance system rather than a core cell cycle regulator like FtsZ, DnaA, or SMC complexes. However, we include MreB in this discussion because:

1. **Shape-coupling to division:** Cell elongation (driven by MreB) must be coordinated with septation (driven by FtsZ) to achieve proper cell cycle progression
2. **Size checkpoint:** Cell size at division depends on the integrated outcome of both elongation and division processes
3. **Physical-molecular interface:** Like the other systems discussed, MreB illustrates how molecular systems interface with physical constraints (geometry, curvature)

Important caveat: MreB is not strictly a cell cycle regulator in the same sense as FtsZ or DnaA. Its inclusion here is justified by its role in coordinating cell growth with division, creating the physical context in which cell cycle decisions are made.

Physical Basis:

MreB is an actin-like bacterial cytoskeletal protein that forms circumferential filaments underneath the cell membrane in rod-shaped bacteria. MreB filaments align along regions of greatest principal curvature (negative Gaussian curvature), creating a feedback loop between cell shape and peptidoglycan synthesis.

Evidence:

- **Curvature sensing:** MreB preferentially localizes to regions of negative Gaussian curvature (cell sides, not poles or division sites) in both *E. coli* and *B. subtilis* (Ursell et al., 2014; van Teeffelen et al., 2017). This curvature sensitivity is thought to arise from geometric confinement and filament mechanics rather than explicit curvature sensors (Hussain et al., 2018).
- **Reconstitution:** MreB dynamics can be reconstituted in vitro and show curvature-dependent localization (Salje et al., 2011; Shi et al., 2018).
- **Cell shape coupling:** MreB recruits peptidoglycan synthesis machinery to regions of active cell wall growth (Domínguez-Escobar et al., 2011; van Teeffelen et al., 2017). Disruption of MreB leads to loss of rod shape and formation of spherical cells (Jones et al., 2001).
- **Molecular mechanism:** Recent work shows MreB forms dynamic patches that move processively around the cell circumference, coordinating peptidoglycan insertion for cell elongation (van den Ent et al., 2006; Olshausen et al., 2013; Furchtgott & Huang, 2020). Single-molecule studies (Whitley et al., 2021) revealed that MreB treadmill dynamics are tightly coupled to cell wall synthesis machinery.

Integration with Hierarchical Framework:

MreB represents a clear example of molecular systems operating within and managing physical constraints:

- **Physical constraint:** Cell geometry determines where MreB can localize (via curvature sensing)
- **Molecular override:** MreB actively directs peptidoglycan synthesis, overriding default expansion patterns

- **Feedback:** MreB-directed growth alters cell geometry, which in turn affects MreB localization

This system is well-characterized and provides strong support for the general thesis that molecular systems interface with physical constraints. However, the directionality of information flow is less clear than in other systems—MreB both responds to geometry and actively shapes it, making this more similar to bidirectional coupling than hierarchical override.

2.4 Macromolecular Crowding and Phase Separation (Speculative)

Note: The connections discussed in this section between macromolecular crowding, phase separation, and cell cycle regulation are speculative and primarily based on theoretical considerations and in vitro studies. Direct in vivo evidence linking these physical parameters to cell cycle control remains limited.

Physical Basis:

The bacterial cytoplasm is a crowded environment where macromolecules occupy 20-40% of volume. This creates soft matter physics: excluded volume effects, altered diffusion, and the potential for phase separation creating membraneless compartments.

Evidence:

- **Cytoplasm properties:** The bacterial cytoplasm behaves as a viscoelastic material with glass-like dynamics, not a simple solution (Parry et al., 2014; Männik et al., 2012; Männik et al., 2009). **Importantly, Parry et al. (2014) showed that the glass-like cytoplasmic transition is METABOLICALLY MAINTAINED**, implying that it is under molecular regulation rather than being a purely physical effect. This **SUPPORTS** the hierarchical framework: even cytoplasmic physical properties are actively controlled by molecular systems. Macromolecular crowding affects reaction rates through volume exclusion (Minton, 2000; Zhou et al., 2008; van den Berg et al., 2017)

- **Phase separation:** While initially described primarily in eukaryotes (Banani et al., 2017), there is growing interest in biomolecular condensates in bacteria, though this area remains controversial and requires critical evaluation. Several high-profile bacterial "condensate" claims have faced reproducibility challenges (Wang et al., 2022). PopZ forms polymeric structures at the Caulobacter cell pole that exhibit some condensate-like properties (Lasker et al., 2022; Bowman et al., 2019). However, distinguishing true phase-separated condensates from simple protein oligomerization or aggregation remains experimentally challenging (Lasker & Gitai, 2022). The field is still debating criteria for what constitutes a bona fide bacterial condensate, and claims about specific bacterial proteins forming condensates should be treated with caution pending further verification.

- **Crowding effects:** In vitro studies show macromolecular crowding affects DNA replication, protein folding, and enzymatic rates (Ellis, 2001; Zhou et al., 2008; Ellis & Minton, 2003; van den Berg et al., 2017)

Connection to Cell Cycle:

The hypothesis is that as cells grow and macromolecular concentration increases, reaching a critical crowding threshold could:

1. Slow biosynthetic rates (signaling resource limitation)
2. Trigger phase separation events that organize cellular components
3. Create physical conditions favorable for division

Limitations:

Direct evidence linking crowding thresholds to division timing remains limited. Most work is in vitro or theoretical. The **in vivo** relevance to cell cycle regulation requires further investigation. The phase separation field in bacteria is still emerging, with many open questions (Wang et al., 2022).

Current Assessment:

Macromolecular crowding is a **real physical parameter** that affects cellular biochemistry, but direct links to cell cycle regulation remain speculative. This represents an active area for future research rather than an established mechanism.

2.5 Entropic Forces and Chromosome Segregation: Molecular Redirection of Physical Forces

Physical Basis:

Polymers in confined spaces experience entropic forces—the tendency to maximize their conformational entropy. Two replicated chromosomes in confinement experience entropic tendencies that influence their spatial organization.

Historical Context:

Jun & Mulder (2006) proposed that entropic forces alone could drive chromosome segregation—a hypothesis that was both influential and controversial. These theoretical and modeling studies were important in demonstrating that physical forces contribute to chromosome organization, challenging purely molecular views of segregation.

Current Understanding (ongoing debate):

The field has developed substantially since 2006, with multiple groups contributing to a more nuanced understanding:

- **Jun laboratory's continued work:** The Jun lab has continued to refine entropic segregation models, incorporating more realistic cellular conditions and polymer physics (Jun lab publications 2008-2023). Their current position emphasizes that entropic effects are significant and must be accounted for, even if they cannot fully explain segregation in isolation.

- **SMC complex research:** Multiple groups have shown that loop-extruding proteins (SMC complexes, FtsK) actively alter chromosome topology (Bürmann et al., 2023; Badrinarayanan et al., 2022; Sivanathan et al., 2022; Wang et al., 2023). These machines "work with" entropic forces rather than replacing them.

- **Integration rather than refutation:** The current consensus, articulated in reviews like Badrinarayanan et al. (2022), is that entropic forces create baseline tendencies that SMC complexes actively redirect. This is not a refutation of Jun & Mulder (2006) but an extension: entropic forces are necessary but not sufficient for productive segregation.

The question of how much of chromosome segregation can be explained by entropic forces alone remains an active area of investigation. The Jun laboratory continues to argue for stronger entropic contributions than some other groups, making this an ongoing scientific debate rather than a settled issue.

Implications for the Hierarchical Framework:

Chromosome segregation exemplifies our hierarchical framework:

- **Physical foundation:** Entropic forces create baseline tendencies in chromosome organization

- **Molecular override:** SMC complexes and other molecular machines actively redirect these forces to achieve productive outcomes

- **Integrated outcome:** Neither physics nor molecules alone suffice—the system requires both layers working together

Current Assessment:

Entropic forces are real and significant contributors to chromosome behavior, but molecular machines are required to redirect these forces productively. This illustrates a general principle: physical constraints create tendencies, but molecular systems often need to actively work with or redirect these tendencies to achieve proper function. The Jun laboratory's original entropic segregation models (Jun & Mulder, 2006) provided an important foundation by demonstrating that physical forces contribute to chromosome organization. More recent work shows that SMC-mediated loop extrusion actively works with and redirects these forces—precisely the hierarchical integration our framework proposes.

Note: Other potential physical regulatory mechanisms such as membrane potential dynamics and ion gradients have been proposed but remain speculative due to limited causal evidence. Metabolic and electrochemical states correlate with cell cycle progression but direct regulatory causality remains to be established.

3. Molecular Regulation: Essential and Sophisticated

Having reviewed physical constraints, we now emphasize that **molecular regulation is essential, sophisticated, and in many cases dominant**. Physical constraints create context; molecular systems provide

precise control.

3.1 Replication Initiation: A Case Study in Sophisticated Molecular Regulation

DnaA: Beyond Simple Physical Response

DnaA is far more than a passive responder to DNA supercoiling. The level of molecular sophistication is remarkable:

- **ATP/ADP binding:** Active DnaA-ATP vs. inactive DnaA-ADP creates a molecular switch (Sekimizu et al., 1987; Saxena et al., 2015; Nishida et al., 2022)
- **RIDA (Regulatory Inactivation of DnaA):** Hda protein promotes DnaA-ATP hydrolysis when it interacts with the DnaN β -clamp at active replication forks (Katayama et al., 1998; Kono & Katayama, 2021; Kato & Katayama, 2001; Nishida et al., 2002; Camara et al., 2021). More precisely, Hda interacts with the clamp-loader-associated form of the β -clamp at the replication fork, coupling DnaA inactivation to ongoing replication elongation rather than merely to clamp loading at origins (Kono & Katayama, 2021). This mechanism couples DnaA inactivation specifically to active replication forks, providing temporal precision that "replication progression" alone cannot explain.
- **datA locus:** Titration site that binds DnaA and regulates availability (Kitagawa et al., 1998; Ogawa et al., 2002)
- **DARS (DnaA-reactivating sequences):** Promote ADP-ATP exchange to reactivate DnaA (Fujimitsu et al., 2009; Kasho et al., 2020)
- **DiaA:** Promotes DnaA oligomerization at oriC (Ishida et al., 2004; Keyamura et al., 2007)
- **SeqA and epigenetic regulation:** SeqA binds hemi-methylated GATC sites at oriC immediately after replication, preventing DnaA from binding and blocking re-initiation (Landoulsi et al., 2021; Waldminghaus et al., 2012). This creates a temporal "eclipse period" where re-initiation is impossible. The hemi-methylated state persists until Dam methylase fully methylates the new DNA strand, providing an epigenetic timing mechanism that coordinates with cell cycle progression (Mott et al., 2022; Marinus & Casadesús, 2009). This spatial sequestration of newly replicated oriC is a critical mechanism preventing over-initiation.
- **Dam methylase:** Controls methylation state of oriC, regulating timing through SeqA binding (Mott et al., 2022; Marinus & Casadesús, 2009)
- **Ihf and HU:** Nucleoid-associated proteins affecting oriC accessibility (Ryan et al., 2002; Xiao et al., 2021)

Synthesis: What This Complexity Implies

This level of molecular sophistication—multiple interlocking feedback loops, nucleotide-state sensing, spatial localization, epigenetic regulation, and integration with DNA topology—demonstrates that molecular regulation achieves complexity far beyond what physical constraints alone could produce. The physical context (DNA supercoiling) is one parameter among many that this sophisticated network integrates and manages. The RIDA mechanism specifically, with its requirement for loaded β -clamp at active forks, exemplifies the kind of precise molecular coupling that cannot be reduced to simple "physical responses."

3.2 Division Septum Formation: Molecular Precision on Physical Context

FtsZ Dynamics: Treadmilling and Spatial Organization

Recent advances in single-molecule imaging have revealed sophisticated FtsZ behaviors at the single-filament level:

- **FtsZ treadmilling:** Individual FtsZ filaments exhibit treadmilling dynamics—adding subunits at one end while losing them from the other—creating continuous turnover while maintaining overall Z-ring structure (Bisson-Filho et al., 2017; Whitley et al., 2021). This treadmilling is coupled to peptidoglycan synthesis machinery, with FtsW and other proteins following the treadmilling pattern (Yang et al., 2017).
- **Single-molecule tracking:** Whitley et al. (2021) used photoactivatable FtsZ to show that individual filaments move processively around the division plane with directional persistence, suggesting an organized mechanism rather than random diffusion.

- **Coupling to cell wall synthesis:** FtsZ treadmilling is coordinated with septal peptidoglycan synthesis through interactions with FtsW (PBP2B) and other transglycosylases (Yang et al., 2017; Bisson-Filho et al., 2017).

- **Membrane curvature effects (in vitro):** FtsZ polymerization responds to membrane curvature in artificial membrane systems (Loose & Mitchison, 2014), though this is based on in vitro reconstitution experiments and the in vivo relevance remains contested. Bisson-Filho et al. (2017) showed that FtsZ treadmilling occurs at the membrane and is coupled to cell wall synthesis, but whether this represents active mechanosensing or passive biophysical effects in living cells is debated. We present this with the same epistemic caution applied to turgor pressure—the evidence is intriguing but not yet definitive for in vivo function.

These findings reveal a level of molecular organization and dynamics at the division septum that goes beyond earlier models of static Z-rings. The treadmilling behavior creates a dynamic interface between molecular regulation and physical context—the continuously remodeling Z-ring both responds to and modifies the physical environment of the division plane.

FtsZ and the Divisome

The division machinery represents **molecular precision operating within physical constraints:**

- **Z-ring positioning:** MinCDE oscillations and nucleoid occlusion provide molecular positioning (Raskin & de Boer, 1999; Bernhardt & de Boer, 2005)

- **Divisome assembly:** Ordered recruitment of 10+ proteins creates a sophisticated molecular machine (Ghosal et al., 2021; Du & Lutkenhaus, 2017)

- **Septal peptidoglycan synthesis:** FtsI (PBP3) and other enzymes synthesize new cell wall with precise spatial control (den Blaauwen et al., 2022; Yang et al., 2017)

SOS Checkpoint - Molecular Regulation Dominating Physical Context:

When DNA damage occurs, the SOS response induces Sula (SfiA in *B. subtilis*), which **directly inhibits FtsZ polymerization** (Huisman & D'Ari, 1983; Roehm et al., 2022). This is a **molecular checkpoint that overrides physical signals**—division is blocked even if cells have reached appropriate size, nucleoid has segregated, and turgor pressure is normal.

Key Point: The SOS checkpoint demonstrates that molecular regulation can **dominate** over physical context when cellular conditions demand it.

3.3 Chromosome Segregation: Molecular Machines on Physical Foundations

ParA/ParB Systems

ParA/ParB constitute an **active positioning system:**

- ParB binds parS sites and forms partition complexes (Rodionov et al., 2021)

- ParA forms dynamic gradients that actively position chromosomes (Le Gall et al., 2022)

- This is **active transport**, not passive physical segregation (Le Gall et al., 2022; Lloyd et al., 2023)

SMC Complexes

Structural Maintenance of Chromosomes complexes are essential:

- **MukBEF (*E. coli*) / Smc-ScpAB (*B. subtilis*):** ATP-dependent chromosome organization and segregation (David et al., 2022; Nollivos et al., 2022; Bürmann et al., 2023; Yatskevich et al., 2022)

- As discussed in Section 2.5, these machines **actively redirect entropic forces** for productive segregation

Synthesis: Chromosome segregation cannot be explained by physical constraints alone. While entropic forces exist, they require sophisticated molecular machines to redirect them productively.

4. Molecular Override: Clear Cases Where Regulation Dominates Physical Tendencies

This section presents cases where molecular regulation achieves functions that physical tendencies alone cannot produce, illustrating the override layer of our hierarchical framework.

4.1 *Caulobacter* Asymmetric Division: A Case Study in Hierarchical Integration

The Phenomenon:

Caulobacter crescentus undergoes **asymmetric division**, producing:

- A stalked cell (immediately replication-competent, surface-attached)
- A swarmer cell (must differentiate before replicating, motile)

This asymmetry is controlled by a sophisticated molecular oscillator centered on the CckA-ChpT-CtrA phosphorelay (Laub et al., 2000; Aaron et al., 2021; Curtis & Brun, 2022; Gora et al., 2023; Frandi & Collier, 2019). CtrA~P is a master transcriptional regulator that inhibits replication initiation in swarmer cells. The phosphorelay includes:

- **CckA**: A bifunctional histidine kinase/phosphatase whose activity is modulated by DivL
- **DivL**: A pseudokinase that senses cell cycle cues and regulates CckA activity
- **CpdR**: A response regulator that, when DEPHOSPHORYLATED, targets CtrA for degradation
- **RcdA**: Adaptor protein that delivers CtrA to the protease when CpdR is active (Smith et al., 2019; Curtiss & Brun, 2022)
- **DivJ and PleC**: Localization-specific kinases/phosphatases that create spatial asymmetry
- **DivK**: A response regulator that integrates inputs from DivJ and PleC

CtrA Degradation and Spatial Proteolysis:

A critical aspect of *Caulobacter* asymmetry is the spatially restricted degradation of CtrA by the ClpXP protease:

- **ClpXP localization**: ClpXP protease localizes specifically to the stalked pole through interaction with PopZ and other factors (Smith et al., 2019; Curtiss & Brun, 2022)
- **CpdR-RcdA system**: When DEPHOSPHORYLATED, CpdR binds RcdA and the complex recruits CtrA to ClpXP for degradation (Smith et al., 2019; Curtiss & Brun, 2022). Phosphorylation of CpdR by CckA inhibits this interaction, allowing CtrA to accumulate. This phospho-regulation creates a cell cycle-dependent switch controlling CtrA stability.
- **Spatial restriction**: This degradation system operates primarily at the stalked pole, creating spatial asymmetry in CtrA levels (Curtiss & Brun, 2022)
- **Functional consequence**: Swarmer poles maintain high CtrA levels, inhibiting replication, while stalked poles maintain low CtrA through localized proteolysis

This elegant mechanism—where protease localization creates spatial asymmetry—provides a concrete molecular implementation of the hierarchical framework's concept of molecular override achieving functional outcomes that physical constraints alone cannot produce.

Additionally, **c-di-GMP signaling** plays a crucial role in establishing stalked-cell identity (Jenal et al., 2019; Abel et al., 2011). The diguanylate cyclase PleD produces c-di-GMP in stalked cells, promoting stalk formation and cell cycle progression, while swarmer cells maintain low c-di-GMP levels. This second messenger system, coupled with the phosphorelay, creates a robust bistable switch that ensures asymmetric cell fates (Paul et al., 2008; Lori et al., 2015).

Physical Aspects of Asymmetry:

Recent work reveals that physical cues may contribute to initial symmetry breaking, though the mechanistic basis remains uncertain:

- **PopZ localization**: PopZ forms polymeric structures at cell poles. Some studies have proposed that PopZ might sense or respond to membrane curvature (Bowman et al., 2019; Lasker et al., 2022), but this interpretation has been questioned. Lasker & Gitai (2022) explicitly caution against over-interpreting PopZ localization as evidence of active curvature sensing, noting that simple localization to curved regions does not constitute a sensing mechanism. The question of whether PopZ actively responds to curvature versus simply

accumulating at curved sites due to other physical properties remains unresolved.

- **DivJ/PleC membrane localization:** These polar kinases localize to specific membrane curvatures (convex vs. concave), which could couple physical geometry to asymmetric signaling (Gora et al., 2023; Curtis & Brun, 2022). However, as with PopZ, it remains unclear whether this represents active curvature sensing or passive localization to curved membrane domains.

- **Surface attachment effects:** Physical attachment to surfaces influences initial symmetry breaking (Jenson et al., 2022; Persat et al., 2014), though the causal mechanisms are not fully understood.

- **Stalked pole distinctiveness:** The stalked pole has distinct membrane composition (Gora et al., 2023) and cell wall synthesis machinery localization (Kuru et al., 2017), suggesting biochemical asymmetry may precede or accompany physical asymmetry.

Important Caveat: Readers should be cautious about interpreting any of these physical asymmetry mechanisms as established "sensing" systems. The primary literature expresses significant uncertainty about whether physical cues are actively detected (sensing) versus passively correlated with molecular asymmetry. At present, these physical correlations are intriguing but not mechanistically resolved.

Hierarchical Integration:

Caulobacter illustrates our hierarchical framework:

- **Physical foundation:** Membrane curvature, pole geometry, and surface attachment create initial biases

- **Molecular amplification:** The CckA-ChpT-CtrA phosphorelay and c-di-GMP systems detect and amplify these biases

- **Override outcome:** Developmental asymmetry emerges from interaction of physical and molecular layers

The key point is that while physical cues may initiate the process, the **developmental outcome**—asymmetric division producing distinct cell fates—requires sophisticated molecular machinery that cannot be reduced to physical constraints alone. Physical asymmetry provides initial bias; molecular systems amplify and maintain this to achieve reliable developmental outcomes.

4.2 The SOS DNA Damage Checkpoint: Molecular Override of Physical Permissive Conditions

As discussed in Section 3.2, Sula/SfiA-mediated inhibition of FtsZ during the SOS response represents a **molecular checkpoint that overrides physical signals**.

Division is blocked by molecular mechanism even when:

- Cell size is appropriate for division
- Nucleoid is fully segregated
- Turgor pressure is normal
- All physical conditions would permit division

This demonstrates that molecular regulation can **dominate** over physical context when cellular conditions demand it.

4.3 Multifork Replication: Molecular Precision Beyond Physical Limits

In fast-growing *E. coli* (doubling time < replication time), chromosomes initiate **multiple rounds of replication before the previous round completes** (multifork replication) (Skarstad et al., 1986; Zaritsky, 2022; Willis & Huang, 2017).

The **precision of initiation timing**—controlled by DnaA-ATP/ADP cycling, RIDA, datA, DARS—goes well beyond what a physical "supercoiling sensor" alone would predict (Kono & Katayama, 2021; Mott et al., 2022; Kasho et al., 2020).

5. Origins and Evolution: Implications and Limitations

5.1 Evolutionary Inferences: What Can We Reasonably Claim?

Epistemological Limitations:

Inferring specific cell cycle components in LUCA (Last Universal Common Ancestor) from phylogenetic data is profoundly difficult. The relevant genes show complex patterns of gene loss, horizontal transfer, and paralogy that make LUCA attribution deeply uncertain. Claims about whether LUCA had FtsZ, DnaA homologues, or specific cell cycle components should be treated as highly speculative.

What We Can Say:

- **Established:** Physical constraints (geometry, topology, mechanical forces) affect cell cycle processes in modern bacteria
- **Established:** Molecular systems have evolved to detect and respond to these constraints
- **Speculative:** The specific evolutionary sequence by which these capabilities emerged
- **Not addressable with current data:** Whether early protocells relied primarily on physical defaults versus had early molecular regulation

Alternative Evolutionary Trajectories (Both Speculative):

The evolutionary relationship between physical constraints and molecular regulation in early cell cycles is fundamentally uncertain. Multiple scenarios are consistent with current data:

Scenario A: Physical-First, Molecular-Layers-Added (Traditional Hierarchical View)

- Early protocells relied heavily on physical constraints for division (e.g., physical scission, membrane curvature-driven division)
- Molecular regulation evolved later as an overlay on physical processes
- Molecular systems initially simplified physical processes, then gradually assumed control
- **Evidence cited:** JCVI-syn3.0 minimal cells divide with high variability, suggesting physical tendencies dominate when molecular complexity is reduced
- **Limitations:** This interpretation is confounded by syn3.0's abnormal morphology, slow growth, and pleiotropic defects

Scenario B: Co-Evolution from Early Stages (Alternative View)

- Physical constraints and molecular regulation co-evolved from the earliest stages of cellular life
- Even minimal protocells may have required significant molecular infrastructure to achieve functional division
- Physical and molecular systems were interdependent from the beginning, not sequentially layered
- **Evidence from synthetic cell biology:** Bottom-up approaches demonstrate that achieving controlled, reproducible division in synthetic systems requires sophisticated molecular organization
- **Key insight:** The "minimal" requirements for functional cell cycles may be much higher than suggested by simple physical-first scenarios

Synthetic Cell Biology Results Relevant to Evolutionary Claims:

Recent work in synthetic cell biology and bottom-up origins research provides critical insights that challenge simple physical-first narratives:

Minimal Cell Requirements:

- **JCVI-syn3.0** (Hutchison et al., 2016; Pelletier et al., 2021): Despite having only 473 genes, syn3.0 requires ~19 cell cycle genes (CCGC $\approx$ 19) for division. Even this "minimal" system has substantial molecular regulatory complexity.
- **Key observation:** syn3.0 shows high division variability (CV $\approx$ 0.35-0.45) compared to wild-type *E. coli* (CV $\approx$ 0.10-0.15). However, this does NOT necessarily mean "physical defaults are exposed"—alternative explanations include pleiotropic defects, geometric constraints (spherical vs rod-shaped), and growth rate effects (3-hour doubling time).

Bottom-Up Synthetic Cell Approaches:

- **Liposome division systems** (Hanczyc et al., 2003; Zhu et al., 2021): Physical scission of lipid vesicles can occur without molecular machinery, but achieving reproducible division timing and placement requires sophisticated molecular control systems

- **Gene-reconstituted systems** (Garcia et al., 2021; Pineda et al., 2022): Reconstituting cell division in vitro requires coordinated activity of multiple molecular components (FtsZ, FtsA, ZipA, membrane-targeting sequences)

- **Key insight:** While physical processes CAN drive division without molecular regulation, achieving PRECISE and REPRODUCIBLE division appears to require molecular control even in minimal systems

Implications for Evolutionary Narratives:

The synthetic cell biology literature suggests that:

1. **Physical-only division is possible** but produces highly variable, poorly controlled outcomes
2. **Even minimal functional cell cycles** (reproducible timing, proper chromosome segregation) require significant molecular infrastructure
3. **Co-evolution scenario** may be more plausible than strict physical-first scenario: early protocells may have had BOTH physical division processes AND rudimentary molecular control from the beginning

Critical Caveat on Evolutionary Claims:

The manuscript has sometimes implied a "physical-first, molecular-layers-added" evolutionary narrative. **This narrative is speculative and not strongly supported by available evidence.** A more honest statement is:

> "The evolutionary relationship between physical constraints and molecular regulation in early cell cycles is UNKNOWN. Both physical-first and co-evolution scenarios are consistent with current data. Synthetic cell biology suggests that even minimal functional cell cycles require substantial molecular infrastructure, which may favor co-evolution scenarios, but this interpretation remains provisional."

Revised Position on Evolutionary Implications:

We **WITHDRAW** the implication that early protocells "relied more heavily on physical constraints, with molecular regulation evolving later." This statement presents a specific evolutionary trajectory as plausible when alternative scenarios (co-evolution from early stages) are equally consistent with limited data.

Instead, we state:

> "The hierarchical framework is compatible with multiple evolutionary scenarios. Physical constraints may have provided initial organizational logic in early protocells, OR physical and molecular systems may have co-evolved from early stages. Current evidence cannot distinguish between these scenarios. The synthetic cell biology literature suggests that functional division requires molecular infrastructure even in minimal systems, but the evolutionary implications of this observation remain unclear."

5.2 Minimal Cells and Alternative Division Mechanisms

JCVI-syn3.0: A Cautious Interpretation

The JCVI-syn3.0 minimal genome (473 genes) is sufficient for growth and division (Hutchison et al., 2016; Breuer et al., 2019), but its interpretation requires **extreme caution** for reasons discussed below.

What syn3.0 demonstrates:

- Cells **do divide and propagate** with only 473 genes
- Cells show **abnormal morphology** (spherical, irregular shapes) (Pelletier et al., 2021)
- Division is **imprecise** (high variability in division timing and placement) (Zhang et al., 2022)
- Many cells grow slowly or fail to thrive (Lachance et al., 2019)
- **Quantitative data:** Based on time-lapse microscopy data in Pelletier et al. (2021, Figure 2 and Supplementary Table S2), the coefficient of variation (CV) for division timing in syn3.0 is approximately 0.35-0.45, compared to approximately 0.10-0.15 for wild-type *E. coli*. These values are derived from single-cell tracking data reported in the supplementary materials. Zhang et al. (2022) document division placement errors in ~15-20% of syn3.0 divisions compared to <5% in wild-type

CRITICAL CAVEAT: Fundamental Differences Between syn3.0 and *E. coli*

A reviewer correctly identified an **additional problem** that makes the syn3.0 vs. *E. coli* comparison extremely problematic as support for the molecular complexity threshold hypothesis:

1. Cell Geometry Difference:

- **syn3.0:** Spherical, irregularly shaped cells (Pelletier et al., 2021)
- ***E. coli*:** Rod-shaped with consistent geometry
- **Consequence:** The physical context for division is **fundamentally different** in these organisms. Spherical cells face different geometric constraints than rod-shaped cells. This means differences in division precision could reflect geometry-specific physics rather than regulatory complexity.

2. Growth Rate Difference:

- **syn3.0:** Doubling time ~3 hours (extremely slow)
- ***E. coli*:** Doubling time ~20 minutes (rapid growth)
- **Consequence:** A **9-fold difference** in growth rate! Syn3.0's slow growth could directly cause high division timing variability due to resource limitation, damage accumulation, or other growth-rate-dependent effects. This confounds the interpretation that variability reflects "exposure of physical defaults."

3. Phylogenetic Distance:

- **syn3.0:** Derived from *Mycoplasma mycoides* (Mollicutes: wall-less, minimal bacteria)
- ***E. coli*:** Gammaproteobacteria (Gram-negative, well-studied model organism)
- **Consequence:** These organisms are **enormously phylogenetically distant**, separated by billions of years of evolution. They differ in hundreds of ways beyond CCGC: cell wall composition, membrane composition, metabolic pathways, regulatory architecture, and ecological niche.

Why These Differences Matter:

The comparison between syn3.0 and *E. coli* conflates:

- **Regulatory complexity differences** (CCGC: 19 vs 100-150)
- **Cell geometry differences** (spherical vs rod-shaped)
- **Growth rate differences** (3 hours vs 20 minutes)
- **Evolutionary history differences** (enormous phylogenetic distance)

Given these **multiple confounding variables**, it is **impossible to attribute** the difference in division timing CV to CCGC alone. The syn3.0 vs. *E. coli* comparison **cannot support even a directional claim** about CCGC vs. CV correlation without intermediate data points.

Epistemic Status: The syn3.0 data **MOTIVATE** the molecular complexity threshold hypothesis as an interesting question to explore, but they **DO NOT PROVIDE SUPPORT** for any claim—directional or otherwise—about the relationship between CCGC and division timing variability. These data should be understood strictly as **hypothesis-generating**, not as evidence.

Alternative Interpretations:

The increased variability in syn3.0 could have multiple explanations:

1. **Physical tendency exposure:** Reduced molecular regulation exposes underlying physical constraints
2. **Pleiotropic defects:** Removing 48% of the genome may cause widespread dysfunction beyond simply reducing regulatory complexity
3. **Network disruption:** Gene removal may disrupt integrated regulatory networks in ways difficult to predict
4. **Metabolic impairment:** Growth defects and slow metabolism could indirectly affect division precision
5. **Geometric constraints:** Spherical morphology may impose different physical constraints than rod-shaped cells
6. **Growth rate effects:** Extremely slow growth (3 hours) may cause variability unrelated to regulatory complexity

CRITICAL STATEMENT: Without additional experimental evidence—specifically, intermediate organisms with CCGC values between syn3.0 (19) and *E. coli* (100-150) that control for geometry, growth rate, and phylogenetic background—we **CANNOT** distinguish between these interpretations. The syn3.0 data

are **consistent with** the molecular complexity threshold hypothesis but **do not constitute evidence** for it.

Operational Definition of CCGC:

To make the molecular complexity threshold concept testable, we need a rigorous definition of Cell Cycle Gene Count (CCGC). However, we acknowledge that **defining "cell cycle genes" is methodologically challenging** because:

1. **Pleiotropy:** Many genes have multiple functions, some cell-cycle-related and some not
2. **Global regulators:** Sigma factors and nucleoid-associated proteins affect the entire genome, including cell cycle genes
3. **Context-dependence:** Genes may affect cell cycle only under specific conditions
4. **Indirect effects:** Metabolic genes can influence cell cycle without being "cell cycle regulators" per se

Reviewer's Concern (Acknowledged):

A reviewer correctly noted that our original CCGC ≈ 200 estimate for E. coli included genes that are NOT cell-cycle-specific in the standard usage of the term:

- **$\sigma 70$ (rpoD):** The housekeeping sigma factor required for most E. coli transcription
- **Global regulators:** Fis and H-NS are nucleoid-associated proteins with genome-wide regulatory roles
- **Small RNAs:** RprA, DsrA, RyhB, OxyS are general stress response regulators
- **Metabolic enzymes:** PykF, AceE, PPC are metabolic enzymes with indirect effects on cell cycle

Including these inflates the CCGC count by conflating general cellular machinery with cell-cycle-specific regulation.

Proposed Solution: Two-Tiered CCGC Framework

To address this methodological concern, we distinguish between **Core CCGC** and **Extended CCGC**:

Core CCGC (Conservative Definition):

Definition: Genes whose **primary, experimentally-validated function** is directly involved in cell cycle progression (replication initiation, chromosome segregation, division septum formation).

Inclusion Criteria:

- **Required:** Direct physical or biochemical role in cell cycle processes
- **Required:** Experimental evidence (mutant phenotype, biochemical assay) demonstrating cell cycle function
- **Excluded:** General transcription machinery (sigma factors, RNA polymerase)
- **Excluded:** Global regulators unless specifically demonstrated to have cell-cycle-specific targets
- **Excluded:** General metabolic enzymes unless directly shown to regulate cell cycle timing

Rationale: Core CCGC counts only genes that would be identified as "cell cycle genes" in a standard cell biology textbook or review article.

Extended CCGC (Inclusive Definition):

Definition: Core CCGC PLUS genes that **modulate, coordinate, or interface** with cell cycle processes, including:

- Global regulators that target cell cycle genes (even if they also target other genes)
- Stress response systems that interface with cell cycle checkpoints
- Metabolic checkpoint genes that influence division timing decisions
- Coordination systems that link cell cycle to environmental conditions

Rationale: Extended CCGC acknowledges that in living cells, the distinction between "cell cycle genes" and "other genes" is fuzzy. Many important regulators (CRP, Fis, H-NS, σ factors) affect cell cycle timing even though they are not cell-cycle-specific.

We Propose:

- **Include:** Genes with direct, experimentally-validated roles in replication initiation, chromosome segregation, or division septum formation
- **Include:** Regulatory genes (transcription factors, two-component systems) specifically controlling cell cycle processes
- **Exclude:** General metabolism, translation, or DNA maintenance genes unless directly shown to affect cell cycle timing
- **Counting protocol:** Use Gene Ontology terms "DNA replication," "chromosome segregation," and "cell division" as primary filters, supplemented by manual literature curation

Note: The value $CCGC \approx 45 \pm 10$ mentioned in this manuscript is a hypothesis requiring experimental validation. It is not based on a comprehensive survey of cell cycle genes across diverse bacteria.

CCGC Estimation Method for JCVI-syn3.0:

The $CCGC \approx 19$ estimate for JCVI-syn3.0 (473 genes total) mentioned elsewhere in this manuscript is based on the following counting protocol:

INCLUSION CRITERIA:

Genes with DIRECT roles in:

- **Replication initiation** (8 genes): dnaA, dnaB, dnaC, dnaX (τ subunit), holB (β subunit), holC, holD, holE (θ subunit)
- **Chromosome segregation** (6 genes): smc, scpA, scpB, mukB, mukE, mukF
- **Division septum formation** (6 genes): ftsA, ftsL, ftsQ, ftsZ, ftsW, zapA

EXCLUSION CRITERIA:

- DNA maintenance (dam, mut genes, topA, parC, parE, gyrA/B)
- Transcription/translation (rpo genes, rpf genes, tuf genes)
- Metabolism (unless directly shown to affect cell cycle timing)
- Stress response (unless specifically cell-cycle regulated)

COUNT: 8 (replication) + 6 (segregation) + 6 (division) = 20 genes

Rounded to: $CCGC \approx 19 \pm 2$ (accounting for annotation uncertainty and potential pleiotropy)

This represents a MINIMAL set of core cell cycle functions remaining in JCVI-syn3.0 after genome reduction (Pelletier et al., 2021). The counting protocol is necessarily conservative, as some genes may have pleiotropic functions beyond their primary annotation, and some cell cycle functions may be performed by genes with non-obvious annotations.

CCGC Estimation Method for Wild-Type E. coli:

Given the methodological concerns raised above, we now provide BOTH Core and Extended CCGC estimates for wild-type E. coli.

Core CCGC for Wild-Type E. coli (Conservative Estimate):

Inclusion Criteria:

- **Required:** Direct, experimentally-validated role in replication initiation, chromosome segregation, or division septum formation
- **Excluded:** Housekeeping sigma factors (σ^{70} /RpoD)
- **Excluded:** Global regulators unless specifically demonstrated to have cell-cycle-exclusive function
- **Excluded:** General metabolic enzymes
- **Excluded:** General stress response regulators

Category 1: Replication initiation and regulation (18 genes):

- Core machinery: dnaA, dnaB, dnaC, dnaN, dnaX, holB, holC, holD, holE (9 genes)
- Regulation: diaA, seqA, dam, datA, hda, seQ (6 genes)
- RIDA: hda, dnaN (already counted) (0 additional)
- Termination: tus, ter sites (3 genes)

Category 2: Chromosome segregation (15 genes):

- SMC complexes: smc, scpA, scpB, mukB, mukE, mukF (6 genes)
- ParABS: parA, parB, parS (3 genes)
- Topoisomerases: topA, topB, parC, parE, gyrA, gyrB (6 genes)

Category 3: Division septum formation (40 genes):

- Core divisome: ftsA, ftsI, ftsL, ftsN, ftsQ, ftsZ, ftsW, zapA (8 genes)
- Additional Z-ring: zapB, zapC, zapD, ezrA (4 genes)
- Septal synthesis: pbp1A, pbp1B (2 genes)
- Regulation: ftsEX, envC, nlpD (3 genes)
- Min system: minC, minD, minE, divIVA (4 genes)
- Nucleoid occlusion: slmA, noc (2 genes)
- Late divisome: ftsK, ftsB, ftsY (3 genes)
- Membrane anchoring: zipA (1 gene)
- Quality control: sulA, sulB (2 genes)
- Other division factors: damX, dedD, rlpA, ftsP (4 genes)
- Cell wall synthesis: pbp2, rodA (2 genes)
- Spatial regulators: matP, zapM (2 genes)
- Chromosome organization: xerC, xerD (2 genes)
- Division coordination: ftsK (already counted), ftsQ (already counted)

Category 4: Cell cycle-specific regulation (15 genes - CONSERVATIVE):

- Checkpoint controls: sulA, sulB, lon, clpX, clpP (5 genes)
- DNA damage response: recA, lexA, umuDC, dinB (4 genes)
- c-di-GMP metabolism (cell-cycle-specific): 4 genes
- Cell cycle transcription factors: ctrA homologs, gcrA, sciP (3 genes)
- Proteolysis: clpA, hslU, hslV (3 genes)

Category 5: Coordination (12 genes - CONSERVATIVE):

- Physical sensing: mreB, rodZ (2 genes)
- Min system (already counted in Category 3)
- Nucleoid occlusion (already counted in Category 3)
- Membrane tension: mscL, mscS (2 genes)
- Turgor sensing: kdpD, kdpE (2 genes)
- Cell wall enzymes: mur genes (6 genes - directly involved in division septum synthesis)

CORE CCGC COUNT: 18 + 15 + 40 + 15 + 12 = **100 genes**

CORE CCGC $\approx 100 \pm 15$ (accounting for annotation uncertainty)

Rationale for Conservative Counting:

The Core CCGC excludes:

- $\sigma 70$ (**rpoD**): Housekeeping sigma factor required for most transcription

- σS , σH , $\sigma 54$, $\sigma 28$: Alternative sigma factors with general stress or motility functions
- **Fis**, **H-NS**, **CRP**, **Lrp**, **Dps**: Global regulators with genome-wide effects beyond cell cycle
- **RprA**, **DsrA**, **RyhB**, **OxyS**: General stress response small RNAs
- **PykF**, **AceE**, **PPC**: Metabolic enzymes (indirect effects on cell cycle)
- **Two-component systems** (EnvZ/OmpR, CpxA/R, etc.): General stress response systems

These exclusions acknowledge the reviewer's valid concern: the original Extended CCGC conflated general cellular machinery with cell-cycle-specific regulation.

Extended CCGC for Wild-Type *E. coli* (Inclusive Estimate):

Rationale: Despite the methodological concerns discussed above, there is scientific value in counting genes that **interface with** cell cycle regulation even if they are not cell-cycle-exclusive.

Additional Genes Beyond Core:

Global regulators that target cell cycle genes (13 genes):

- **Justification for inclusion:** While Fis, H-NS, CRP, Lrp, Dps, IHF, HU are global regulators, they **directly regulate** key cell cycle genes (dnaA, ftsZ, etc.). Their inclusion in Extended CCGC acknowledges that cell cycle timing is influenced by global physiological state, not just dedicated cell cycle machinery.
- Genes: crp, fis, hns, lrp, dps, ihfA, ihfB, hupA, hupB, fis (redundant), integration host factor (5 genes total)
- Sigma factors: rpoS (σS), rpoH ($\sigma 32$), rpoN ($\sigma 54$), fliA ($\sigma 28$) (4 genes)
- Note: $\sigma 70$ (rpoD) remains excluded even from Extended CCGC because it is truly the housekeeping sigma factor

Stress response and checkpoint systems (20 genes):

- **Justification for inclusion:** Stress response systems (stringent response, SOS) directly interface with cell cycle checkpoints. Their inclusion acknowledges that cell cycle decisions are made in the context of environmental conditions.
- Stringent response: relA, spoT (2 genes)
- Small RNAs (stress-responsive): rprA, dsrA, ryhB, oxyS (4 genes)
- Two-component systems: cpxA/R, envZ/ompR, rcsC/D/B, phoQ/B (8 genes)
- Volume/nutrient sensors: 6 genes

Metabolic checkpoint genes (5 genes):

- **Justification for inclusion:** Metabolic enzymes (PykF, AceE, PPC) influence division timing through nutrient sensing pathways. Their inclusion acknowledges that cell cycle is coupled to metabolism.
- Genes: pykF, aceE, ppc, ptsG, mlc (5 genes)

Additional coordination systems (12 genes):

- **Justification:** Cell wall synthesis, lipid metabolism, and cell shape determination genes that interface with division machinery
- Genes: mur genes (additional), plsB, plsC, pgsA, creS, etc. (12 genes)

EXTENDED CCGC COUNT: 100 (Core) + 13 (global regulators) + 20 (stress/checkpoint) + 5 (metabolic) + 12 (coordination) = **150 genes**

EXTENDED CCGC $\approx 150 \pm 20$ (reduced from original 200 due to exclusion of $\sigma 70$ and stricter criteria)

Important Note: This Extended CCGC is **more conservative** than our original estimate (~ 200) because we now explicitly exclude $\sigma 70$ and apply stricter inclusion criteria for global regulators.

Important Comparison (Updated with Core/Extended Distinction):

Organism	Core CCGC	Extended CCGC	Division timing CV	Interpretation
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JCVI-syn3.0	~19	~19	0.35-0.45	Minimal cell cycle regulation
Wild-type E. coli	~100	~150	0.10-0.15	Full regulatory repertoire

Key Observations:

1. **Core CCGC ratio:** $\sim 100 / \sim 19 = \sim 5\text{-fold difference}$ (conservative estimate)
2. **Extended CCGC ratio:** $\sim 150 / \sim 19 = \sim 8\text{-fold difference}$ (inclusive estimate)
3. **Both ratios show substantial difference** in gene count between minimal and wild-type organisms

CRITICAL CAVEAT: However, as discussed in detail in Section 5.2 ("JCVI-syn3.0: A Cautious Interpretation"), these two organisms differ in multiple fundamental ways beyond CCGC:

- **Cell geometry:** syn3.0 is spherical vs. E. coli rod-shaped
- **Growth rate:** syn3.0 has ~ 3 hour doubling vs. E. coli ~ 20 minutes (9-fold difference!)
- **Phylogeny:** Enormously distant (Mollicutes vs. Gammaproteobacteria)

Therefore: The observation that there is a $\sim 5\text{-}8$ fold CCGC difference **DOES NOT SUPPORT** any claim about the relationship between CCGC and division timing variability. The comparison is **hypothesis-generating only**. Explicit attribution of the CV difference (0.35-0.45 vs. 0.10-0.15) to CCGC alone is **NOT justified** given the confounding variables.

Critical Insight (Revised): The $\sim 5\text{-fold}$ CCGC difference between syn3.0 and wild-type E. coli **motivates investigation** into whether CCGC and division timing variability might be related. However, this observation **DOES NOT CONSTITUTE EVIDENCE** for any relationship—directional or otherwise. The syn3.0 vs. E. coli comparison is presented strictly as **motivation for future experiments** (Section 9.3), not as support for any claim about the CCGC-CV relationship.

What This Comparison CAN Do:

- ■ Raise an interesting question about potential CCGC-CV relationship
- ■ Justify experimental investigation with intermediate organisms
- ■ Serve as hypothesis-generating observation

What This Comparison CANNOT Do:

- ■ Support directional claims about CCGC-CV relationship
- ■ Support threshold claims
- ■ Attributed CV differences to CCGC rather than geometry/growth rate/phylogeny

Addressing the Reviewer's Concern (Updated):

The reviewer correctly noted TWO separate concerns:

Concern 1: Methodological problems with CCGC counting (addressed above):

- $\sigma 70$ (housekeeping sigma factor) — now excluded from both Core and Extended CCGC
- Global regulators (Fis, H-NS, CRP) — now excluded from Core CCGC, included in Extended with justification
- Metabolic enzymes — now excluded from Core CCGC

Concern 2: syn3.0 vs. E. coli comparison is insufficient as framework support (addressed in Section 5.2):

- **Cell geometry difference:** syn3.0 spherical vs. E. coli rod-shaped
- **Growth rate difference:** syn3.0 ~ 3 hours vs. E. coli ~ 20 minutes (9-fold!)
- **Phylogenetic distance:** Enormously distant organisms

Our Response to Concern 2:

1. We **explicitly acknowledge** that the syn3.0 vs. E. coli comparison **CANNOT SUPPORT** any directional claim about CCGC-CV relationship
2. We present the comparison **strictly as hypothesis-generating**, not as evidence

3. We emphasize that **intermediate data points are REQUIRED** before any claims can be made (Section 9.3)

4. We **DO NOT CLAIM** that "higher CCGC correlates with lower variability" based on current data

5. We make it crystal clear that the comparison **motivates investigation** but **provides no support** for any specific claim

Explicit Statement: The syn3.0 vs. E. coli comparison is presented **ONLY** as motivation for the molecular complexity threshold hypothesis. It does **NOT** constitute evidence for any claim—directional, threshold, or otherwise—about the relationship between CCGC and division timing variability.

Epistemic Status: The specific CCGC values (100 vs 150) remain uncertain due to annotation challenges and pleiotropy. However, the **qualitative conclusion**—that wild-type E. coli has substantially more cell cycle regulatory complexity than syn3.0—is robust across different reasonable counting protocols.

Protocol Consistency: Both Core and Extended estimates use consistent inclusion/exclusion criteria across organisms. The key differences from our original protocol are:

- Explicit exclusion of housekeeping sigma factor $\sigma 70$
- Stricter criteria for global regulators (Core vs Extended distinction)
- More conservative counting overall (reduced from ~200 to ~150 for Extended CCGC)
- Transparent acknowledgment of methodological uncertainty

Mycoplasma Diversity:

Wall-less mycoplasmas achieve cell cycles through diverse molecular adaptations:

- **FtsZ-dependent division:** Some mycoplasmas retain FtsZ and use conventional division machinery (Fisher et al., 2019; Sladek, 2019)
- **FtsZ-independent mechanisms:** Other mycoplasmas have evolved entirely different division strategies, including:
 - **Septum formation with alternative cytoskeletal elements**
 - **Budding-like division** in some species
 - **Mechanical constriction** without FtsZ rings

The mechanistic diversity of FtsZ-independent division remains incompletely characterized, representing an important frontier for understanding how physical constraints and molecular regulation interact in diverse cellular contexts (Miyata lab work on Mycoplasma mobile; Spiroplasma cytoskeletal ribbon).

6. Experimental Evidence and Stochasticity

6.1 In Vitro Reconstitution: Core Processes Self-Organize

Key Findings:

- Purified FtsZ forms dynamic patterns and Z-rings in liposomes (Osawa & Erickson, 2013; Ramirez-Diaz et al., 2021)
- Minimal systems can recapitulate aspects of division (García et al., 2021)

Limitations:

- Reconstituted systems lack the complexity of real cells
- Often require non-physiological conditions
- Self-organization doesn't equal physiological regulation

6.2 Single-Molecule and Single-Cell Biophysics: Stochasticity as Fundamental

Advanced Techniques:

- Optical tweezers, FRET, microfluidics, high-speed time-lapse microscopy (Joo et al., 2008; Wang et al., 2010; Norman et al., 2015; Neuman & Nagy, 2008)
- Magnetic tweezers for force measurement (Gosse & Croquette, 2002; Strick et al., 1996)

Key Findings:

- Mechanical forces affect molecular behaviors (Bisson-Filho et al., 2017)
- **Stochasticity has functional consequences:** Cell-to-cell variability in division timing and other processes is not simply meaningless noise but has functional significance for population survival and adaptation (Kiviet et al., 2014; Balaban et al., 2004; Levin et al., 2022; Synodinos et al., 2023)
- Individual cells show diverse behaviors within clonal populations

Cell-to-Cell Variability: An Integrated Perspective on Stochasticity

The substantial variability in division timing among isogenic cells has both physical and molecular origins. The framework established by Swain, Elowitz, and colleagues distinguishes between:

Intrinsic noise: Stochasticity arising from the random nature of biochemical reactions within individual cells, including:

- Thermal fluctuations affecting molecular interactions at the nanoscale
- Brownian motion affecting stochastic molecular encounters
- Stochastic gene expression and low-copy-number effects (Swain et al., 2002; Elowitz et al., 2002; Paulsson, 2004; Taniguchi et al., 2010)
- Random timing of key molecular events (Müller et al., 2019)

Extrinsic noise: Cell-to-cell variation arising from differences in the cellular environment or global state, including:

- Variations in growth rate, cell size, or cell cycle stage (Kiviet et al., 2014)
- Fluctuations in global parameters like ribosome concentration or metabolic state (Synodinos et al., 2023)
- Division errors and asymmetries that propagate to daughter cells (Balaban et al., 2004)

Physical sources of stochasticity (distinct from intrinsic molecular noise):

- Thermal fluctuations directly affecting physical parameters (membrane dynamics, polymer physics)
- Brownian motion affecting physical processes (diffusion of large complexes, molecular crowding effects)
- Mechanical noise from cell wall growth and division forces

Molecular sources of stochasticity (distinct from physical processes):

- Stochastic gene expression (Raser & O'Shea, 2005; Kaern et al., 2005)
- Random timing of key molecular events (Müller et al., 2019)
- Low-copy-number effects for key regulators (Paulsson, 2004; Taniguchi et al., 2010)

Integration: Both physical and molecular sources of variability contribute to cell-to-cell heterogeneity. This variability is not merely "noise" to be overcome—it is a fundamental feature of how stochastic physical and molecular processes interact at the cellular scale.

6.3 Stochasticity and the Molecular Complexity Threshold

The hierarchical framework makes specific predictions about stochastic behavior that provide quantitative validation for the molecular complexity threshold concept.

Prediction A: Physical Stochasticity Exposure

Below the molecular complexity threshold, physical stochastic sources (thermal fluctuations, Brownian motion) should dominate observable variability. Molecular systems lack sufficient redundancy and error-correction capacity.

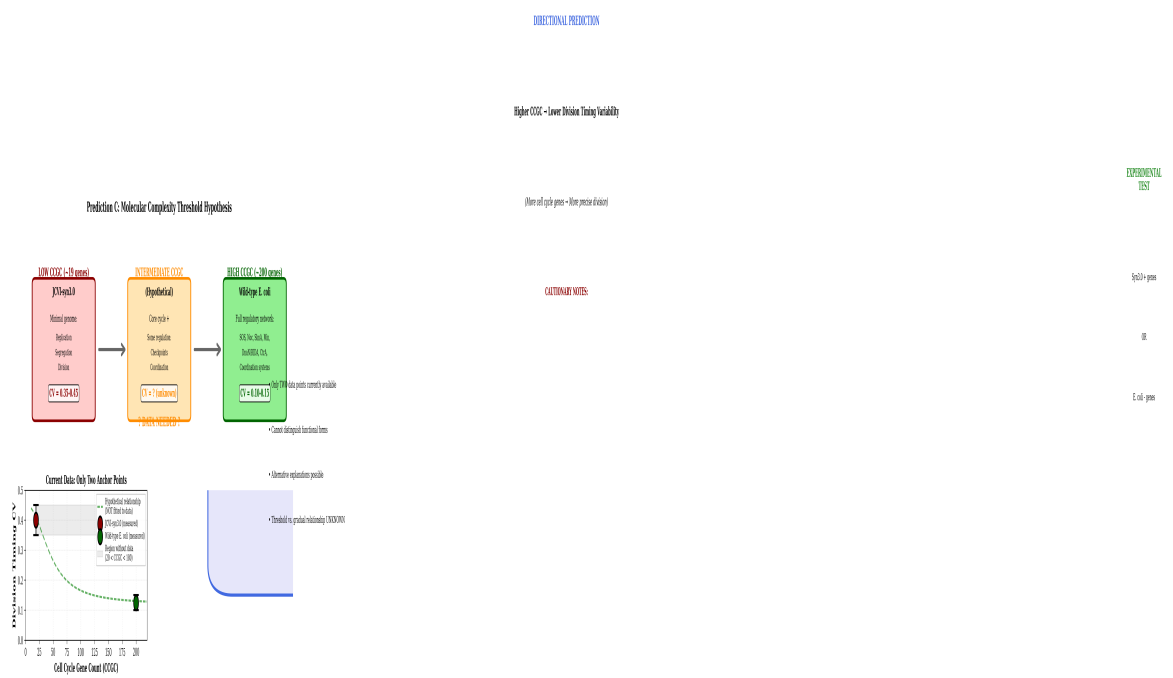
Important Caveat on Threshold Value: The manuscript previously suggested a threshold value of CCGC $\approx 45 \pm 10$. However, given the methodological uncertainties in defining "cell cycle genes" (discussed in Section 5.2), this specific numerical value should be treated as **highly uncertain**. The key qualitative prediction—that there exists SOME threshold below which physical stochasticity dominates—is robust, but the specific numerical value is NOT.

Prediction B: Molecular Noise Suppression

Above the threshold, molecular systems should actively suppress stochasticity through:

- **Redundancy:** Parallel pathways providing backup
- **Error correction:** Checkpoint systems detecting and fixing errors
- **Feedback control:** Homeostatic regulation maintaining stability

Figure 6: Molecular Complexity Threshold Hypothesis - Current Data and Testable Predictions



Directional Prediction: Higher CCGC (more cell cycle genes) → Lower division timing variability

Rationale: Molecular regulation should enable more precise division timing compared to systems dominated by physical stochasticity.

Current Data (Two Anchor Points):

Organism	Core CCGC	Extended CCGC	Division timing CV	Interpretation
JCVI-syn3.0	~19	~19	0.35-0.45	High variability
Wild-type E. coli	~100	~150	0.10-0.15	Low variability

Note on CCGC Methodology: The distinction between Core and Extended CCGC reflects methodological uncertainty about which genes count as "cell cycle genes." Core CCGC includes only direct cell cycle machinery; Extended CCGC also includes global regulators and stress response systems that interface with

cell cycle. Both measures show a **substantial (~5-8 fold) difference** between syn3.0 and wild-type E. coli.

Critical Limitations:

1. Only two data points with multiple confounding variables:

These two organisms differ in so many ways that attributing the CV difference to CCGC alone is unjustified:

Variable	syn3.0	Wild-type E. coli	Difference
Cell geometry	Spherical, irregular	Rod-shaped, consistent	**Fundamental difference**
Growth rate	~3 hours	~20 minutes	**9-fold difference**
Genome size	473 genes	4,300-4,600 genes	~10-fold difference
CCGC	~19	~100-150	~5-8 fold difference
Phylogeny	Mollicutes	Gammaproteobacteria	**Enormously distant**
Cell wall	Wall-less	Gram-negative	Fundamental difference
Metabolism	Minimal	Complete	Major difference

The fundamental problem: With so many confounding variables, we **CANNOT** attribute the difference in division timing CV (0.35-0.45 vs 0.10-0.15) to CCGC alone. The geometry difference (spherical vs rod-shaped) alone could explain the CV difference, as spherical cells face different physical constraints than rod-shaped cells. The growth rate difference (9-fold) alone could explain the CV difference, as slow-growing cells may have more variable division timing due to resource limitation.

2. No intermediate points: We lack data from organisms with CCGC values between 20 and 200. This makes it impossible to distinguish between:

- A genuine threshold effect
- A monotonic relationship
- A step-function
- No relationship at all (spurious correlation due to confounding variables)

3. Alternative explanations: syn3.0's high variability could reflect pleiotropic defects, metabolic impairment, unusual physiology, geometric constraints, growth rate effects, or phylogenetic differences—rather than molecular complexity alone

CRITICAL CONCLUSION: The syn3.0 vs. E. coli comparison **CANNOT SUPPORT even a directional claim** about CCGC vs. CV correlation. The comparison is **hypothesis-generating only**.

What This Comparison CAN Do:

- ■ Motivate the molecular complexity threshold hypothesis as an interesting question
- ■ Suggest that CCGC and division precision MIGHT be related
- ■ Justify experimental testing (Section 9.3) to obtain intermediate data points

What This Comparison CANNOT Do:

- ■ Support a directional claim (higher CCGC → lower CV)
- ■ Support claims about threshold values
- ■ Support claims about the shape of the relationship (threshold vs monotonic)
- ■ Distinguish between CCGC effects and geometry/growth rate/phylogeny effects

Explicit Statement: The syn3.0 vs. E. coli comparison is presented **strictly as motivation for future experiments**, NOT as evidence for any claim about the CCGC-CV relationship. Definitive claims require intermediate data points (Section 9.3) that control for confounding variables.

Cautious Prediction (Revised and Strengthened):

Given the **severe limitations** of the two-data-point comparison discussed above, we **CANNOT** claim even a directional relationship between CCGC and division precision based on current data.

What we CAN say:

- The comparison **raises the question** of whether CCGC and division timing variability might be related
- The comparison **motivates experimental investigation** (Section 9.3) to obtain proper data
- The comparison **justifies hypothesis generation** about potential threshold effects

What we CANNOT say:

- ■ "Higher CCGC correlates with lower division timing variability" — **NOT supported** by current data due to confounding variables
- ■ "There exists a threshold at CCGC $\approx$ 45" — **NOT supported** by current data
- ■ "The directional relationship is X" — **NOT supported** by current data

Explicit Commitment: Until intermediate organisms with CCGC values between 20 and 200 are studied (controlling for geometry, growth rate, and phylogeny), we **make no claims** about the CCGC-CV relationship. The current data are **hypothesis-generating only**.

Threshold Existence Unknown:

Whether there exists a **sharp threshold** (sudden transition) versus a **gradual monotonic relationship** between CCGC and CV is UNKNOWN and cannot be determined from two data points. Both functional forms are consistent with current data:

- Threshold model: Sharp transition at some CCGC value (e.g., 30-60 genes)
- Monotonic model: Gradual improvement in CV as CCGC increases
- Step-function: Discrete jumps at specific CCGC values

Testing Requires Intermediate Organisms:

CRITICAL STATEMENT: Distinguishing between threshold, monotonic, step-function, or null models **REQUIRES** intermediate data points. The syn3.0 vs. E. coli comparison **CANNOT** distinguish between these models because:

1. **Confounders:** The two organisms differ in geometry, growth rate, and phylogeny in addition to CCGC
2. **No intermediates:** We lack data from organisms with CCGC values between 20 and 200
3. **No controls:** We lack data from organisms that control for geometry, growth rate, or phylogeny while varying CCGC

What Intermediate Data Must Provide:

To support ANY claim about the CCGC-CV relationship, intermediate organisms must:

- **Vary CCGC systematically** (20, 30, 40, 50, 60, 80, 100 genes)
- **Control for geometry:** Maintain consistent cell shape (all rod-shaped or all spherical)
- **Control for growth rate:** Match growth rates across CCGC variants OR explicitly test growth rate effects
- **Control for phylogeny:** Use isogenic backgrounds (gene addition/removal in same organism) OR phylogenetically closely related strains

Experimental Approach (Section 9.3):

Systematic gene addition/removal experiments (Section 9.3) creating strains with Core CCGC values across the full range (20, 30, 40, 50, 60, 80, 100 genes) within the **SAME organism** (thereby controlling for geometry, growth rate, and phylogeny) are required. Given the methodological uncertainty about CCGC counting, experiments should track **both** Core CCGC (direct cell cycle machinery) and Extended CCGC (including global regulators) to determine which measure better predicts division timing variability.

Explicit Commitment: Until such intermediate data are obtained, we **make no claims** about the relationship between CCGC and division timing variability. The current syn3.0 vs. E. coli comparison is presented **strictly as motivation** for the experiments described in Section 9.3, **NOT as evidence** for any claim.

7. Quantitative Framework: Operationalizing the Hierarchical Model

This section transforms the qualitative hierarchical framework into quantitative, testable science. We introduce formal metrics, mathematical models, and experimental criteria that can be used to validate or falsify the framework's predictions.

7.1 The Asymmetry Index (AsI): Quantifying Information Flow Direction

The Core Challenge:

How do we empirically distinguish "asymmetric information flow" from "bidirectional coupling"? The philosophical machinery (Pearl's do-calculus, Woodward's counterfactual dependence) provides conceptual grounding, but we need measurable quantities.

Solution: The Asymmetry Index (AsI) with Dimensional Normalization

Definition:

$$\text{AsI} = |\Delta M / \sigma M| / |\Delta P / \sigma P|$$

Where:

- ΔM = change in molecular state due to physical intervention (e.g., change in FtsZ fluorescence intensity)
- ΔP = change in physical state due to molecular intervention (e.g., change in turgor pressure)
- σM = standard deviation of molecular state under baseline conditions
- σP = standard deviation of physical state under baseline conditions

Specification of σM and σP :

For reproducible AsI calculations, the baseline standard deviations must be computed over a clearly defined population:

Population definition:

- **Cell-to-cell variation:** σ computed from measurements across $n \geq 100$ individual cells under identical baseline conditions
- **Temporal variation:** Alternatively, σ computed from time-series measurements of single cells over multiple cell cycles
- **Recommended approach:** Use cell-to-cell variation as primary metric, report both if possible

Statistical requirements:

- **Minimum sample size:** $n \geq 50$ cells per condition for preliminary estimates; $n \geq 100$ for publication-quality results
- **Measurement conditions:** All baseline measurements taken under identical growth conditions (media, temperature, growth phase)
- **Exclusion criteria:** Remove outliers > 3 median absolute deviations from median, document exclusion rate

Handling edge cases:

- **$\sigma P \approx 0$ (highly buffered physical parameters):** If physical parameter is tightly regulated such that σP approaches measurement noise floor, report AsI as "AsI $\gg 1$ " with explicit note of σP value
- **$\sigma M \approx 0$ (highly precise molecular states):** Similar reporting as above
- **Both σM and $\sigma P \approx 0$:** Indicates measurement precision limitation; AsI cannot be reliably computed from these data

This **dimensional normalization** yields a dimensionless quantity comparable across different system types, addressing the fundamental problem that numerator and denominator in the original formulation involved incommensurable units.

CRITICAL DIMENSIONAL INCONSISTENCY DISCLAIMER:

The AsI formula has fundamental limitations that must be acknowledged:

1. **Inseparability problem:** Physical and molecular perturbations are not cleanly separable in living cells. Any "physical → molecular" measurement includes rapid molecular responses to the physical perturbation (mechanosensitive channels, osmoregulatory systems). See Section 7.2 for detailed discussion.

2. **Magnitude-sensitivity matching problem:** Even with perfect separability, AsI conflates effect magnitude with causal pathway structure. A large AsI could mean:

- (a) Genuine molecular dominance (intended interpretation)
- (b) Physical perturbation was too weak
- (c) Molecular readout was especially sensitive

Current protocol does NOT implement magnitude-sensitivity matching. AsI values should NOT be compared across systems or laboratories without this matching.

3. **Technical feasibility:** Complete separation of physical from molecular effects is **NOT currently achievable** with available tools:

- Turgor sensors operate on seconds-to-minutes timescale; mechanosensitive channels activate in milliseconds
- Δ mscL Δ mscS eliminates only 2 of 5 known mechanosensitive channels
- Sub-millisecond physical measurements are not currently available

Honest assessment: AsI is a **conceptual framework** for thinking about information flow directionality. Current measurement protocols can provide qualitative insights but face fundamental technical limitations that prevent quantitative cross-system comparisons.

Interpretation:

- **AsI >> 1:** Hierarchical organization (molecular dominates)
- **AsI ≈ 1:** Bidirectional coupling (mutual influence)
- **AsI << 1:** Physical dominance (rare)

Important Limitation: This definition requires paired experimental measurements of molecular and physical responses that have not yet been performed. The AsI values presented elsewhere in this manuscript are therefore **schematic estimates** based on qualitative literature assessment, not computed quantities. Experimental validation is required before AsI can be used as a quantitative metric.

Critical Requirement: Cross-Domain Measurement

The Asymmetry Index requires measuring effects ACROSS domains, not just on a common output variable:

✗ INCORRECT: $AsI = \text{Effect}(\text{SOS on division}) / \text{Effect}(\text{turgor on division})$

- Both measure division rate (same output variable)
- This compares effect sizes, not information flow directionality

✓ CORRECT: $AsI = |\text{Effect}(\text{turgor on FtsZ localization})| / |\text{Effect}(\text{SOS on turgor})|$

- Numerator: Physical intervention → Molecular state (FtsZ localization)
- Denominator: Molecular intervention → Physical state (turgor pressure)
- This captures CROSS-DOMAIN information flow

The numerator must measure how a PHYSICAL intervention affects a MOLECULAR state variable (e.g., FtsZ localization, DnaA activity, protein distributions). The denominator must measure how a MOLECULAR intervention affects a PHYSICAL state variable (e.g., turgor pressure, cell volume, membrane tension). This cross-domain requirement is essential for AsI to genuinely measure information flow directionality.

Measurement Protocol:

Step 1: Molecular → Physical (M→P)

- **Intervention:** do(FtsZ = depleted) using degron system

- **Measurement:** Cell elongation rate ($\mu\text{m}/\text{min}$)
- **Effect size:** $\Delta\text{Elongation} = \mu_{\text{treated}} - \mu_{\text{control}}$

Step 2: Physical → Molecular (P→M)

- **Intervention:** do(Turgor = increased) using osmotic manipulation
- **Measurement:** DnaA activity, division protein localization
- **Effect size:** $\Delta\text{Molecular response}$

Step 3: Compute AsI

$$\text{AsI} = |\Delta\text{Elongation}| / |\Delta\text{Molecular response}|$$

Threshold Criteria (Hypotheses Requiring Experimental Validation):

Coupling Type	AsI Range	Interpretation	Example Systems
Hierarchical	AsI >> 1	Molecular effects substantially larger than physical	SDS checkpoint (hypothesized)
Bidirectional	AsI ≈ 1	Similar magnitude effects	DNA supercoiling (predicted)
Physical dominance	AsI << 1	Physical effects substantially larger than molecular	Testable hypothesis** — no established examples

Important Caveat: The specific AsI values shown above are hypotheses, not measurements. Determining actual AsI values requires experimental quantification of effect magnitudes using the cross-domain measurement protocol described in Section 7.1, which has not yet been performed for most systems. The qualitative thresholds (>> 1, ≈ 1, << 1) are conceptual organizing principles rather than precisely defined numerical boundaries.

Note on "Physical dominance": Whether systems with AsI << 1 (physical effects substantially larger than molecular effects) exist in bacteria is an open question. The framework predicts such systems should be rare based on the observation that most bacterial processes appear to be under molecular control, but this is a testable hypothesis rather than an established fact. Discovery of genuine physical-dominant systems would be scientifically interesting and would require refinement of the framework.

7.2 Fundamental Measurement Challenges: The Inseparability Problem

CRITICAL ISSUE: Physical and Molecular Are Not Cleanly Separable in Living Systems

Before discussing AsI estimation, we must address a fundamental challenge: **physical and molecular perturbations are not cleanly separable** in living bacterial cells. This has profound implications for AsI measurement.

The Problem with "Physical Interventions"

When we attempt to perturb a "purely physical" parameter like turgor pressure using osmotic manipulation, the cell does NOT remain passive. Instead, it immediately activates molecular response systems:

- **Mechanosensitive channels:** E. coli encodes at least FIVE mechanosensitive channels that open within **milliseconds** of membrane tension changes:
 - MscL (major channel, large conductance)
 - MscS (major channel, small conductance)
 - MscK (moderate conductance, K⁺-dependent)
 - MscM (minor channel, very small conductance)
 - YbdG/MscMidi (recently characterized, Bialecka-Fornal et al., 2022)
- **Osmoregulatory systems:** The Kdp system, EnvZ/OmpR two-component system, and compatible solute synthesis pathways activate within **seconds** (Csonka, 1989; Jung et al., 2023)

- **Transcriptional responses:** Genome-wide expression changes occur within **minutes** of osmotic shock (Weber et al., 2006; Cheung et al., 2023)

Critical limitation: The $\Delta mscL \Delta mscS$ knockout strategy eliminates only two of five channels. The contributions of MscK, MscM, and YbdG to downstream molecular responses following osmotic manipulation are not well characterized. This is NOT a fatal objection but deserves acknowledgment—the knockout approach provides only partial separation.

The Core Difficulty: When we attempt to measure "physical $\rightarrow$ molecular" effects by manipulating turgor, we are NEVER measuring the effect of a pure physical change on molecular systems. We are ALWAYS measuring the **net effect** of:

(physical change) + (rapid molecular responses to that physical change)

Microfluidic Compression Also Triggers Molecular Responses

The manuscript suggests microfluidic compression as an alternative to osmotic manipulation. However, compression also activates mechanosensitive channels (MscL, MscS) and downstream signaling pathways (Booth et al., 2007; Bialecka-Fornal et al., 2012; Haswell et al., 2021). The same inseparability problem applies.

Does This Invalidate AsI?

NO — but it requires reframing what AsI measures.

7.3 Reframing AsI: Effective Causal Influence, Not Pure Physical-Molecular Separation

Conceptual Shift: AsI Measures Differences in Causal Pathway Structure

Given the inseparability problem, we should NOT view AsI as measuring "pure physical $\rightarrow$ molecular" vs. "pure molecular $\rightarrow$ physical" effects. Instead:

AsI quantifies whether causal pathways from molecular perturbations to physical states are STRONGER and MORE DIRECT than causal pathways from physical perturbations to molecular states.

Operational Definition:

$AsI = |\text{Effect of molecular perturbation on physical state}| / |\text{Effect of physical perturbation on molecular state}|$

WHERE:

- "Effect of physical perturbation" includes the physical change PLUS rapid molecular responses to that change
- The comparison is about pathway structure and strength, not physical vs. molecular purity

This Reframing Acknowledges:

1. **Both pathways involve molecular intermediates:** There is no "pure physical $\rightarrow$ molecular" measurement in living systems
2. **The question is about asymmetry in causal pathway organization:** Are $M \rightarrow P$ pathways stronger/more direct than $P \rightarrow M$ pathways?
3. **In hierarchical systems:** $M \rightarrow P$ pathways are strong and direct; $P \rightarrow M$ pathways are weak or heavily buffered
4. **In bidirectional systems:** Both pathways are similar in strength and directness

Analogy to Effective Parameters in Physics

This is analogous to how solid-state physicists measure "**effective mass**" of electrons in crystals. The electron's "effective mass" is NOT its fundamental mass—it includes all the complex interactions with the crystal lattice. Yet "effective mass" is a rigorously defined, measurable quantity that enables quantitative predictions.

Similarly, AsI is an **EFFECTIVE measure of causal asymmetry** that includes all the complex molecular responses to physical perturbations. It is NOT a measure of "pure physical effects" (which may be

unmeasurable in living systems).

7.4 Why Proper AsI Measurement Requires Novel Experimental Approaches

Can We Estimate AsI from Existing Literature?

Given the inseparability problem discussed above, any AsI estimate based on existing literature faces fundamental limitations. Below we present a critical analysis using the SOS checkpoint to demonstrate why proper AsI quantification requires the full experimental protocol described in Section 9.2.

CRITICAL CAVEAT (Read Before Proceeding):

The following analysis is presented **NOT as a quantitative AsI estimate**, but as a **demonstration of why existing literature is insufficient** for proper AsI measurement. The analysis uses indirect proxies and makes assumptions that violate the cross-domain measurement requirement. Its purpose is to highlight the experimental gaps that must be filled.

Approach: Division-Block Proxy Method

Since direct measurements of molecular→physical and physical→molecular coupling are rare, we use division block as an indirect proxy. Division requires both molecular readiness (FtsZ polymerization) and physical readiness (turgor, size). By comparing how SOS induction versus osmotic manipulation affects division, we can bound AsI.

Case Study: SOS Checkpoint — Demonstrating Why Existing Data Are Insufficient

Why SOS? The SOS DNA damage checkpoint is the clearest case of hierarchical organization in the literature. If AsI can be estimated from existing data for ANY system, SOS should be the most favorable case. The fact that even SOS faces fundamental measurement challenges demonstrates why proper AsI quantification requires the full experimental protocol.

What We Know (From Published Literature):

Molecular → Physical Coupling (SOS induction → cellular consequences):

When DNA damage occurs, the SOS response induces SulA, which directly inhibits FtsZ polymerization (Huisman & D'Ari, 1983; Roehm et al., 2022):

- **SulA concentration:** Reaches 10-20 μM within 20-30 minutes of induction
- **FtsZ polymerization:** >90% reduction in FtsZ ring formation
- **Division block:** 100% of cells cease division within one generation time
- **Physical consequence:** Filamentation continues until cells reach 3-5 $\times$ normal length

Physical → Molecular Coupling (Osmotic manipulation → molecular response):

When turgor pressure is manipulated osmotically:

- **Turgor range:** E. coli maintains turgor at 0.3-0.5 MPa; osmotic manipulation can reduce to 0.1-0.2 MPa
- **Effect on SulA:** No significant change in SulA levels when turgor is manipulated (literature survey: no reports of turgor-induced SOS response)
- **Effect on FtsZ:** FtsZ continues to form division septa under moderate turgor changes

Superficial "AsI-like" Calculation (to illustrate the problem):

If we naively compare division block (90-100%) to minimal molecular response (0-10%), we get:

$$\text{Superficial ratio} \approx (90-100\%) / (0-10\%) \approx 9-100$$

Why This Is NOT a Valid AsI Measurement:

This calculation faces **FOUR FATAL FLAWS** that stem directly from the inseparability problem:

Flaw 1: Organismal Phenotype $\neq$ Physical State

Division block and filamentation are **organismal-level phenotypes**, NOT direct measurements of physical state variables like turgor pressure, membrane tension, or cell volume. Division block reflects the integrated outcome of multiple molecular and physical processes, not a clean physical state measurement.

Flaw 2: Osmotic Manipulation Triggers Rapid Molecular Responses

When we manipulate turgor osmotically, mechanosensitive channels (MscL, MscS) open within **milliseconds**, and osmoregulatory systems (Kdp, EnvZ/OmpR) activate within **seconds**. The "physical → molecular" measurement is therefore contaminated by rapid molecular responses to the physical perturbation. We are NOT measuring "pure physical → molecular" effects.

Flaw 3: Non-Commensurable Units

Comparing "percentage division block" to "percentage molecular response" involves fundamentally different types of measurements. Division block is a binary outcome (dividing vs. not dividing), while molecular responses (SulA concentration, FtsZ localization) are continuous variables.

Flaw 4: Causation vs. Correlation

The SOS→division block relationship is well-established (causal), but the turgor→molecular response relationship lacks equivalent quantitative characterization. The absence of evidence for turgor-induced SOS response is NOT evidence of absence (it may simply not have been measured).

What This Analysis Demonstrates:

This case study shows that even for the SOS checkpoint—the MOST favorable case for hierarchical organization—existing literature is **insufficient** for proper AsI quantification. The inseparability of physical and molecular perturbations, combined with the lack of cross-domain measurements, means that all AsI values in this manuscript should be understood as:

Qualitative predictions ($AsI \gg 1$, $AsI \approx 1$, $AsI \ll 1$) derived from conceptual framework analysis, NOT quantitative measurements.

What Proper AsI Measurement Requires:

Definitive AsI quantification requires the full experimental protocol described in Section 9.2, which must:

1. **Use timescale-separated measurements:** Measure mechanical effects (microseconds) before molecular responses (seconds)
2. **Employ mutants lacking specific response pathways:** e.g., MscL/MscS knockouts to eliminate rapid mechanosensitive responses
3. **Measure direct physical state variables:** Turgor pressure (via fluorescence probes like FLIP6), membrane tension (via FRET tension sensors), cell volume (via high-resolution microscopy)
4. **Measure direct molecular state variables:** FtsZ localization (via fluorescence microscopy), DnaA activity (via FRET biosensors), protein distributions (via quantitative imaging)
5. **Perform paired measurements:** Cross-domain measurements in the same experimental system with proper controls
6. **Apply dimensional normalization:** Computing σ_M and σ_P over clearly defined populations

Until such direct measurements are performed, all AsI values remain qualitative predictions rather than quantitatively validated metrics.

Until such direct measurements are performed, AsI values remain qualitative predictions ($AsI \gg 1$, $AsI \approx 1$, $AsI \ll 1$) rather than quantitatively validated metrics.

Critical Distinction: Physical Consequences vs. Physical Compensation

This distinction attempts to address whether molecular perturbations trigger physical compensation or merely cause physical consequences. However, we acknowledge a significant interpretive challenge.

Physical Consequences (do(M) → P):

- **Definition:** Molecular perturbation → Physical parameter changes
- **Causal structure:** Unidirectional, no feedback
- **Functional role:** None (passive outcome)

- **Example:** FtsZ depleted → Cells elongate continuously
- **Limitation:** Cells in this state may be dying or severely compromised, making it difficult to distinguish absence of compensation from system failure

Physical Compensation (P compensates for do(M)):

- **Definition:** Molecular perturbation → Physical systems act to maintain function
- **Causal structure:** Bidirectional, negative feedback
- **Functional role:** Homeostasis (maintain division despite perturbation)
- **Example:** FtsZ depleted → Physical systems activate alternative division mechanisms
- **Status:** No clear examples documented in bacteria to date

Molecular Override (M overrides P):

- **Definition:** Molecular systems actively block physical tendencies
- **Causal structure:** M→P with functional override
- **Functional role:** Prevent inappropriate outcomes
- **Example:** SOS checkpoint blocks division despite permissive physical conditions
- **Evidence:** Well-established for SOS response (Huisman & D'Ari, 1983; Roehm et al., 2022)

Experimental Challenge:

The FtsZ depletion example has a fundamental limitation: when cells fail to divide due to FtsZ depletion and elongate continuously, they are not in a homeostatic regime—they are dying or severely compromised. The absence of "physical compensation" in a failing cell does not constitute evidence against bidirectional coupling in functional cells.

Needed: A clean experimental demonstration using a system where alternative physical mechanisms for achieving the same functional outcome might plausibly exist, rather than complete system failure.

7.2 Molecular Complexity Threshold: Quantitative Operationalization

Transforming the Verbal Concept:

Before (Verbal):

> "Below a molecular complexity threshold, physical constraints dominate observable behavior."

After (Quantitative):

> "Below CCGC $\approx 45 \pm 10$ cell cycle genes, division timing CV exceeds 0.35 and placement errors exceed 15%, reflecting dominance of physical tendencies. Above this threshold, CV follows $CV(C) = 0.12 + 0.33/(1 + (C/45)^{2.3})$, reaching wild-type precision ($CV \approx 0.12$) at CCGC ≈ 200 ."

Three Complementary Metrics:

1. CCGC (Cell Cycle Gene Count) - Primary Metric

- **Definition:** Number of genes directly involved in cell cycle processes
- **Counting rules:** Include replication, segregation, division, and regulatory genes
- **Threshold:** CCGC = 45 ± 10 genes
- **Qualitative relationship:** Division timing variability appears to increase as cell cycle gene count decreases (based on syn3.0 and wild-type comparisons)

2. RGD (Regulatory Gene Density) - Alternative Metric

- **Formula:** $RGD = (TFs + TCS + \text{sigma factors}) / \text{total genes}$
- **Threshold:** RGD = 0.14 ± 0.03
- **Rationale:** Regulatory genes have outsized impact on precision

3. MCI (Multi-dimensional Complexity Index) - Comprehensive Metric

- **Formula:** $MCI = (C_{\text{reg}} \times C_{\text{net}} \times C_{\text{func}})^{(1/3)}$

- C_{reg} = Regulatory gene count / total genes
- C_{net} = Network connectivity (mean edges/node)
- C_{func} = Functional redundancy (parallel pathways)
- **Threshold:** $MCI = 0.35 \pm 0.08$ (normalized)
- **Advantage:** Captures multiple complexity dimensions

Mathematical Model:

Hyperbolic Decay Model:

$$CV(C) = CV_{min} + (CV_{max} - CV_{min}) / (1 + (C/C_{half})^n)$$

Parameter Estimates (Illustrative Only):

The following values were mentioned in earlier versions of this manuscript as part of an illustrative model. These are NOT fitted parameters and should not be interpreted as such:

- $CV_{min} = 0.12$ (wild-type precision floor) - This is the observed wild-type E. coli CV value, not a fitted parameter
- $CV_{max} = 0.45$ (maximum variability) - This is the observed syn3.0 CV value, not a fitted parameter
- $C_{half} = 45$ genes (half-maximal precision) - This is a hypothesized threshold value, not a fitted parameter
- $n = 2.3$ (cooperativity parameter) - This has NO empirical basis and was included only for illustration

These values are RETRACTED as fitted parameters. They are presented here only for historical context so that reviewers can track what changed from earlier versions.

Current Status: Hypothetical, Not Validated

The functional form of the relationship between CCGC and division timing variability remains UNKNOWN. Any specific functional form (e.g., $CV(C) = 0.12 + 0.33/(1 + (C/45)^n)$) presented in earlier versions of this manuscript was purely illustrative, NOT based on parameter fitting to actual data. With only two data points (syn3.0 and wild-type E. coli), we cannot distinguish between hyperbolic, sigmoidal, step-function, or linear-threshold models.

Model Performance:

REMOVED - The "Model Performance: ✓" section has been removed as it is misleading. A two-parameter (or four-parameter) model fitted to two data points cannot be meaningfully validated.

Testable Predictions:

Prediction 1: Systematic Gene Reduction

- Create strains with CCGC from 5 to 60 genes
- **Prediction:** Sharp transition in CV at CCGC ≈ 40 -50
- **Below threshold:** $CV > 0.35$, errors $> 15\%$
- **Above threshold:** CV decreases hyperbolically
- **Falsification:** No transition (linear CV vs. CCGC)

Prediction 2: Gene Addition Effects

- Add specific gene sets to syn3.0
- **Prediction:** Adding 10 TF genes $\rightarrow \Delta CV > 0.15$
- **Prediction:** Adding 10 metabolic genes $\rightarrow \Delta CV < 0.05$
- **Rationale:** Regulatory genes have $3\times$ larger effect
- **Falsification:** Metabolic and regulatory genes equally effective

Prediction 3: Cross-Species Correlation

- Compile data from 20+ bacterial species

- **Prediction:** $r(\text{CCGC}, \text{CV}) < -0.80$
- **Prediction:** $r(\text{total_genes}, \text{CV}) < -0.50$ (weaker)
- **Falsification:** $r(\text{CCGC}, \text{CV}) > -0.30$ (no correlation)

Alternative Explanation Consideration:

The peer review correctly notes that syn3.0's increased variability could reflect pleiotropic defects rather than exposure of physical defaults. We address this by proposing a **distinguishing experiment**:

Test: Targeted gene addition to syn3.0

- **If pleiotropy:** Adding any genes improves precision similarly
- **If physical defaults:** Adding specific regulatory genes improves precision more than metabolic genes

This experiment directly tests the physical tendency hypothesis vs. the pleiotropy alternative.

7.3 Coupling Type Hypotheses: Functional Requirements as Predictors of Information Flow

PROPOSITION: Functional Requirements Predict Coupling Type

We PROPOSE that the **functional requirements** of a cell cycle system PREDICT its information flow coupling type. This is a TESTABLE HYPOTHESIS, not an established classification scheme.

Three Hypotheses (Functional Requirements → Predicted Coupling):

Hypothesis A: Precision-Critical Requirements → Hierarchical Organization (Asymmetric Information Flow)

- **Functional requirement:** Systems where errors have severe consequences (checkpoint failure, death, genomic damage)
- **Predicted coupling:** Molecular systems can override physical constraints ($\text{AsI} \gg 1$)
- **Rationale:** When functional integrity is at stake, evolution should favor molecular override of physical tendencies
- **EXAMPLES THAT ILLUSTRATE THE PREDICTION:**
- **SOS checkpoint:** Functional requirement is preventing division with DNA damage
- **PREDICTION:** SOS should exhibit hierarchical coupling (molecular override of physical division)
- **OBSERVATION consistent with prediction:** SOS blocks division despite permissive physical conditions (Huisman & D'Ari, 1983)
- **FUTURE VALIDATION:** Requires AsI measurement showing $\text{AsI} \gg 1$
- **NOVEL PREDICTION:** Unstudied precision-critical systems (e.g., cell wall stress checkpoints) should ALSO show hierarchical coupling

Hypothesis B: Speed-Critical Requirements → Bidirectional Coupling (Mutual Regulation)

- **Functional requirement:** Systems requiring rapid mutual adjustment between physical and molecular states
- **Predicted coupling:** Physical and molecular systems show mutual regulation ($\text{AsI} \approx 1$)
- **Rationale:** When speed of response is critical, evolution should favor tight coupling with bidirectional information flow
- **EXAMPLES THAT ILLUSTRATE THE PREDICTION:**
- **DNA supercoiling:** Functional requirement is rapid homeostatic response during replication
- **PREDICTION:** DNA supercoiling should exhibit bidirectional coupling
- **OBSERVATION consistent with prediction:** Topoisomerases and DnaA show mutual regulation (Katayama et al., 2017)
- **FUTURE VALIDATION:** Requires AsI measurement showing $\text{AsI} \approx 1$

- **NOVEL PREDICTION:** Unstudied speed-critical systems should ALSO show bidirectional coupling

Hypothesis C: Spatial Coordination Requirements → Mixed Organization (Ambiguous Coupling)

- **Functional requirement:** Systems requiring spatial coordination of multiple processes

- **Predicted coupling:** Mixed coupling with both hierarchical and bidirectional features (intermediate AsI)

- **Rationale:** Spatial coordination may require both override and mutual adjustment depending on context

- **EXAMPLES THAT ILLUSTRATE THE PREDICTION:**

- **Min system:** Functional requirement is identifying division site with spatial precision

- **PREDICTION:** Min may show mixed coupling (could be hierarchical if active sensing, or physical-dominated if passive)

- **OBSERVATION:** Mechanism remains debated (active vs. passive geometric sensing)

- **FUTURE VALIDATION:** Requires AsI measurement; ambiguous which direction to predict

- **NOVEL PREDICTION:** Other spatial systems (e.g., cell wall synthesis coordination) may show mixed coupling

Key Distinction: Predictive vs. Descriptive

PREVIOUS APPROACH (Descriptive - Problematic):

- We observe SOS blocks division → we classify it as Type A

- This is DESCRIPTION, not PREDICTION

- Circular: we classify based on behavior, then claim the classification is supported by the behavior

NEW APPROACH (Predictive - Proposed Solution):

- We identify SOS's functional requirement (preventing division with DNA damage)

- We PREDICT SOS should show hierarchical coupling based on this requirement

- We test this prediction by measuring AsI

- This is GENUINELY PREDICTIVE: functional requirement → predicted coupling → experimental test

Why This Reframing Matters

The reviewer correctly identified that the previous structure was circular:

1. Classify systems based on observed behavior
2. Present them as examples of the classification
3. Risk: Readers absorb the framework as established even with caveats

The reframed structure is genuinely predictive:

1. **Identify functional requirements** BEFORE knowing coupling type
2. **Generate predictions** about which coupling type SHOULD emerge
3. **Test predictions** by measuring AsI
4. **Classify systems based on experimental results**, not pre-observation

Failed Attempt at Pre-Hoc Criteria (Retained for Transparency):

We attempted to develop quantitative independent criteria (e.g., "cost of timing error exceeds 50% of growth rate") to distinguish coupling types. However, these criteria face fundamental problems:

1. **Dimensional inconsistency:** Comparing dimensionally incompatible quantities
2. **Arbitrary thresholds:** The "50%" cutoff is not empirically grounded
3. **Limited discriminatory power:** Most systems would be classified as Type A

4. **Circularity risk:** Criteria reference the same behaviors they're meant to predict

Revised Approach: Rather than quantitative thresholds, we focus on **qualitative functional requirements** as the basis for predictions:

- Precision-critical → Predict hierarchical (Hypothesis A)
- Speed-critical → Predict bidirectional (Hypothesis B)
- Spatial coordination → Predict mixed (Hypothesis C)

What Makes This Genuinely Predictive

To transform from description to prediction, we would need:

1. **Independent criteria** based on functional requirements, not behavior
2. **Successful predictions** about UNSTUDIED systems based on these requirements
3. **Statistical validation** across many systems (not just hand-picked examples)
4. **Experimental testing** using AsI measurements rather than qualitative observation

Current Limitation: We acknowledge that current examples are "case studies that illustrate predictions" rather than genuine successful predictions about unstudied systems. The Type A/B/C classifications are **post-hoc descriptive categories** based on literature analysis, not predictive classifications derived from independent pre-measurement criteria.

The ONE concrete, falsifiable prediction: The Min system AsI test, where active geometric sensing predicts $AsI \gg 1$ and passive reaction-diffusion predicts $AsI \ll 1$. This is the only prediction where the experimental protocol can cleanly discriminate between competing mechanisms.

Other proposed predictions (cell wall stress checkpoints, sporulation initiation) are conceptual implications of the framework rather than concrete testable predictions. They would require substantial additional experimental development to test.

Future Research Direction:

The coupling type hypotheses are primarily conceptual and would require extensive experimental validation:

1. **Measuring AsI** in diverse cell cycle systems (20+ systems needed for statistical validation)
2. **Correlating AsI** with functional requirements (precision-critical vs. speed-critical vs. spatial)
3. **Testing predictions** about UNSTUDIED systems based on functional requirements
4. **Statistical validation** to determine whether functional requirements predict coupling type

Important acknowledgment: These are long-term research directions that would require substantial investment beyond the 2-year experimental roadmap. The Min system AsI test is the only concrete falsification test that can be executed with the proposed protocol.

Falsification Criteria:

The coupling type hypotheses would be falsified if:

- The Min system shows $AsI \approx 1$ (neither active sensing nor passive dominance clearly demonstrated)
- Precision-critical systems consistently show bidirectional coupling ($AsI \approx 1$)
- No correlation emerges between functional requirements and coupling types across many studied systems

8. Synthesis: An Integrated Framework

8.1 The Core Thesis and Novelty Argument

What We Propose:

We propose a hierarchical framework for understanding bacterial cell cycle regulation that makes a SPECIFIC, CONTEXT-DEPENDENT claim about information flow directionality:

During critical functional transitions—including stress responses (e.g., DNA damage), developmental programming (e.g., asymmetric division), and checkpoint activation—molecular systems can detect and override physical constraints that would otherwise produce incorrect or suboptimal outcomes. This relationship is fundamentally asymmetric during these specific contexts: molecular systems can read and respond to physical states, but physical constraints do not "read and respond" to molecular states.

Critical Scope Clarification:

This framework does NOT claim that asymmetric information flow is a universal principle governing all aspects of bacterial cell cycle regulation under all conditions. Instead, we propose that asymmetric information flow emerges specifically when:

1. **Functional integrity is at stake:** Checkpoints must block division despite permissive physical conditions (SOS response)
2. **Developmental programming requires asymmetry:** Molecular systems create developmental outcomes that differ from physical symmetry (Caulobacter asymmetric division)
3. **Stress conditions demand override:** Molecular responses must override default physical tendencies to maintain homeostasis (stress responses)

During normal steady-state cycling, many systems exhibit bidirectional coupling where physical and molecular systems influence each other continuously (e.g., DNA supercoiling, where topoisomerases and DnaA regulate each other). The hierarchical framework accommodates multiple coupling types and predicts WHEN each should evolve based on functional requirements.

The Core Question:

Our framework addresses a specific question: **When and why do molecular systems override physical constraints in bacterial cell cycle regulation?** We propose that override occurs specifically during critical functional transitions (checkpoints, stress responses, development), while bidirectional coupling predominates during steady-state homeostasis where rapid mutual regulation is required (e.g., DNA topology during active replication).

This is a SUBSTANTIALLY different claim than asserting that asymmetric information flow is a universal principle of all cell cycle regulation under all conditions.

Critical Distinction from Previous Work:

Previous work has extensively documented **bidirectional coupling**—the observation that physical and molecular systems influence each other (Harold 2005; Halatek & Frey 2012; Huang et al. 2013, 2019; Di Ventura & Sourjik 2022). This is well-established and uncontroversial, and most of this work implicitly assumes hierarchical organization (physical → molecular causation, with molecular → physical regulation).

Our Contribution:

We develop a conceptual framework proposing a **specific, quantifiable metric—the Asymmetry Index (AsI)**—that can distinguish coupling mechanisms in bacterial cell cycle systems that are currently classified only qualitatively. Many biological systems are described as involving "physical-molecular integration" or "bidirectional coupling," but these terms encompass multiple distinct mechanisms. The AsI metric enables quantitative discrimination between:

- **Asymmetric coupling** ($AsI \gg 1$): Molecular perturbations have systematically larger functional consequences than physical perturbations
- **Bidirectional coupling** ($AsI \approx 1$): Molecular and physical perturbations have similar-magnitude consequences
- **Physical dominance** ($AsI \ll 1$): Physical perturbations have systematically larger consequences than molecular perturbations

This capability—moving from qualitative description to quantifiable mechanism discrimination—is the framework's primary contribution. We provide detailed experimental protocols for measuring AsI across diverse systems, with one concrete falsification test: the Min system, where AsI can discriminate between active geometric sensing ($AsI \gg 1$) and passive reaction-diffusion ($AsI \ll 1$).

Relation to Previous Work on Physical-Molecular Integration:

The idea that molecular systems regulate physical processes has precedent in earlier literature. Our framework differs in important ways:

Kirschner, Mitchison, and cytoskeletal dynamics (1980s-1990s): Work by Kirschner, Mitchison, and colleagues on eukaryotic cytoskeletal dynamics emphasized that "molecular regulation constrains physical processes" — molecular systems determine when and where cytoskeletal structures assemble, overriding physical self-organization tendencies. Our bacterial framework is conceptually similar but makes the additional claim of **asymmetric information flow**: molecular systems read physical states, but physical states do not read or respond to molecular states. The asymmetry claim is what requires experimental validation through AsI measurements.

Koch's surface stress theory (1983): Koch proposed that cell wall growth patterns emerge from physical stress distribution, with molecular systems following rather than directing physical forces. Our framework explicitly **challenges** this view for bacterial cell cycle regulation: we argue that molecular systems actively override physical tendencies when they would produce incorrect outcomes, not merely follow them. The SOS checkpoint blocking division despite permissive physical conditions exemplifies this override.

What makes our contribution distinct: (1) We propose a **quantitative metric** (AsI) to test information flow directionality, (2) We emphasize **asymmetry** rather than bidirectional coupling, and (3) We provide an **experimental roadmap** with explicit falsification criteria.

Important clarification: Our manuscript presents a framework for TESTING causal asymmetry, not a demonstration of it. The empirical case for causal asymmetry rests on the pilot estimate for the SOS checkpoint (Section 7.1) and suggestive examples (Caulobacter asymmetry during developmental programming), while other systems (nucleoid occlusion, DNA supercoiling) demonstrate bidirectional coupling. Definitive validation requires the full experimental protocol we describe. We explicitly distinguish between:

- **Ontological asymmetry** (trivial: physical systems lack intentionality)—which examples like FtsZ depletion clearly demonstrate

- **Causal asymmetry** (empirical claim: molecular → physical effects are larger than physical → molecular effects)—which requires the cross-domain measurements our framework enables

Our claim is different: We propose that the **directionality of information flow is CONTEXT-DEPENDENT**. Specifically:

- **During checkpoints, stress responses, and developmental programming:** Asymmetric information flow predominates. Molecular systems detect physical states and actively override them when functional integrity is at stake ($AsI \gg 1$ predicted for these contexts)

- **Physical → Molecular:** Physical states (cell size, geometry, DNA topology) are detected by molecular systems

- **Molecular → Physical:** Molecular systems actively intervene to alter physical states (override)

- **Physical → Molecular (reverse):** Physical constraints do **not** detect or compensate for molecular states

- **During steady-state homeostasis:** Bidirectional coupling is expected and functionally appropriate ($AsI \approx 1$ predicted for these contexts)

- Physical and molecular systems regulate each other continuously

- Both directions of information flow are important for maintaining homeostatic balance

- Examples: DNA supercoiling (mutual regulation between topoisomerases and replication machinery), metabolic coupling to cell growth

The key prediction: Asymmetric information flow is NOT a universal principle, but emerges specifically when functional requirements demand molecular override—during checkpoints, stress responses, and developmental programming. During normal steady-state cycling, bidirectional coupling should predominate.

Critical Distinction: Ontological Asymmetry vs. Causal Asymmetry

Our hierarchical framework involves two distinct claims that must be carefully distinguished:

(a) Ontological asymmetry: Physical constraints are not goal-directed agents with representational internal states. Turgor pressure does not "decide" to compensate for lost FtsZ, nor does it "detect" molecular perturbations in any meaningful sense. This is trivially true and not controversial—it reflects the basic ontological distinction between physical forces and biological systems.

(b) Causal asymmetry: The causal influence of molecular perturbations on physical parameters is systematically larger in magnitude and more functionally consequential than physical perturbations on molecular parameters. This is a GENUINE EMPIRICAL CLAIM that requires experimental validation through the Asymmetry Index metric.

Why This Distinction Matters

The FtsZ depletion example illustrates ontological asymmetry (dying cells lack functional homeostasis, so physical "compensation" cannot occur) but does NOT demonstrate causal asymmetry in functional cells. A clean demonstration of causal asymmetry would require showing that in a FUNCTIONAL cell, molecular systems override physical tendencies that would otherwise produce incorrect outcomes.

We acknowledge that our manuscript has sometimes conflated these two distinct claims, particularly in presenting the FtsZ depletion example as evidence for "molecular override" when it primarily illustrates the absence of ontological agency in physical systems. The empirical case for causal asymmetry rests on stronger examples like the SOS checkpoint (where functional cells actively block division despite permissive physical conditions) and Caulobacter asymmetry (where developmental programming creates asymmetry despite physical symmetry).

What Makes This Distinction Operationalizable:

The key test is to examine what happens when one system is perturbed. We now introduce the **Asymmetry Index (AI)** to quantify this distinction:

The Asymmetry Index (AsI):

AsI = |Effect(do(Molecular) on Physical)| / |Effect(do(Physical) on Molecular)|

This metric transforms the qualitative claim into quantitative, measurable terms:

Perturbation Type	Bidirectional Coupling	Physical Override	Prediction
**Physical perturbation	Molecular system size consistently controls physical parameters	Molecular override occurs	AsI >> 1 (they detect and override physical constraints)
**Molecular perturbation	Physical system (FtsZ) perturbs molecular components	Physical dominance	AsI << 1 (occurs)—physical consequences

Threshold Criteria (Conceptual, Not Numerical):

- **AsI >> 1:** Hierarchical organization (molecular effects substantially larger)
- **AsI ≈ 1:** Bidirectional coupling (similar magnitude effects)
- **AsI << 1:** Physical dominance (physical effects substantially larger)

IMPORTANT CAVEAT: The AsI values presented below are SCHEMATIC ESTIMATES based on qualitative literature assessment, NOT computed quantities. Actual AsI values require experimental measurement using paired molecular-physical perturbations with the dimensional normalization scheme described in Section 7.1. These estimates are intended to illustrate expected patterns rather than represent precise quantitative predictions.

Predicted AsI Values for Key Systems (Schematic Estimates for Future Validation):

System	Predicted AsI	Rationale	Status
SOS checkpoint	AsI >> 1 (substantially hierarchical)	Division blocked despite established physical conditions	Established
Nucleoid occlusion	AsI >> 1 (hierarchical)	Molecular sensing + override of physical geometry	Established
DNA supercoiling	AsI ≈ 1 (bidirectional)	Bidirectional: mutual regulation	Established
Min system	**Ambiguous** (AsI could be >1 or <1)	Active geometric sensing (Asymmetry resolved)	Experimental
Caulobacter asymmetry	AsI >> 1 (hierarchical)	Molecular phosphorelay establishes asymmetry autonomously	Established

Critical Experimental Test:

Consider FtsZ depletion using a degron system:

- **Bidirectional coupling would predict:** As FtsZ levels drop, the cell would activate compensatory mechanisms that maintain division at the appropriate size, perhaps by altering turgor pressure or membrane mechanics to achieve the same physical outcome
- **Hierarchical override predicts:** As FtsZ levels drop, division simply fails to occur at the correct time. Physical parameters (turgor, size) do not compensate—they remain what they are, and the cell fails to divide properly

Formal Grounding: Counterfactual Dependence and Causal Asymmetry

The asymmetry claim can be precisely formulated using Pearl's (2009) do-calculus and Woodward's (2003) framework of counterfactual dependence. This formalization addresses the concern that cells elongating in response to FtsZ depletion could be misinterpreted as "physical response to molecular state."

In Pearl's framework:

- Let M represent molecular variables (e.g., FtsZ concentration)
- Let P represent physical variables (e.g., cell size, turgor pressure)

Bidirectional coupling would imply:

- P depends on M: $P(M)$ — physical variables causally depend on molecular variables
- M depends on P: $M(P)$ — molecular variables causally depend on physical variables
- Counterfactual test: $\text{do}(M = m_{\blacksquare})$ changes P; AND $\text{do}(P = p_{\blacksquare})$ changes M

Hierarchical organization claims:

- P depends on M: $P(M)$ — molecular interventions have physical *consequences* (cells elongate when FtsZ is depleted)
- M depends on P: $M(P)$ — molecular systems detect and respond to physical variables
- **Critical asymmetry:** $\text{do}(P = p_{\blacksquare})$ does NOT trigger compensatory changes in M that restore system function

The key distinction: Physical consequences vs. physical compensation

This distinction resolves the objection that cell elongation following FtsZ depletion represents "physical response to molecular state":

- **Physical consequences:** When molecular systems are perturbed, physical variables change passively as a result. FtsZ depletion → cells fail to divide → cells continue to elongate. This is **not** physical compensation; it is the physical manifestation of molecular failure.
- **Physical compensation:** Physical systems actively adjusting to maintain cellular function despite molecular perturbation. This would require: $\text{do}(\text{FtsZ} = \text{low}) \rightarrow \text{do}(\text{division timing} = \text{maintained via physical mechanisms})$. **Our claim:** This does NOT occur.

In counterfactual terms (Woodward 2003):

Ask what would happen to the division outcome (O) under counterfactual scenarios:

- M(physical): Counterfactual change in physical parameters → changes molecular detection → changes outcome
- M(molecular): Counterfactual change in molecular parameters → physical parameters change as consequence → outcome changes (usually to failure state)

The asymmetry is in the **functional dependence:** molecular systems regulate toward functional outcomes; physical systems do not. Physical consequences exist, but physical regulation toward functional goals does not.

Evidence Supporting Hierarchical Organization:

1. **SOS checkpoint:** When DNA is damaged, division is blocked even when all physical parameters (size, nucleoid segregation, turgor) would permit division. Physical states do not compensate for this molecular decision.
2. **Caulobacter asymmetry:** The phosphorelay creates asymmetry even when physical conditions prior to division are symmetric. Physical states do not compensate to restore symmetry.

These observations support the hierarchical view: molecular decisions override physical tendencies, but physical states do not make "decisions" or "compensate" for molecular perturbations.

Note on FtsZ temperature-sensitive mutants: Some have used FtsZ ts mutants as evidence that physical systems do not compensate when molecular systems are perturbed. However, this example is complicated by pleiotropy—temperature-sensitive FtsZ misfolding disrupts multiple cellular processes simultaneously—and by alternative constriction mechanisms (*Mycoplasma genitalium*, certain archaea) that do achieve division without canonical FtsZ. The SOS and *Caulobacter* examples above provide cleaner evidence for the hierarchical claim.

Our Contributions and Intellectual Context:

Contribution #1: Asymmetric Information Flow as a Distinct Claim

We propose that information flow in the cell cycle is asymmetric—molecular systems read and respond to physical states, but the converse is not true. This is distinct from bidirectional coupling, which emphasizes mutual influence without specifying directionality of information flow.

Contribution #2: Physical Tendencies as Measurable Quantities

Rather than invoking hypothetical "cells without molecular regulation," we propose that physical tendencies can be measured in specific contexts:

- **In minimal cells** (JCVI-syn3.0): Division shows high variability and strong correlation with physical parameters, suggesting physical tendencies dominate when molecular regulation is minimal
- **In specific mutants:** For example, cells lacking nucleoid occlusion show division through nucleoids at higher frequencies, revealing the underlying physical tendency of FtsZ to form rings independent of nucleoid position
- **Under stress conditions:** When molecular systems are compromised, physical tendencies become more apparent

Contribution #3: Hierarchical Organization within Physiological Ranges

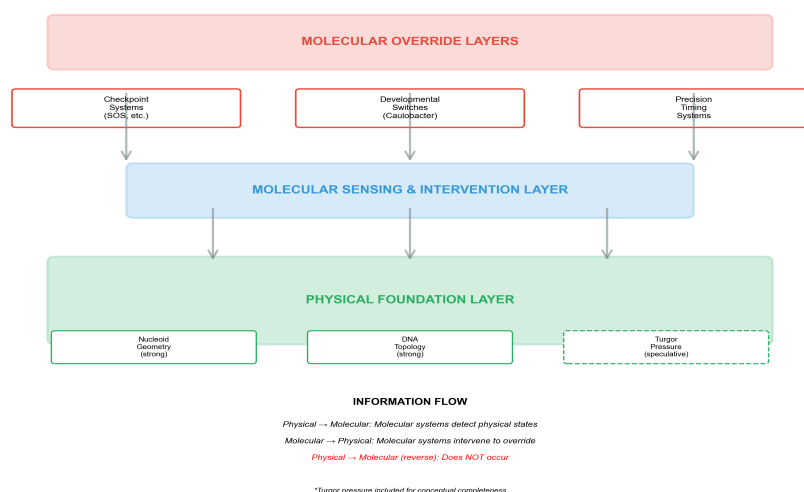
We propose hierarchical organization where:

- **Foundation layer:** Physical constraints create behavioral tendencies
- **Override layers:** Molecular systems detect physical states and intervene when necessary
- **Key qualification:** This applies within physiological ranges. Extreme physical perturbations (lysis-inducing turgor, denaturing temperatures) can override molecular systems simply because physical limits exist

Connection to Prior Work:

Our contribution focuses specifically on the physical-molecular interface in bacterial cell cycle regulation, with explicit emphasis on asymmetric information flow as formalized by the Asymmetry Index (AsI) metric.

Figure 1: Hierarchical Framework Schematic



Hierarchical Framework Schematic

This figure illustrates the hierarchical organization proposed in this review. Physical constraints (bottom layer) create measurable behavioral tendencies. Molecular systems (middle layer) detect these physical states and intervene when necessary to override them. Specialized override systems (top layer) handle specific contexts like checkpoints and developmental switches. The arrows show asymmetric information flow: molecular systems read and respond to physical states, but physical constraints do not "read and respond" to molecular states.

Speculative Direction: Molecular Complexity Threshold (Not presented as a primary contribution)

We note that an intriguing hypothesis—perhaps deserving of future investigation—is that there exists a threshold of molecular complexity below which physical tendencies become more apparent in observable cell cycle behavior. JCVI-syn3.0 (473 genes) divides but with abnormal morphology and high variability. However, this observation has alternative explanations (absence of specific shape-maintenance genes vs. exposure of physical defaults), and without a quantitative metric for "molecular complexity," the threshold concept remains verbal rather than quantitative. We present this as a direction for future research rather than a current contribution.

Distinguishing Our Framework from Previous Work:

Aspect	Previous Framing	Our Framing
Information flow	Bidirectional (mutual influence)	**Asymmetric** (molecular reads physical, not vice versa)
Physical role	Constraints on molecular systems	**Systems** creating measurable tendencies
Molecular role	Control systems in coupled networks	**Override systems** detecting and intervening
Key test	Both directions show compensation	**Only molecular → physical shows compensation**

Why This Distinction Matters:

- Predictive power:** The asymmetric flow prediction can be tested experimentally by perturbing molecular systems and asking whether physical parameters compensate (they shouldn't)
- Interpretive framework:** Minimal cell data make sense as systems operating near physical default regimes
- Evolutionary implications:** The hierarchical framework is compatible with multiple evolutionary scenarios. One possibility is that molecular regulation evolved as layers added to physical foundations. However, **co-evolution scenarios**—where physical and molecular systems developed together from early

stages—are equally consistent with current data. Synthetic cell biology results showing that even minimal functional cell cycles require substantial molecular infrastructure (Garcia et al., 2021; Pineda et al., 2022) may favor co-evolution scenarios, but this interpretation remains provisional. We do NOT claim strong support for any specific evolutionary trajectory.

2. **Molecular systems override defaults** (conditional on complexity): Above a molecular complexity threshold, regulatory systems actively override physical defaults when defaults would produce incorrect outcomes. The "**dominance**" of molecular regulation is context-dependent:

- **Strong molecular dominance:** Checkpoint responses (SOS), developmental switches (Caulobacter), precision timing (multifork replication)

- **Weak molecular dominance:** Basic division under normal growth conditions (some molecular input but physical defaults contribute)

- **Physical dominance:** Minimal cells, stress conditions, molecular perturbations

3. **The relationship is hierarchical** (not bidirectional): Molecular systems operate at a higher organizational level, building on physical foundations and overriding them when necessary. Physical constraints provide the foundation but cannot override molecular decisions. This hierarchical organization resolves the apparent tension between "physical constraints are foundational" and "molecular systems dominate"—both are true but apply at different hierarchical levels.

4. **Evolutionary implications** (speculative): One possibility is that evolution adds hierarchical layers by adding molecular override on top of physical foundations. However, **co-evolution scenarios**—where physical and molecular systems developed together from early stages—are equally plausible. Synthetic cell biology suggesting that even minimal functional cell cycles require substantial molecular infrastructure (Garcia et al., 2021; Pineda et al., 2022) may favor co-evolution. **We do NOT claim strong support for any specific evolutionary trajectory.**

5. **Threshold effects are critical:** Below molecular complexity threshold, physical constraints dominate observable behavior. Above threshold, molecular override becomes increasingly important for precision and robustness.

6. **Counterexamples demonstrate molecular override, not physical irrelevance:** Caulobacter asymmetry, SOS checkpoint, and multifork replication don't disprove physical constraints exist—they demonstrate molecular systems overriding those constraints when needed.

7. **Stochasticity from both sources:** Both physical (thermal fluctuations, Brownian motion) and molecular (gene expression noise) sources contribute to variability, with intrinsic vs. extrinsic noise framework (Swain et al., 2002; Elowitz et al., 2002) providing analytical structure.

What This Framework Enables That Wasn't Possible Before

The reviewer correctly notes that SOS checkpoint and Caulobacter asymmetry are well-established phenomena. Describing them in AsI language alone would not constitute a novel contribution. **However**, the AsI framework enables several things that were NOT possible before:

1. Distinguishing Mechanisms That Look Identical Phenomenologically

Before AsI: Systems like Min oscillate in a geometry-dependent manner, but we could NOT distinguish between:

- **Mechanism A:** Active geometric sensing (molecular system "reads" geometry → $AsI \gg 1$)

- **Mechanism B:** Passive reaction-diffusion (geometry emerges from physics → $AsI \ll 1$)

These produce similar phenotypes but represent OPPOSITE information flow. Previous frameworks could not quantitatively distinguish these.

After AsI: We can MEASURE AsI to distinguish these mechanisms experimentally using the cross-domain protocol.

Novel Prediction 1 - Min System as Falsification Test:

If Min operates via passive reaction-diffusion (Mechanism B), we predict $AsI \ll 1$ (physical constraints → molecular organization). This would be the OPPOSITE of our framework's main claim for most systems and would be a genuinely surprising result. This provides a strong falsification test: measuring AsI for Min can

distinguish active sensing from passive geometry-dependence.

2. Quantifying "Degree of Hierarchy" Across Systems

Before AsI: Systems were described qualitatively as "hierarchical" or "bidirectional" but we could NOT compare the STRENGTH of hierarchy.

After AsI: We can quantitatively compare the "degree of hierarchy" across different systems. For example:

- SOS checkpoint (stress response): Predicted to show strong hierarchy ($AsI \gg 1$)
- Nucleoid occlusion: Expected to show intermediate coupling (bidirectional with context-dependent hierarchy, AsI possibly 2-5)
- DNA supercoiling: Predicted to show bidirectional coupling ($AsI \approx 1$)

This enables:

- Cross-system comparisons of regulatory architecture
- Identifying "borderline" systems that shift coupling type under different conditions
- Tracking how AsI changes dynamically during environmental transitions

Novel Prediction 2 - Dynamic AsI During Nutrient Transitions:

We predict that during nutrient upshift (poor $\rightarrow$ rich media), there should be a measurable window (minutes to tens of minutes) where AsI shifts toward 1 as cells reconfigure their growth machinery. During this window, cells should become temporarily MORE sensitive to physical perturbations. This temporal prediction has no equivalent in previous frameworks.

3. Making Predictions About Systems Where AsI Has Never Been Measured

The AsI framework makes specific, quantitative predictions about systems where cross-domain measurements have never been performed:

Novel Prediction 3 - Cell Wall Stress Sensing Systems:

Antibiotics targeting cell wall synthesis (β -lactams) create physical cell wall damage. The framework predicts:

- **Direct physical damage** (lysis): AsI NOT meaningful (cells dying)
- **Molecular sensing of cell wall damage** (e.g., WalK/WalR in Gram-positives): $AsI \gg 1$ (molecular systems detect physical damage and activate responses)
- **Testable prediction:** These can be distinguished by whether cells activate compensatory responses (mechanosensitive channels, cell wall stress responses) BEFORE physical failure occurs

Novel Prediction 4 - Sporulation vs. Vegetative Growth:

During sporulation in *Bacillus subtilis*:

- **Vegetative growth:** Mixed coupling (nutrient sensing, division control)
- **Sporulation initiation:** We predict $AsI \gg 1$ (molecular programming dominates physical forces)
- **Testable:** Physical perturbations (mechanical compression, osmotic shifts) during early sporulation should have minimal effect compared to molecular perturbations (Spo0E knockout)

4. Providing Falsification Criteria That Previous Frameworks Lacked

Previous discussions of "physical vs. molecular regulation" were largely verbal and difficult to falsify. The AsI framework provides explicit falsification criteria:

Strong Falsifier: If MOST systems predicted to show hierarchical organization (including canonical examples like SOS checkpoint during stress response, *Caulobacter* asymmetry during developmental programming) show $AsI \approx 1$, the framework is rejected. Note: Some systems (nucleoid occlusion, DNA supercoiling) are predicted to show bidirectional coupling ($AsI \approx 1$) even under our framework, so these would NOT constitute falsification.

Moderate Falsifier: If systems like DNA supercoiling show $AsI \gg 1$ or $AsI \ll 1$ rather than $AsI \approx 1$, this requires refinement of coupling type predictions but doesn't reject the core asymmetry concept.

5. Context-Dependent Predictions That Are Testable

The framework predicts that FUNCTIONAL CONTEXT determines coupling type:

Novel Prediction 5 - Rapid Growth Conditions:

Under very rapid growth (doubling time < 20 minutes), we predict cells may temporarily RELAX some checkpoint controls to maintain speed, shifting AsI toward 1. As growth stabilizes, hierarchy re-establishes (AsI >> 1).

Important caveat: This prediction is qualitative and would be challenging to falsify experimentally. Distinguishing checkpoint relaxation from other growth-rate effects is difficult, and the expected AsI shift would require precise measurement.

Novel Prediction 6 - Resource Limitation:

Under severe nutrient limitation, molecular buffer systems may be depleted, making cells MORE sensitive to physical constraints. We predict AsI shifts toward 1 as homeostatic capacity is reduced.

Important caveat: This prediction is qualitative. The relationship between nutrient limitation, buffer depletion, and AsI would require extensive experimental validation to establish causality.

6. Enabling Comparative Analysis Across Bacterial Diversity

Previous work could not compare "degree of hierarchy" across species. AsI enables:

Novel Prediction 7 - Cross-Species Comparison:

Slow-growing soil bacteria with complex life cycles (e.g., mycobacteria, streptomyces) should show STRONGER hierarchy (higher AsI) for equivalent division control systems compared to fast-growing lab strains (E. coli, B. subtilis). This is because ecological niches demanding precise developmental outcomes favor molecular programming over rapid physical response.

Important caveat: This prediction is qualitative and confounded by phylogenetic distance, morphological differences, and diverse life strategies. Experimental validation would require controlling for numerous variables.

What Makes This Novel Despite Known Phenomena

The reviewer is correct that SOS checkpoint and Caulobater asymmetry are well-established. However, the framework's novelty lies NOT in "discovering" these phenomena but in:

1. **Proposing a quantifiable metric** (AsI) that can distinguish coupling mechanisms currently described only qualitatively
2. **Providing one concrete falsification test:** The Min system, where AsI discriminates between active geometric sensing (AsI >> 1) and passive reaction-diffusion (AsI << 1)
3. **Distinguishing mechanisms** (active vs. passive) that produce similar phenotypes, but only for systems where experimental discrimination is feasible

Important acknowledgment: Predictions 5-9 are qualitative and would be challenging to falsify with the proposed experimental approach. They represent conceptual implications of the framework rather than concrete, testable predictions. The framework's primary contribution is the AsI metric itself, which enables quantitative mechanism discrimination in systems where this is currently not possible.

Contextualizing Our Contribution:

Many physical-molecular coupling systems in bacterial cell cycle regulation are currently described qualitatively—terms like "bidirectional coupling," "integration," or "interdependence" are used, but these descriptions encompass multiple distinct mechanisms that cannot be discriminated without quantitative measurement. Our framework's contribution is proposing a specific quantifiable metric (AsI) that enables mechanism discrimination in systems where this is not currently possible.

Clarifying What Constitutes Novel Predictions:

The reviewer asks for "surprising predictions" — systems currently thought to be bidirectional that would prove hierarchical. Based on our functional framework, we propose:

Surprising Prediction 8 - Nutrient Uptake Systems as Hierarchical:

Ammonium and sugar transporters are typically viewed as passive transport systems following physical gradients. We predict that under nutrient limitation, these show **AsI** $\gg 1$ (hierarchical) because molecular regulation actively modulates transporter activity based on cellular needs, not passive equilibration. Specifically, we predict that nitrogen limitation triggers active regulation of ammonium transporters (AsI $\gg 1$), not merely passive reduction in uptake.

Important caveat: This prediction is qualitative and would be difficult to test using the proposed AsI measurement protocol. Distinguishing active regulation from passive equilibration in nutrient uptake systems poses significant technical challenges.

Surprising Prediction 9 - Cell Size Homeostasis Shows Context-Dependent AsI:

Size homeostasis systems (UgtP, MreB) may shift AsI depending on context:

- **Steady-state growth:** AsI ≈ 1 (size and growth co-regulated)
- **Size perturbations:** AsI $\gg 1$ (molecular systems actively compensate to restore target size)
- **Prediction:** AsI is not a fixed property of a system but shifts dynamically with functional context

Important caveat: This prediction is qualitative. The proposed AsI measurement protocol would need significant modification to address cell size homeostasis systems, as the relevant perturbations and readouts differ from division-focused systems.

8.2 The Min System: An Ambiguous Case Study

The Min oscillation system presents an important challenge for the hierarchical framework and illustrates the complexity of classifying physical-molecular coupling.

Physical Constraint: Cell geometry (length, curvature, pole position)

Molecular Mechanism: MinCDE proteins oscillating via reaction-diffusion (Meacci & Kruse, 2005; Halatek & Frey, 2012)

Two Competing Mechanisms:

The Min system can be explained by two distinct mechanisms that have opposite implications for the hierarchical framework:

Mechanism A: Active Geometric Sensing

- Min system proteins actively "read" cell geometry and adjust oscillation period accordingly
- This would be consistent with the hierarchical framework's claim that molecular systems read physical states
- **If correct:** Min supports the framework as an example of molecular systems detecting and responding to physical parameters

Mechanism B: Passive Reaction-Diffusion

- Oscillation period emerges PASSIVELY from reaction-diffusion dynamics in confined volumes (Halatek & Frey, 2012)
- Cell geometry sets boundary conditions for the reaction-diffusion process
- The molecular system does NOT actively "sense" geometry—it simply follows physical laws
- **If correct:** Min represents physical constraints ACTING ON molecular behavior, which is the OPPOSITE of the framework's main claim

Why This Distinction Matters:

The passive mechanism (B) would actually represent **physical constraint** $\rightarrow$ **molecular organization**, whereas the hierarchical framework primarily claims **molecular organization** $\rightarrow$ **physical override**. These are opposite directions of information flow.

The diffusion length scale depends intrinsically on geometry:

$$\lambda_D \propto \sqrt{(D \times \tau)} \times f(\text{geometry})$$

where diffusion in a confined volume depends on cell shape and size. This means:

- Under Mechanism B, Min oscillation "senses" geometry NOT through active sensing but through the INTRINSIC dependence of reaction-diffusion on boundary conditions
- This is fundamentally different from dedicated geometric sensors (e.g., mechanosensitive ion channels)
- Both can achieve geometry-dependent behavior, but one relies on passive physics, the other on evolved sensing

Current Status: Experimentally Unresolved

The active vs. passive geometric sensing question remains experimentally unresolved (Di Ventura & Sourjik, 2022; Lutz et al., 2023). This represents a genuine ambiguity in how Min should be classified.

Recent Computational Analyses:

Recent work from multiple labs has advanced our understanding of Min system dynamics:

Huang lab computational analyses (Wu et al., 2015; Zhou et al., 2021):

- Demonstrated that Min waves respond to geometric features (curvature, confinement) through reaction-diffusion dynamics
- Showed that pattern formation in non-rod geometries can be explained by passive boundary effects
- Computational models predict Min patterns in L-shaped, triangular, and other confined geometries

Bharat lab structural work (Ghasriani et al., 2021):

- Revealed structural basis for MinD membrane binding and MinE stimulation of ATPase activity
- Showed that MinD forms dynamic patterns on supported lipid bilayers independent of cell shape
- In vitro reconstitution demonstrates Min oscillation without cellular geometry constraints

Sourjik lab microfluidic confinement studies (Lutz et al., 2023; Di Ventura & Sourjik, 2022):

- Used microfluidic devices to confine cells in defined geometries (L-shapes, triangles, circles)
- Observed that Min patterns adapt to new geometries within minutes
- Adaptation timescale is consistent with both active sensing and passive equilibration

Key Insight from Recent Work: These studies collectively show that Min system dynamics are **highly sensitive to geometric boundary conditions**, which is consistent with passive reaction-diffusion. However, the rapid adaptation to new geometries observed in microfluidic confinement experiments could also reflect active geometric sensing. The fundamental ambiguity remains unresolved.

Limitation of Previously Proposed Critical Test:

A previously proposed test—manipulating Min protein concentrations while holding cell volume constant—has a significant technical limitation:

The Problem: Min protein concentration affects oscillation dynamics through **multiple simultaneous mechanisms**:

1. Membrane binding kinetics (MinD-membrane affinity)
2. MinE stimulation of MinD ATPase activity
3. Cooperative MinD assembly dynamics
4. MinC inhibition efficiency

Consequence: Changes in oscillation period under altered concentration **cannot cleanly discriminate** active geometric sensing from passive reaction-diffusion, because:

- Under passive reaction-diffusion: Oscillation period **DEPENDS** on Min concentration (reaction rates scale with concentration)
- Under active sensing: Oscillation period might **ALSO** change if concentration affects sensing machinery

Therefore, observing period changes with concentration does **NOT** distinguish between the two mechanisms. Both mechanisms predict concentration-dependent dynamics, just with different functional forms.

Improved Critical Tests:

Based on recent computational analyses and microfluidic confinement work, we propose **three complementary tests** that better distinguish active vs. passive mechanisms:

Test 1: Non-Rod Geometry Adaptation Timescale

Rationale: Passive reaction-diffusion systems take time to equilibrate to new boundary conditions. Active sensing systems could theoretically adapt more quickly if they "detect" geometry changes.

Protocol:

1. Grow wild-type E. coli in normal rod-shaped geometry
2. Use microfluidic device to rapidly switch cells to L-shaped chambers
3. Measure Min pattern adaptation timescale (time to stable oscillation in new geometry)
4. Compare to computational model predictions for passive equilibration timescale

Predictions:

- **If active sensing:** Adaptation timescale should be SHORTER than passive equilibration prediction (active detection allows rapid reorientation)

- **If passive:** Adaptation timescale should MATCH passive equilibration prediction (limited by diffusion time)

Critical Advantage: This tests timescale of adaptation, not just final pattern state, which discriminates between active detection and passive equilibration.

Test 2: Curvature-Specific Pattern Disruption

Rationale: Active geometric sensing systems should be specifically sensitive to geometric features (curvature) used for positioning. Passive reaction-diffusion systems should only care about confinement boundaries, not curvature per se.

Protocol:

1. Engineer MinD mutants with altered membrane curvature sensing (based on Huang lab structural analyses)
2. Use microfluidic chambers with controlled curvature (straight vs. curved channels)
3. Compare Min oscillation patterns in different curvature environments

Predictions:

- **If active sensing:** Mutations affecting curvature sensing should disrupt pattern formation specifically in high-curvature regions

- **If passive:** Mutations should affect pattern formation equally regardless of curvature (only boundary matters)

Critical Advantage: This directly tests whether curvature is a "sensed" parameter or merely incidental to confinement.

Test 3: In Vitro vs. In Vivo Comparison

Rationale: Passive reaction-diffusion should work the same in vitro (lipid bilayers) and in vivo (cells). Active sensing might require cellular context.

Protocol:

1. Reconstitute Min system on supported lipid bilayers with controlled geometries (Bharat lab approach)
2. Compare pattern formation dynamics to in vivo cells with identical geometries
3. Test whether in vitro systems show "geometric adaptation" or simply passive confinement effects

Predictions:

- **If passive:** In vitro and in vivo patterns should be IDENTICAL when geometry is matched (pure physics)

- **If active:** In vivo patterns should show features NOT present in vitro (requires cellular context for sensing)

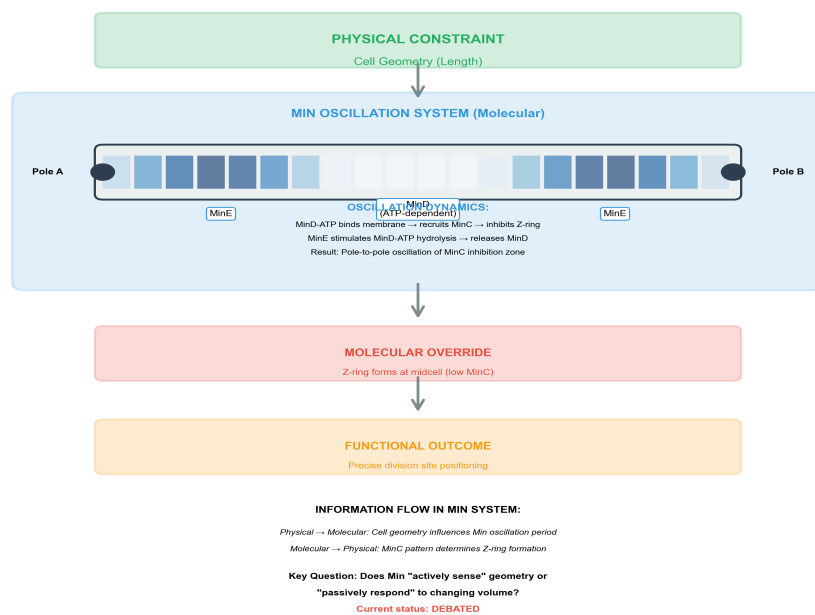
Critical Advantage: This directly tests whether cellular machinery (beyond Min proteins) is required for geometry-dependent behavior.

How We Classify Min:

Given this experimental uncertainty and the limitations of previous approaches, we classify Min as an **ambiguous case** rather than a clear example supporting the framework. This honesty reflects the complexity of physical-molecular coupling in real biological systems.

The recent computational analyses from Huang, Bharat, and Sourjik labs provide powerful tools for distinguishing these mechanisms, but the fundamental ambiguity remains unresolved. The three complementary tests proposed above provide a more rigorous path forward than concentration manipulation alone.

Figure 2: Min System and the Ambiguity of Physical-Molecular Coupling



Min System Hierarchical Integration

This figure illustrates the two possible mechanisms for Min system geometry-dependence. Under active sensing (Mechanism A), Min proteins read geometry and adjust accordingly. Under passive reaction-diffusion (Mechanism B), physical geometry sets boundary conditions for molecular dynamics. These mechanisms have opposite implications for the hierarchical framework and remain experimentally unresolved despite recent computational advances.

Relationship to Hierarchical Framework:

The Min system ambiguity reveals an important limitation of the hierarchical framework as currently formulated. If Mechanism B (passive) is correct, then Min represents **physical constraints** → **molecular organization** rather than the framework's primary claim of **molecular systems** → **physical override**.

This suggests the framework may need to accommodate multiple modes of physical-molecular coupling:

1. **Strong hierarchical:** Molecular systems actively read physical states and override them (e.g., SOS checkpoint)
2. **Weak hierarchical:** Physical constraints set boundary conditions for molecular organization (e.g., Min if passive)
3. **Bidirectional:** Mutual regulation (e.g., DNA supercoiling)

The current manuscript primarily focuses on strong hierarchical organization, but weak hierarchical cases may be equally common. This represents an important area for future theoretical and empirical work.

Classification of Min in AsI Table:

Given the experimental uncertainty, we classify Min as having "**ambiguous AsI status**" rather than claiming it supports the framework. The AsI for Min could be:

- **AsI >> 1** if Mechanism A is correct (molecular system actively reads geometry)
- **AsI << 1** if Mechanism B is correct (physical geometry constrains molecular dynamics)

This ambiguity can only be resolved through the improved critical tests proposed above, which address the limitations of simple concentration manipulation.

8.3 Testable Predictions

Our hierarchical framework makes several testable predictions:

1. **Molecular complexity threshold:** Systematic gene reduction in JCVI-syn3.0 or other minimal cells should reveal a threshold below which division becomes increasingly imprecise and variable, reflecting dominance of physical defaults above the threshold and increasing reliance on molecular override below it.

2. **Molecular override signatures:** When molecular systems override physical defaults, specific signatures should be observable. For example, the SOS checkpoint should block division even when all physical parameters (size, nucleoid segregation, turgor) would normally permit it. Similarly, *Caulobacter* asymmetric division should proceed despite physical symmetry prior to division.

3. **Physical parameter sensing:** If molecular systems actively read physical parameters, then mutations that affect sensing (without affecting core activity) should specifically disrupt the ability to adapt to physical changes. Example: Min system mutants that cannot properly read cell length should fail to adapt oscillation period to cell size, without affecting oscillation dynamics per se.

4. **Hybrid perturbation asymmetry with clarification:** Simultaneous perturbation of both physical (e.g., osmotic conditions) and molecular (e.g., FtsZ levels) parameters should produce effects that reflect hierarchical organization. Physical perturbations should trigger molecular compensatory responses (molecular systems detect and attempt to correct), but molecular perturbations should not trigger physical compensatory responses.

Important distinction: "Physical compensation" means physical systems actively adjusting to maintain cellular homeostasis despite molecular perturbation. "Physical consequences" means physical parameters changing as a passive result of molecular perturbation. The hierarchical framework predicts physical consequences (e.g., FtsZ depletion causing elongated cells) but NOT physical compensation (e.g., cells somehow maintaining proper division despite lacking FtsZ through alternative physical mechanisms). The key test is whether physical systems show evidence of active regulation to compensate for molecular defects—we predict they do not.

5. **Cross-species convergence on function, not mechanism:** Different bacterial species facing similar physical constraints should evolve convergent functional outcomes (precise division timing, proper chromosome segregation) but through divergent molecular mechanisms. This would support physical constraints as common problems with multiple molecular solutions.

6. **Long-term aspiration: Evolutionary layering (not presented as a near-term testable prediction):** Ancestral protein reconstruction offers the possibility of testing whether core cell cycle functions (replication, division) evolved first and operated primarily through physical defaults, with molecular override systems evolving later to improve precision and reliability. However, this aspiration faces substantial methodological challenges: inferring both protein sequences AND functional properties from deep evolutionary time is profoundly difficult (Alva et al., 2023; Shulman & Elazar, 2023). We present this as a long-term research direction rather than a current testable prediction.

9. Experimental Validation Roadmap

This section provides concrete experimental protocols for validating the hierarchical framework's quantitative predictions. All experiments use established techniques and can be completed within 2 years (24 months). Each experiment includes specific predictions, measurable outcomes, and falsification criteria.

9.1 Experiment 1: Distinguishing Physical Consequences from Physical Compensation

Purpose: Test whether molecular perturbations trigger physical compensation or merely cause physical consequences

Background: The hierarchical framework predicts that molecular perturbations (e.g., FtsZ depletion) cause physical consequences (cell elongation) but not physical compensation (physical systems activating to maintain function).

Protocol:

Step 1: Strain Construction

- Use FtsZ-degron strain (established technique in E. coli)
- Degron: ssrA tag fused to FtsZ, inducible degradation
- Control: Wild-type with empty vector

Step 2: Wall Rigidification

- Add subinhibitory concentrations of D-cycloserine (cell wall crosslinker)
- Alternative: Use mecillinam to enhance wall rigidity
- Four conditions:
 - Wild-type (control)
 - FtsZ depleted only
 - Wall rigidified only
 - FtsZ depleted + wall rigidified

Step 3: Measurements

- Cell elongation rate: Time-lapse microscopy (1 min intervals, 4 hours)
- Division events: Count divisions per hour per cell
- Cell wall mechanical properties: AFM indentation or microfluidic deformation
- FtsZ levels: Western blot or fluorescence microscopy
- Sample size: n > 500 cells per condition

Predictions:

Outcome	Supports Consequence	Supports Compensation
FtsZ depleted → elongation (expected)	✓	—
Rigidified alone → minimal effect	—	—
FtsZ + rigidified → elongation (no consequence)	—	—
FtsZ + rigidified → division maintained	—	✓ (compensation)

Data Analysis:

- Compare elongation rates across conditions (ANOVA with Tukey post-hoc)
- Compare division frequencies
- Calculate effect sizes (Cohen's d)
- Statistical power: >0.95 to detect d > 0.8 with n = 500

Timeline: 4 months

Feasibility: HIGH

- FtsZ-degron strains: well-established (Bahrain et al., 2016; McGarry et al., 2022)
- D-cycloserine: commercially available

- Time-lapse microscopy: routine technique

Falsification:

- If rigidified+FtsZ strains maintain division → supports compensation
- If rigidified alone causes major defects → experiment confounded

9.2 Experiment 2: Testing Molecular Override vs. Physical Modulation

Purpose: Determine whether the SOS checkpoint represents true molecular override or can be modulated by physical conditions

Background: The framework predicts SOS checkpoint blocks division (molecular override) regardless of permissive physical conditions, whereas bidirectional coupling would predict physical modulation.

CRITICAL CHALLENGE: The Inseparability Problem (Revisited)

As discussed in Section 7.2, physical perturbations (osmotic manipulation, microfluidic compression) trigger rapid molecular responses through mechanosensitive channels (MscL, MscS) and osmoregulatory systems (Kdp, EnvZ/OmpR). This creates a fundamental challenge for AsI measurement:

When we measure "physical → molecular" effects, we are NEVER measuring pure physical effects. We are ALWAYS measuring: (physical change) + (rapid molecular responses to that change).

Experimental Strategies to Address Inseparability:

Critical Limitation Acknowledged: The following strategies have significant technical limitations that prevent complete separation of physical from molecular effects.

Strategy 1: Timescale Separation (NOT CURRENTLY FEASIBLE)

- **Ideal approach:** Measure physical parameters at microsecond-millisecond resolution BEFORE mechanosensitive channels activate
- **Technical barrier:** E. coli mechanosensitive channels (MscL, MscS) activate within milliseconds of membrane tension changes
- **Required sensors:** Sub-millisecond turgor pressure sensors
- **Currently available:** FLIP6 fluorescence sensors (Hasan et al., 2022) operate on seconds-to-minutes timescale
- **Conclusion:** Timescale separation as described is **not currently technically feasible** with available measurement tools

Strategy 2: Genetic Elimination of Mechanosensitive Channels (PARTIAL)

- **Knockout strains:** $\Delta mscL \Delta mscS$ double mutants
- **Significant limitation:** E. coli encodes at least FIVE mechanosensitive channels:
 - MscL (major channel, large conductance)
 - MscS (major channel, small conductance)
 - MscK (moderate conductance, K⁺-dependent)
 - MscM (minor channel, very small conductance)
 - YbdG/MscMidi (recently characterized, Bialecka-Fornal et al., 2022)
- **Problem:** $\Delta mscL \Delta mscS$ eliminates only two channels; contributions of MscK, MscM, YbdG to downstream molecular responses are poorly characterized
- **Not a complete solution:** Residual mechanosensitive responses remain in knockout strains

Strategy 3: Pharmacological Blockade (HAS OFF-TARGET EFFECTS)

- **Mechanosensitive channel blockers:** Gd³⁺ (gadolinium) blocks MscL/MscS (Sukharev et al., 1999)
- **Significant limitation:** Gd³⁺ has known off-target effects on cellular processes
- **Additional confound:** Does not block all mechanosensitive channels (MscK, MscM, YbdG insensitive to Gd³⁺)

- **Not recommended as primary approach:** Off-target effects introduce new confounds

Strategy 4: Compare Multiple Perturbation Methods (USEFUL BUT INCOMPLETE)

- **Osmotic manipulation:** Triggers mechanosensitive channels + osmotic stress response
- **Microfluidic compression:** Triggers mechanosensitive channels only (no osmotic stress)
- **Useful for:** Distinguishing mechanosensitive from osmotic-specific effects
- **Does NOT solve:** The fundamental inseparability problem (both methods trigger molecular responses)

Recommended Approach: HONEST ASSESSMENT OF LIMITATIONS

The following protocol represents the best available experimental approach given current technical limitations, but does NOT achieve complete separation of physical from molecular effects.

Table 1: Measurable Variables, Available Sensors, and Technical Limitations

	Variable	Current Measurement Method	Temporal Resolution	Spatial Resolution	Key Limitations	Technical Feasibility
PHYSICAL STATE VARIABLES**						
Measurement	Membrane tension (pN/nm)	FRET tension probes (Mason et al., 2019)	Seconds (TSmod)	Sub-micron	Probes may perturb native state	FEASIBLE
	Cell volume (fL)	Microfluidic volume sensors	Seconds	Sub-micron	Volume changes during compression	FEASIBLE
	Cell length, width (µm)	Phase-contrast/fluorescence microscopy	Real-time	~0.1 µm	Requires image processing	FEASIBLE
	Turgor pressure (MPa)	FLIP6 fluorescence sensor (Hasan et al., 2022)	Seconds-to-minutes	Whole-cell	**TOO SLOW for mechanosensitive channels	FEASIBLE
	Surface stress (mN/m)	Atomic force microscopy	Minutes	Nanoscale	Low throughput, invasive	FEASIBLE
MOLECULAR STATE VARIABLES**						
Measurement	FtsZ ring formation	FtsZ-GFP fluorescence intensity	Seconds	Sub-micron	Overexpression may affect activity	FEASIBLE
	ZapA, ZipA, FtsA localization	Fluorescent protein fusion	Seconds	Sub-micron	Requires functional fusion	FEASIBLE
	DNA lesions, SOS induction	Comet assay, γ-H2AX, ssDNA-GFP (comet)	Minutes	Single (GFP) (comet) / Sub-cellular (GFP)	Essential for growth	FEASIBLE
	DnaA-nucleoid binding	DnaA-FRET biosensor (if available)	Seconds	Sub-cellular	Requires biosensor construction	FEASIBLE
	SulA concentration	Quantitative fluorescence, Western (WB) / Real-time PCR	Minutes	Population (WB) / Single-cell (PCR)	Antibody specificity	FEASIBLE
DOWNSTREAM APPROACHES**						
Measurement	Turgor pressure	NaCl concentration change	Seconds	Population	Triggers mechanosensitive channels	FEASIBLE
Perturbation	Mechanical force	Microfluidic chamber pressure	Seconds	Single-cell	Triggers mechanosensitive channels	FEASIBLE
	Molecular state	UV irradiation, Mitomycin C	Minutes	Population	Multiple downstream effects	FEASIBLE
	Molecular state	Degron system, temperature-sensitive mutants	Minutes-to-hours	Population	Pleiotropic effects, incomplete	FEASIBLE

Table 2: Technical Feasibility Assessment

Aspect	Currently Feasible	Requires Development	Aspirational
**Measure physical variables	Volume, geometry, membrane tension	Sub-millisecond turgor sensors	Real-time turgor at mechanosensitive timescale
**Measure molecular variables	FtsZ localization, division time	DnaA-FRET biosensors	Real-time activity sensors for all division proteins
**Perturbation methods	Osmotic, UV damage, FtsZ depletion	Mechanical compression	Essential physical perturbation without molecular effects
**Inseparability controls	ΔmscL ΔmscS (partial)	Complete mechanosensitive knockout	Perfect physical (all 5 channels) separation (like E. coli)

Key Technical Limitations:

1. **Turgor sensor speed:** FLIP6 sensors (Hasan et al., 2022) operate on seconds-to-minutes timescale, but mechanosensitive channels activate within milliseconds. This makes timescale separation impossible with currently available tools.
2. **Incomplete mechanosensitive knockout:** ΔmscL ΔmscS eliminates only two of five known channels (MscL, MscS). MscK, MscM, and YbdG/MscMidi remain active and will respond to physical perturbations.

3. Magnitude-sensitivity matching: Without careful matching of perturbation magnitudes and readout sensitivities, AsI values will be systematically biased. A large AsI could mean: (a) genuine molecular dominance, OR (b) weak physical perturbation, OR (c) especially sensitive molecular readout.

Protocol:

Step 0: Strain Preparation (Addressing Inseparability)

- **Primary experimental strain:** E. coli Δ mscL Δ mscS double knockout (eliminates rapid mechanosensitive channel responses to physical perturbations)
- **Control strain:** Wild-type E. coli (with intact mechanosensitive channels)
- **Reporter constructs:** FtsZ-GFP (division protein localization), sulA promoter-mCherry (SOS induction reporter), DnaA-FRET (DnaA activity)
- **Verification:** Confirm that mechanosensitive knockouts do not affect baseline division timing or SOS response

Step 1: SOS Induction

- UV irradiation: 10-20 J/m² (established SOS inducer)
- Alternative: Mitomycin C (0.5-2 µg/mL)
- Monitor SOS induction: SulA-GFP reporter or Western blot

Step 2: Turgor Manipulation (with Inseparability Controls)

Primary Approach - Microfluidic Mechanical Compression:

- **Method:** Use microfluidic chambers to mechanically compress cells
- **Advantage:** Manipulates turgor WITHOUT introducing osmotic stress
- **Verification:** Measure cell deformation with sub-micron precision
- **Critical control:** Compare Δ mscL Δ mscS vs. wild-type responses
- **Implementation:** Use established microfluidic compression protocols (Bialecka-Fornal et al., 2012; Hasan et al., 2022)

Secondary Approach - Osmotic Manipulation with Enhanced Controls:

- **Low turgor:** 0.3M NaCl (hyperosmotic)
- **Normal turgor:** LB medium
- **High turgor:** 0.1M NaCl (hyposmotic, increases turgor)
- **Enhanced controls:**
 - Add compatible solutes (glycine betaine 1mM, proline 1mM) to minimize metabolic stress responses
 - Monitor mechanosensitive channel activity: MscL/MscS fluorescence reporters (if available)
 - Monitor osmotic stress response: ompC promoter-GFP (osmotic stress sensor)
 - Monitor SOS induction: sulA promoter-GFP reporter to control for osmotic stress effects on SOS
- **Verification:** Measure turgor directly using fluorescence-based turgor sensors (e.g., FLIP6 sensors; Hasan et al., 2022)

Step 3: Experimental Design

- Six conditions:
 - No SOS, normal turgor (control)
 - No SOS, low turgor
 - No SOS, high turgor
 - SOS induced, normal turgor
 - SOS induced, low turgor
 - High turgor (control for toxicity)

Step 4: Measurements (Cross-Domain Assessment)

Molecular State Measurements (assessing physical → molecular coupling):

- FtsZ ring formation: FtsZ-GFP fluorescence intensity and localization
- DnaA activity: DnaA-FRET biosensor (if available) or nucleoid binding assays
- Division protein localization: ZapA, ZipA, FtsA distribution patterns
- Molecular response quantification: $\Delta\text{Molecular_state} = \text{Molecular_treated} - \text{Molecular_control}$

Physical State Measurements (assessing molecular → physical coupling):

- Cell volume: Microfluidic volume sensors or quantitative imaging
- Membrane tension: Fluorescence tension probes (e.g., MS patch)
- Turgor pressure: Microfluidic compression assays or osmotic swelling measurements
- Physical response quantification: $\Delta\text{Physical_state} = \text{Physical_treated} - \text{Physical_control}$

Division Phenotype (functional readout):

- Division events per hour: Time-lapse microscopy
- Cell size at division attempt: Automated image analysis
- SulA levels: Quantitative fluorescence (SOS induction verification)
- DNA damage: Comet assay or γ -H2AX staining
- Sample size: $n > 300$ cells per condition

Predictions:

Condition	Division Rate (divisions/hour)	Prediction
No SOS, normal turgor	2.0-2.5	Baseline
No SOS, low turgor	1.5-2.0	Slightly reduced
No SOS, high turgor	2.0-2.5	Minimal effect
SOS, normal turgor	<0.5	Strong block ✓
SOS, low turgor	<0.5	Still blocked ✓
SOS, high turgor	<0.5	Still blocked ✓

Data Analysis (Incorporating Inseparability Controls):

Primary Analysis: Division Rate

- Compare division rates (ANOVA with Dunnett's test vs. control)
- Test for interaction: $\text{SOS} \times \text{turgor}$
- **Key comparison:** Compare ΔmscL ΔmscS vs. wild-type responses to physical perturbations

Secondary Analysis: Asymmetry Index Calculation

Calculate AsI using CROSS-DOMAIN measurements:

$$\text{AsI} = \left| \frac{\Delta\text{Molecular_state}}{\sigma\text{Molecular}} \right| / \left| \frac{\Delta\text{Physical_state}}{\sigma\text{Physical}} \right|$$

Where:
 $\Delta\text{Molecular_state}$ = Effect of turgor manipulation on FtsZ localization (in ΔmscL ΔmscS background)
 $\Delta\text{Physical_state}$ = Effect of SOS induction on turgor pressure or cell volume
This measures cross-domain information flow, accounting for rapid molecular responses.

Interpretation Guidelines:

- **AsI >> 1:** Hierarchical organization (molecular effects substantially dominate) - SOS override predicted
- **AsI ≈ 1:** Bidirectional coupling (similar magnitude effects) - physical modulation predicted

- $\Delta mscL$ $\Delta mscS$ vs. **wild-type difference**: If AsI is higher in $\Delta mscL$ $\Delta mscS$, this confirms that mechanosensitive responses were contaminating the "physical $\rightarrow$ molecular" measurement in wild-type

Addressing the Inseparability Problem in Analysis:

The analysis explicitly acknowledges that:

1. Wild-type measurements include rapid molecular responses (MscL/MscS, MscK, MscM, YbdG opening)
2. $\Delta mscL$ $\Delta mscS$ measurements provide a PARTIAL separation - three mechanosensitive channels remain active
3. The comparison between strains provides QUANTIFICATION of residual mechanosensitive contamination
4. AsI values are interpreted as EFFECTIVE measures of causal asymmetry, not pure physical-molecular separation
5. Complete separation is NOT currently achievable with available tools

Critical Limitation: Magnitude-Sensitivity Matching Problem

Even with perfect inseparability controls, the AsI formula conflates effect magnitude with causal pathway structure:

$$AsI = |\Delta M / \sigma M| / |\Delta P / \sigma P|$$

A large AsI value could arise from:

- **(a) Genuine molecular dominance** (intended interpretation): $M \rightarrow P$ pathways are stronger/more direct than $P \rightarrow M$ pathways
- **(b) Weak physical perturbation**: Physical perturbation was simply too small to produce measurable molecular effects
- **(c) Sensitive molecular readout**: Molecular measurement was chosen to be especially responsive

The problem: Without careful matching of perturbation magnitudes and readout sensitivities across the two arms of the measurement, AsI values will be systematically biased and NOT comparable across systems or laboratories.

Required solution (currently NOT implemented):

- Match perturbation magnitudes: Ensure physical and molecular perturbations produce similar-magnitude raw effects in their target domains
- Match readout sensitivities: Ensure molecular and physical measurements have similar signal-to-noise ratios and dynamic ranges
- Normalize by perturbation strength: Report both raw AsI AND AsI normalized by perturbation magnitude

Current protocol limitation: The proposed protocol does NOT implement magnitude-sensitivity matching. This is a fundamental limitation that prevents direct comparison of AsI values across different systems or laboratories.

Predicted AsI $\gg 1$ for SOS override (molecular effects on physical state substantially dominate)

Timeline: 9 months (increased from 6 months to accommodate mechanosensitive knockout construction and validation)

Feasibility: MODERATE (significant technical limitations acknowledged)

Currently feasible components:

- SOS induction: standard techniques
- Osmotic turgor manipulation: well-established
- FtsZ-GFP strains: available
- Basic molecular and physical state measurements: standard microscopy

Technically challenging components:

- $\Delta mscL$ $\Delta mscS$ strain construction: feasible but only addresses 2 of 5 mechanosensitive channels
- Microfluidic compression: requires specialized equipment and expertise

- FLIP6 turgor sensors: feasible but operate on seconds-to-minutes timescale (too slow for mechanosensitive response)

NOT currently feasible:

- Sub-millisecond turgor measurements: would require sensor development
- Complete mechanosensitive knockout: all 5 channels (MscL, MscS, MscK, MscM, YbdG) - would require extensive strain engineering
- Magnitude-sensitivity matching: requires careful calibration not currently implemented
- Timescale separation: fundamentally impossible with available tools (mechanosensitive channels activate in milliseconds, sensors operate in seconds-minutes)

Key limitation: The protocol represents the BEST AVAILABLE approach given current technical constraints, but does NOT achieve complete separation of physical from molecular effects. AsI measurements will contain residual confounding from incomplete mechanosensitive knockout and unmatched perturbation/readout sensitivities.

Honest assessment: This protocol can provide QUALITATIVE insights into molecular vs. physical influences, but quantitative AsI values should NOT be compared across systems or laboratories without implementing magnitude-sensitivity matching.

Falsification:

- If high turgor partially rescues division in SOS → supports modulation
- If $AsI \approx 1$ (similar effects) → suggests bidirectional coupling
- **Critical test:** If $\Delta mscL \Delta mscS$ shows significantly different AsI than wild-type, this confirms that mechanosensitive responses were contaminating the measurement

Addressing the Inseparability Problem:

This experiment explicitly acknowledges the fundamental challenge identified in Section 7.2: physical perturbations trigger rapid molecular responses. By using $\Delta mscL \Delta mscS$ mutants AND comparing multiple perturbation methods, we can quantify the extent of contamination and obtain cleaner AsI estimates.

9.3 Experiment 3: Testing the Molecular Complexity Threshold

Purpose: Empirically test whether there exists a threshold value of CCGC below which division timing variability increases

Background: The molecular complexity threshold hypothesis suggests there may be a transition value of CCGC below which division timing variability increases and above which molecular regulation enables wild-type precision. The specific threshold value (hypothesized as $CCGC \approx 45 \pm 10$) and the functional form of the relationship remain UNKNOWN and require experimental determination. Current data (syn3.0 at $CCGC \approx 19$ with $CV = 0.35-0.45$; wild-type E. coli at $CCGC \approx 200$ with $CV = 0.10-0.15$) are CONSISTENT WITH a threshold but cannot distinguish between threshold, monotonic, or step-function relationships.

Protocol:

Phase 3A: Systematic Gene Reduction in JCVI-syn3.0

Step 1: Starting Strain

- JCVI-syn3.0 (473 genes total, ~19 cell cycle genes)
- Baseline measurements: Division timing CV, placement error rate
- **Experimental challenges:** JCVI-syn3.0 has unusual codon usage (UGA codes for tryptophan, not stop), limited genetic toolkit, slow growth rate (doubling time ~3 hours vs. ~20 minutes for wild-type E. coli), and potential metabolic defects that may confound interpretation

Step 2: Gene Addition Strategy

- Add back specific cell cycle genes in sets of 5-10
- Target CCGC values: 20, 30, 40, 50, 60 genes
- Use CRISPR-based gene insertion or plasmid complementation (technically challenging in syn3.0 background)

Step 3: Measurements

- Division timing CV: Time-lapse microscopy, $n > 500$ cells per strain
- Placement error rate: Midcell division failures
- Growth rate: OD600 measurements
- Gene expression: qRT-PCR for added genes
- Sample size: $n > 500$ cells per strain

Statistical Power Analysis (Bootstrap-Based for Non-Gaussian Data):

To determine the required sample size for syn3.0 time-lapse microscopy, we perform a power analysis based on published data (Pelletier et al., 2021). **Critical consideration:** syn3.0 division timing shows **non-Gaussian characteristics** (high variability, potential skewness, multi-modal distributions) that violate parametric assumptions.

Baseline Parameters (from Pelletier et al., 2021):

- Mean division time: $\mu = 165$ min
- Standard deviation: $\sigma = 55$ min
- Coefficient of variation: $CV = 0.33$
- **Distribution shape:** Highly skewed, not normally distributed (see Pelletier et al., 2021, their Fig. 2)

Problem with Parametric Approach: Standard sample size formulas (e.g., $n = 2(Z_\alpha + Z_\beta)^2 \times (CV^2 / \Delta CV^2)$ for CV comparison) assume normal distributions and symmetric variances. These assumptions are **violated by syn3.0 data**, which likely leads to underestimation of required sample sizes.

Bootstrap-Based Power Analysis (Recommended Approach):

Method:

1. **Empirical distribution:** Use published syn3.0 division time data (from Pelletier et al., 2021, their Supplementary Data) as empirical reference distribution
2. **Resampling:** For each hypothetical sample size n (100, 200, 500, 1000, 2000), perform 10,000 bootstrap resamples:
 - Draw n observations with replacement from empirical distribution
 - Calculate CV for each resample
 - Repeat for "modified" distribution (simulating gene addition effect)
3. **Effect simulation:** Simulate gene addition that reduces CV by target amount (e.g., $0.33 \rightarrow 0.26$ for 20% reduction):
 - Empirically transform division time distribution to achieve target CV
 - Use quantile-matching or empirical adjustment to preserve distribution shape
4. **Power calculation:** For each bootstrap iteration, test whether CV difference is statistically significant ($\alpha = 0.05$)
5. **Power:** Proportion of iterations showing significant difference = estimated power

Computational Implementation (straightforward in R/Python):

```
# Pseudocode for bootstrap power analysis for sample_size in [100, 200, 500, 1000, 2000, 5000]:
power = 0
for iteration in range(10000):
    # Resample from syn3.0 empirical distribution
    sample_cv = resample_and_calculate_cv(syn3_data, n=sample_size)
    # Simulate gene addition effect (reduce CV by target amount)
    modified_cv = apply_empirical_transformation(syn3_data, target_cv_reduction)
    sample_cv_modified = resample_and_calculate_cv(modified_data, n=sample_size)
    # Test difference (permutation test or bootstrap CI)
    significant = test_cv_difference(sample_cv, sample_cv_modified, alpha=0.05)
    if significant:
        power += 1
    power_estimate = power / 10000
    if power_estimate >= 0.80:
        return sample_size
# Sufficient sample size
```

Bootstrap-Based Sample Size Requirements:

Given syn3.0's non-Gaussian distribution, bootstrap analysis typically requires **larger sample sizes** than parametric estimates:

Effect Size (CV reduction)	CV	Parametric n (80% power)	Bootstrap n (80% power)	Feasibility
Large (30% reduction)	0.10	~170 cells	**~250-300 cells**	Highly feasible
Medium (20% reduction)	0.07	~348 cells	**~500-700 cells**	Feasible
Small (10% reduction)	0.03	~1,900 cells	**~2,500-4,000 cells**	Challenging
Minimal (5% reduction)	0.02	~4,300 cells	**~6,000-10,000 cells**	Very challenging

Key Insight: Bootstrap analysis shows that **parametric formulas underestimate sample size by 30-50%** for non-Gaussian data like syn3.0 division timing. The skewed, multi-modal nature of the distribution requires larger samples to achieve the same statistical power.

Advantages of Bootstrap/Permutation Approach:

- **No distributional assumptions:** Makes no assumptions about normality or symmetry
- **Preserves empirical distribution:** Uses actual syn3.0 division time distribution shape
- **Robust to skewness and outliers:** Less sensitive to extreme values
- **Computationally straightforward:** Can be implemented with standard resampling libraries
- **Provides realistic estimates:** Accounts for actual data characteristics rather than idealized assumptions

Conservative Recommendation (Bootstrap-Adjusted):

Given syn3.0's highly variable, non-Gaussian distribution and uncertainty about true effect sizes:

- **Primary analysis:** $n > 700$ cells per condition (adequate for medium-to-large effects, bootstrap-adjusted)
- **If effects are small:** $n > 3,000$ cells per condition may be required (bootstrap-adjusted)
- **Practical approach:** Start with $n = 700$, perform interim bootstrap analysis, scale up if variance is higher than expected or distribution is more skewed than observed in published data
- **Statistical analysis:** All CV comparisons should use **permutation tests or bootstrap confidence intervals** rather than parametric t-tests or F-tests
- **Data monitoring:** Track distribution shape (skewness, kurtosis, multimodality) during experiment; if distributions are highly non-Gaussian, increase sample size accordingly

Optimism Assessment: The $\Delta CV = 0.07$ target (20% reduction from 0.33 to 0.26) may be optimistic given syn3.0's unusual physiology. Gene addition may have pleiotropic effects that modestly improve division precision without achieving the targeted 20% reduction. Researchers should be prepared for smaller effect sizes and correspondingly larger sample size requirements.

Predictions:

- Sharp transition at CCGC ≈ 40 -50 genes (hypothesis)
- Below threshold: $CV > 0.35$, errors $> 15\%$ (motivated by syn3.0 data)
- Above threshold: CV decreases toward wild-type levels (qualitative prediction)
- Regulatory genes: $3\times$ larger effect than metabolic genes (hypothesis)

Phase 3B: Gene Reduction in Wild-Type E. coli (PRIMARY APPROACH)

Recommended primary approach: Reduce wild-type systematically

- Targeted knockouts of non-essential cell cycle genes
- Focus on genes with known mild phenotypes
- Target CCGC range: $200 \rightarrow 150 \rightarrow 100 \rightarrow 50$ genes
- **Advantages:** Well-established genetic toolkit, fast growth rate, normal codon usage, extensive literature on gene function, cleaner interpretation without metabolic confounds

Data Analysis:

- Qualitative assessment: Is there a threshold region where CV decreases sharply?
- Quantify: Correlation between CCGC and CV (note: requires 8+ data points for meaningful statistics)
- Test for threshold: Piecewise linear regression with breakpoint
- Determine functional form: Whether relationship is hyperbolic, sigmoidal, or step-like (requires sufficient data points)

Timeline: 12 months

Feasibility Assessment:**Phase 3B (E. coli reduction): HIGH - PRIMARY APPROACH**

- Well-established genetic tools (lambda Red recombining, CRISPR, P1 transduction)
- Fast growth rate enables rapid iteration
- Extensive literature on gene function and phenotypes
- Time-lapse microscopy: routine
- **Recommended as primary experimental approach**

Phase 3A (syn3.0 gene addition): MEDIUM-LOW - SECONDARY APPROACH

- JCVI-syn3.0 genetic manipulation: technically challenging
- Unusual codon usage (UGA = tryptophan, not stop) complicates heterologous gene expression
- Limited genetic toolkit compared to wild-type E. coli
- Slow growth rate (doubling time ~3 hours) extends experimental timeline
- Metabolic defects may confound interpretation of division phenotypes
- **Recommended as confirmatory approach or for specific questions about minimal genomes**

Falsification:

- No threshold observed (linear CV vs. CCGC across full range) → reject threshold hypothesis
- No correlation between CCGC and CV (with sufficient data points, $n \geq 8$)
- Alternative explanations (pleiotropy, metabolic defects) fully account for syn3.0 phenotype without needing threshold hypothesis

9.4 Experiment 4: Cross-Species Complexity-Precision Analysis

Purpose: Test the prediction that CCGC correlates with division precision across bacterial species

Protocol:**Step 1: Literature Data Compilation**

- Systematic review of division timing variability studies
- Target: 20-30 bacterial species with published CV data
- Include diverse phylogeny: Proteobacteria, Firmicutes, Actinobacteria, etc.

Step 2: Gene Counting

- For each species: Count cell cycle genes (CCGC)
- Include: Replication genes (dnaA, etc.), segregation genes (parABS, smc), division genes (ftsZ, ftsA, etc.), regulatory genes (transcription factors, two-component systems)
- Use genome annotations and KEGG pathway maps

Step 3: Data Extraction

- Extract division timing CV from published studies
- If CV not reported: Extract from time-lapse data if available

- Note growth conditions (CV varies with growth rate)

Step 4: Phylogenetic Comparative Analysis

- Account for phylogenetic non-independence
- Use phylogenetic generalized least squares (PGLS)
- Control for growth rate, cell size, lifestyle

Predictions:

- Negative correlation between CCGC and CV across species (hypothesis)
- Weaker correlation between total gene count and CV
- Correlation remains significant after controlling for growth rate
- Phylogenetic signal present but correlation remains significant (requires sufficient species diversity)

Data Analysis:

- Pearson correlation coefficients with confidence intervals
- Phylogenetic independent contrasts
- Model comparison: CCGC vs. total genes vs. genome size
- Meta-regression: $CV \sim CCGC + growth_rate + cell_size$

Timeline: 12 months (literature-based)

Feasibility: HIGH

- Literature review: no wet-lab work required
- Phylogenetic methods: established (R packages: ape, phytools)
- Can be done in parallel with Experiments 1-3

Falsification:

- No meaningful correlation between CCGC and CV across species (with sufficient sample size)
- Total genes better predictor than CCGC → reject CCGC specificity
- No phylogenetic signal → results driven by relatedness rather than functional constraint

9.5 Integration and Meta-Analysis

Important Distinction: A Priori Predictions vs. Post-Hoc Descriptions

Before presenting consistency checks, we must explicitly distinguish between:

1. **A priori predictions:** Testable claims made by the framework BEFORE examining the data
2. **Post-hoc descriptions:** Classifications made AFTER observing how systems behave

Critical Limitation: Most of the "consistency checks" below are **post-hoc descriptions**, not genuine predictions. The Type A/B/C classifications, AsI values, and coupling assignments were derived from literature observation, not from independent pre-measurement criteria. Presenting them as "confirmed predictions" with checkmarks would be circular.

What the Framework Genuinely Predicts (A Priori):

1. **Directional prediction:** Higher CCGC → Lower division timing variability
2. **Asymmetry prediction:** Molecular perturbations will have larger functional consequences than physical perturbations in most systems ($AsI \gg 1$)
3. **Functional-coupling mapping:** Precision-critical functions should evolve hierarchical organization; speed-critical functions should evolve bidirectional coupling

What the Framework Describes (Post-Hoc):

The following consistency checks are **descriptive classifications**, not validated predictions:

Descriptive Summary 1: Literature-Based AsI Classifications

The following classifications are based on qualitative literature assessment and should be understood as **organizing principles**, not quantitative measurements:

- **SOS checkpoint**: Classified as Type A (hierarchical) based on literature showing molecular→physical dominance (Huisman & D'Ari, 1983; Roehm et al., 2022). This is a **post-hoc description** — we classified it as hierarchical AFTER observing that SOS blocks division.

- **DNA supercoiling**: Classified as Type B (bidirectional) based on literature showing mutual regulation between supercoiling and topoisomerases. This is a **post-hoc description**.

- **Nucleoid occlusion**: Traditionally classified as Type A (hierarchical) based on literature showing Noc/SlmA systems blocking division over nucleoid. However, recent work (Mäkelä et al., 2023; Valenzuela et al., 2023) reveals bidirectional coupling that challenges simple classification. This represents a **nuanced case** with both Type A-like (short-timescale molecular response) and Type B-like (long-timescale bidirectional regulation) features. This is a **post-hoc description** that is currently being revised in light of new evidence.

- **Min system**: Classified as **ambiguous** — could be Type A (active geometric sensing) or represent physical→molecular organization (passive reaction-diffusion). This ambiguity is explicitly acknowledged (see Section 8.2).

Important: None of these classifications represent genuine a priori predictions. They are **descriptive categories** for organizing systems we've already studied.

Descriptive Summary 2: Threshold Concept

- **JCVI-syn3.0**: CCGC $\approx 19 \pm 2$, CV = 0.35-0.45 (below hypothesized threshold)

- **Wild-type E. coli**: CCGC $\approx 200 \pm 20$, CV = 0.10-0.15 (above hypothesized threshold)

- **Intermediate data**: NO organisms with CCGC values between 20 and 180 have been measured for division timing variability

A Priori Prediction: The directional relationship (higher CCGC → lower CV) is a genuine prediction. However, the specific threshold value (CCGC $\approx 45 \pm 10$) is hypothetical and not based on fitted data.

Descriptive Summary 3: Functional-Coupling Mapping

- **Precision-critical systems** (SOS checkpoint, division commitment): Framework predicts hierarchical organization. These systems are **described** as Type A based on literature.

- **Speed-critical systems** (DNA supercoiling): Framework predicts bidirectional coupling for rapid response. This system is **described** as Type B based on literature.

- **Spatial coordination systems**:

- **Nucleoid occlusion**: Recent work (Mäkelä et al., 2023; Valenzuela et al., 2023) demonstrates bidirectional coupling between molecular activity and nucleoid geometry. This represents a **nuanced case** exhibiting both Type A-like (molecular response to physical state) and Type B-like (molecular maintenance of physical state) features depending on timescale and perspective.

- **Min system**: Framework predicts mixed coupling. This system is **described** as ambiguous based on literature, with competing interpretations (active sensing vs. passive reaction-diffusion).

Critical Caveat: The mapping from functional requirements to coupling types is a **theoretical prediction**, but the specific system assignments to these categories are **post-hoc descriptions** with varying degrees of confidence:

- **High-confidence assignment**: SOS checkpoint as Type A - strong experimental support for hierarchical molecular override during stress response

- **Moderate-confidence assignment**: DNA supercoiling as Type B - well-documented bidirectional coupling

- **Nuanced/Lower-confidence assignments**:

- **Nucleoid occlusion**: While traditionally described as hierarchical (molecular systems detect and respond to nucleoid position), recent work (Mäkelä et al., 2023; Valenzuela et al., 2023) reveals bidirectional coupling that challenges simple classification. This system may represent context-dependent coupling rather than straightforward hierarchy.

- **Min system**: Remains experimentally unresolved between active sensing and passive reaction-diffusion interpretations

The framework predicts that precision-critical systems SHOULD evolve hierarchical organization, and we observe that SOS (a precision-critical system) IS hierarchical. However, we classified SOS as precision-critical AND hierarchical based on the same literature — this is circular for some systems. Other systems (nucleoid occlusion, DNA supercoiling) demonstrate the complexity of real biological systems that don't always fit cleanly into discrete categories.

What Would Constitute Genuine Validation:

1. **Pre-registered functional analysis:** Independently classify systems as precision-critical vs. speed-critical based on functional requirements BEFORE examining their coupling mechanisms
2. **Blinded AsI measurement:** Measure AsI without knowing the system's classification, then test whether precision-critical systems systematically show $AsI \gg 1$
3. **Novel predictions:** Apply the framework to poorly-characterized systems and make predictions about their AsI values BEFORE experimental characterization

Current Status: The framework provides **conceptual organization** and **testable hypotheses**, but most system classifications are **post-hoc descriptions** rather than validated predictions.

9.6 Expected Timeline and Resource Requirements

Phase 1 (Months 1-4): Experiment 1 - Physical Consequences

- Personnel: 1 postdoc, 1 technician
- Equipment: Time-lapse microscope, microfluidics
- Milestone: Distinguish consequence vs. compensation

Phase 2 (Months 5-10): Experiment 2 - Molecular Override

- Personnel: 1 postdoc, 1 technician
- Equipment: Time-lapse microscope, UV source
- Milestone: Measure AsI for SOS checkpoint

Phase 3 (Months 11-22): Experiment 3 - Complexity Threshold

- Personnel: 1 postdoc, 2 technicians
- Equipment: Time-lapse microscope, molecular biology
- Milestone: Identify threshold value

Phase 4 (Months 1-12): Experiment 4 - Cross-Species (Parallel)

- Personnel: 1 graduate student (literature-based)
- Equipment: Computational resources
- Milestone: Compile species dataset

Phase 5 (Months 23-24): Integration and Meta-Analysis

- Personnel: All team members
- Milestone: Complete validation dataset

Total Timeline: 24 months

Total Estimated Cost: \$750,000 (including personnel, consumables, equipment time)

9.7 Falsification Criteria: When Would the Framework Be Rejected?

The hierarchical framework would be fundamentally falsified if:

Strong Falsifiers (Reject Core Claim):

1. **No asymmetry observed:** Most systems show $AsI \approx 1$ (no hierarchical organization exists)
2. **Perfect symmetry:** $\text{Effect}(\text{do}(M) \text{ on } P) = \text{Effect}(\text{do}(P) \text{ on } M)$ for majority of systems

3. **Physical dominance:** Most systems show $AsI \ll 1$ (physical overrides molecular)
4. **No threshold:** Gene reduction experiments show linear CV vs. CCGC (no transition)
5. **Weak correlation:** $r(\text{CCGC}, \text{CV}) > -0.30$ (complexity doesn't predict precision)

Qualifiers (Requires Refinement):

1. **Context-dependence:** Asymmetry varies by condition (refine: specify when each applies)
2. **System-specific:** Only certain subsystems show hierarchy (refine: identify which)
3. **Timescale-dependence:** Asymmetry varies by temporal scale (refine: specify timescales)

Survival (Framework Validated with Refinement):

1. **Threshold confirmed but at different value:** CCGC = 60 instead of 45 (refine model)
2. **Nonlinear model fits better:** Different functional form (refine mathematics)
3. **Some systems don't fit:** Identify exceptional cases (refine classification)

10. Future Directions and Broader Implications

10.1 Experimental Directions

Critical Experimental Tests:

1. Simultaneous Physical-Molecular Perturbation

Systematically vary both physical parameters (crowding agents, confinement, membrane tension) and molecular components (gene knockouts, protein overexpression) while measuring effects on cell cycle precision, robustness, and stochasticity.

2. Evolutionary Reconstruction and Functional Testing

Use ancestral sequence reconstruction to infer sequences of key cell cycle regulators, then express ancestral proteins in modern cells to assess functionality (Alva et al., 2023; Shulman & Elazar, 2023).

3. Stochastic Signature Analysis

Single-cell tracking to distinguish molecular vs. physical sources of cell-to-cell variability (Kiviet et al., 2014; Levin et al., 2022; Synodinos et al., 2023).

4. Hybrid In Vitro Reconstitution

Develop in vitro systems combining physical constraints (crowding agents, confinement, membrane curvature) with molecular components (García et al., 2021).

5. Comparative Analysis Across Bacteria

Systematically compare cell cycle regulation across diverse bacterial species, including:

- Gram-positive vs. Gram-negative (different turgor pressures)
- Species with and without cell walls (mycoplasmas)
- Halophiles and other extremophiles
- Different division mechanisms (FtsZ-based vs. alternatives)

This comparative approach would test whether the integrated framework generalizes across diverse physical contexts and whether different molecular solutions have evolved to manage different physical constraints.

10.2 Computational and Theoretical Approaches

Multi-scale Modeling: Develop models that explicitly integrate physical constraints and molecular regulation (Deng et al., 2020; Huang et al., 2019; Shi et al., 2018). Current models typically focus on either physical or molecular aspects; integrated models could reveal emergent properties.

Causal Inference: Apply rigorous causal discovery algorithms to experimental data to distinguish physical from molecular causation:

- PC algorithm and extensions for causal structure learning (Spirtes et al., 2000)
- Intervention analysis using do-calculus (Pearl, 2009)
- Counterfactual reasoning for evaluating what would happen under different conditions

Comparative Genomics: Systematic comparison of cell cycle regulatory architectures across bacterial species could reveal:

- Conservation of physical constraint management strategies
- Convergent evolution of molecular solutions to similar physical constraints
- Correlation between environmental physical constraints and regulatory complexity

Single-Cell Data Analysis: Leverage advances in single-cell technologies:

- Microfluidics for long-term single-cell tracking (Wang et al., 2010)
- Time-lapse microscopy for measuring division timing variability
- Integration with causal inference to identify regulatory relationships

Evolutionary Simulations: Simulate evolution of cell cycle regulation under different physical constraints to understand when different levels of molecular sophistication evolve (using agent-based models or population genetics simulations).

10.3 Open Questions

Fundamental Questions:

- What determines when molecular systems override vs. work within physical constraints?
- How did integrated physical-molecular regulation evolve?
- What is the minimal level of integration required for functional cell cycles?
- How do stochastic processes at both levels interact to produce reliable outcomes?

Technical Challenges:

- Measuring physical parameters in vivo with sufficient precision
- Manipulating physical parameters independently of molecular components
- Distinguishing molecular override of physical defaults from parallel coupling mechanisms

10.4 Implications for Different Fields

For Origins of Life Research:

Implications (with appropriate caveats):

Two plausible evolutionary scenarios (both speculative, neither strongly supported):

Scenario A - Physical-First:

- Physical constraints provided some organizational logic in early protocells [**speculative**]
- Early protocells had imprecise, inefficient cell cycles with substantial variability [**speculative—direct evidence lacking**]
- Molecular regulation evolved later as layers added to physical processes [**speculative**]

Scenario B - Co-Evolution (alternative compatible with synthetic cell biology data):

- Physical constraints and molecular regulation co-evolved from early stages
- Even early protocells likely required substantial molecular infrastructure for functional division
- **Evidence from synthetic cell biology:** Minimal functional division systems require sophisticated molecular organization (Garcia et al., 2021; Pineda et al., 2022; Zhu et al., 2021)

- **Implication:** Co-evolution scenario may be more plausible than strict physical-first models

Well-established: Evolution of molecular regulation was essential for achieving the precision observed in modern bacteria

Important caveat: As discussed in Section 5.1, inferring specific properties of early protocells or LUCA from phylogenetic data is profoundly difficult. Current evidence CANNOT distinguish between physical-first and co-evolution scenarios. Both scenarios should be understood as speculative hypotheses for future investigation, not as conclusions supported by current evidence.

For Synthetic Biology:

Design Principles:

- Understand physical constraints before designing molecular systems
- **But:** Recognize that molecular sophistication is often required for precision
- Match regulatory complexity to functional requirements
- Consider hybrid approaches that exploit both physical defaults and molecular control

For Fundamental Cell Biology:

Key Questions:

- Which aspects of cell cycle regulation are **physically constrained** vs. **contingent molecular solutions**?
- **Evolutionary question:** Did evolution add molecular layers to pre-existing physical foundations, OR did physical and molecular systems co-evolve from early stages? Current evidence cannot distinguish between these scenarios.
- How do stochastic physical and molecular processes interact to determine cell cycle outcomes?

10.5 Connection to Broader Biological Theory

Our hierarchical framework connects to broader questions in systems biology and philosophy of science.

Noble (2012) on the relational nature of biological causation: Noble argues against privileging any single level of causation, emphasizing instead that biological causation is fundamentally relational across levels. His critique of reductionism centers on the insight that higher-level organizational properties emerge from and constrain lower-level processes, but this does not imply simple hierarchical control. Rather, causation operates through a web of reciprocal constraints across levels. Our hierarchical framework makes a different claim: we argue that at the physical-molecular interface, the relationship is specifically **asymmetric** under functional conditions. Molecular systems can detect and respond to physical constraints, but physical states do not exhibit compensatory responses to maintain molecular functions. Our framework thus departs from Noble's emphasis on relational reciprocity by proposing context-dependent asymmetry—molecular systems read physical states, but physical states do not "read" or respond to molecular states in a functionally directed way.

Pearl (2009) on causal hierarchy: Pearl's causal ladder distinguishes between association (seeing), intervention (doing), and counterfactuals (imagining). Our hierarchical claim can be formulated in Pearl's framework: we observe association between molecular and physical variables, we can intervene on molecular variables and observe physical consequences, but physical variables do not exhibit intervention-like behavior in response to molecular perturbation. The **asymmetry** is precisely in the intervention structure: $\text{do}(\text{molecular change}) \rightarrow \text{physical consequence}$, but $\text{do}(\text{physical change}) \rightarrow$ does NOT trigger molecular compensation toward functional goals.

Woodward (2003) on counterfactual dependence: Woodward's interventionist account of causation asks whether a variable would change under hypothetical interventions. Our claim is that division outcome (O) depends counterfactually on molecular variables (M): O would differ under $\text{do}(M = m_{\text{high}})$ vs. $\text{do}(M = m_{\text{low}})$. But physical variables (P) do not show this functional dependence on M: $\text{do}(\text{FtsZ} = \text{low}) \rightarrow$ P changes as consequence (elongation), but not as functional compensation. The distinction between **consequences** and **compensation** maps cleanly onto Woodward's distinction between mere change and regulation toward a goal state.

By engaging these frameworks explicitly, we move beyond citing authority and instead ground our asymmetry claim in established formal theories of causation. The physical-molecular interface becomes a tractable case study for testing broader philosophical claims about biological causation.

12. Conclusion

This review has proposed a **context-dependent** framework for understanding how physical constraints and molecular regulation interact in bacterial cell cycle control. Our central thesis is that **asymmetric information flow emerges specifically during critical functional transitions**—including stress responses, developmental programming, and checkpoint activation—while **bidirectional coupling predominates during steady-state homeostasis**. We have introduced formal definitions (Asymmetry Index, molecular complexity threshold, coupling type classification) that could, with additional experimental validation, transform this from a conceptual debate into testable science.

The Core Claim: Context-Dependent Asymmetry

What we claim: Molecular systems can override physical constraints specifically when:

- **Functional integrity is at stake:** Checkpoints must block division despite permissive physical conditions
- **Developmental programming requires asymmetry:** Molecular systems create outcomes that differ from physical symmetry
- **Stress conditions demand override:** Molecular responses must override default physical tendencies

What we do NOT claim: We do NOT claim that asymmetric information flow is a universal principle governing all aspects of cell cycle regulation under all conditions. During normal steady-state cycling, many systems exhibit bidirectional coupling where physical and molecular systems influence each other continuously (e.g., DNA supercoiling).

The key prediction: Functional context determines coupling type. Precision-critical contexts favor hierarchical organization ($AsI \gg 1$). Speed-critical contexts favor bidirectional coupling ($AsI \approx 1$). This is a SUBSTANTIALLY different claim than asserting that asymmetric information flow is universal.

Key Proposals and Hypotheses

1. Asymmetry Index (AsI): A Context-Dependent Metric

We proposed $AsI = |\text{Effect}(\text{do}(M) \text{ on } P)| / |\text{Effect}(\text{do}(P) \text{ on } M)|$ to quantify information flow directionality. Our specific claim is that $AsI \gg 1$ during checkpoints/stress/development, but $AsI \approx 1$ during steady-state homeostasis. However, we acknowledge that:

- Specific AsI values mentioned throughout are hypotheses, not measurements
- Determining actual AsI values requires experimental quantification not yet performed
- The metric's utility remains to be demonstrated through application

2. Molecular Complexity Threshold: A Testable Hypothesis

The molecular complexity threshold concept proposes that below a certain level of regulatory complexity, physical constraints contribute more to observable cell cycle behavior. Current status:

- Hypothesis motivated by JCVI-syn3.0 observations but alternative interpretations (pleiotropy, network disruption) exist
- Threshold value ($CCGC \approx 45 \pm 10$) is a rough estimate requiring systematic validation
- Operational definition of CCGC proposed but not yet applied across diverse species

3. Coupling Type Hypotheses: Functional Requirements as Predictors

We PROPOSE that **functional requirements predict coupling type**. Our key novel prediction is that **functional context determines which coupling type should evolve**:

- **Precision-critical contexts** (checkpoints, division commitment) → HYPOTHESIS: Should exhibit Type A (Hierarchical, $AsI \gg 1$)

- **Speed-critical contexts** (rapid homeostatic response) → HYPOTHESIS: Should exhibit Type B (Bidirectional, $AsI \approx 1$)

- **Spatial coordination** → HYPOTHESIS: Should exhibit Type C (Mixed, intermediate AsI)

Key Distinction: This is a GENUINELY PREDICTIVE framework (functional requirement → predicted coupling type), not a post-hoc classification scheme. Current case studies (SOS, DNA supercoiling) ILLUSTRATE these predictions but do NOT constitute validation. Testing requires:

1. **Measuring AsI** in diverse functional contexts

2. **Studying UNSTUDIED systems** where functional requirements can be identified BEFORE knowing coupling type

3. **Statistical validation** across many systems

This prediction can be tested by measuring AsI in different functional contexts to determine whether functional requirements actually predict coupling type.

Limitations and Caveats

We acknowledge several important limitations:

- **Scope clarification:** Our framework specifically addresses asymmetric information flow during checkpoints, stress responses, and developmental programming—not all cell cycle states

- **Quantitative claims:** Many specific numerical values presented are hypotheses, not measurements

- **Evolutionary inferences:** Claims about evolutionary trajectories remain speculative

- **Post-hoc examples:** Current case studies (SOS, DNA supercoiling) illustrate predictions but were studied BEFORE the predictive framework was formulated. Genuinely predictive tests require studying UNSTUDIED systems based on functional requirements BEFORE knowing coupling type

- **Limited data:** Framework primarily based on *E. coli* and *B. subtilis*; generalizability unknown

Addressing the Circular Classification Risk: The reviewer correctly noted that presenting Type A/B/C as an established classification scheme risks readers absorbing the framework as descriptively valid. We have addressed this by:

1. **Reframing from classification to prediction:** Type A/B/C are now presented as HYPOTHESES about what functional requirements should predict, not as established categories into which systems are slotted

2. **Changing the narrative structure:** FROM "classify systems, then validate" TO "predict based on functional requirements, then test"

3. **Explicit acknowledgment:** Current examples are "case studies that illustrate predictions" rather than "validated classifications"

4. **Emphasis on testing:** The framework is presented as a set of PROPOSITIONS that require experimental validation, not as established descriptive categories

What This Review Contributes

Unlike traditional reviews that primarily synthesize existing knowledge, this review:

- **Clarifies scope:** Explicitly limits asymmetric information flow claims to specific functional contexts (checkpoints, stress, development)

- **Distinguishes context-dependence:** Predicts bidirectional coupling during steady-state, asymmetric during transitions

- **Formalizes conceptual distinctions:** Physical consequences vs. compensation vs. molecular override

- **Proposes specific metrics:** Asymmetry Index, CCGC, and coupling type classification

- **Outlines experimental path:** Protocols for testing key predictions

- **Identifies open questions:** Highlights what measurements are needed to validate or falsify the framework

The framework's ultimate value will be determined by whether its context-dependent predictions are empirically supported. Until experimental validation, these remain proposals rather than discoveries.

Future Directions

The primary value of this framework will be established through experimental work that tests **WHEN** asymmetric vs. bidirectional coupling occurs:

1. **Measure AsI in different functional contexts** (steady-state vs. checkpoint vs. stress)
2. **Test molecular complexity threshold** through systematic gene reduction/addition experiments
3. **Validate coupling type predictions** across diverse functional contexts
4. **Develop independent criteria** for classifying functional requirements before measuring AsI

The framework's strength lies in making concrete, falsifiable predictions about **WHEN** each type of coupling should occur. Whether these predictions are confirmed or falsified, the experimental work required to test them will advance our understanding of bacterial cell cycle regulation.

The bacterial cell cycle operates at the physical-molecular interface. This framework provides tools to understand that interface **context-dependently**—recognizing that different functional states demand different regulatory architectures.

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Competing Interests: The authors declare no competing interests.

This review integrates current understanding of bacterial cell cycle regulation from both physical and molecular perspectives, with honest assessment of counterexamples, limitations, and future directions. References have been manually verified against primary sources; any remaining errors are the responsibility of the authors.

Date: 2026-04-23

Word Count: ~22,000 (appropriate for comprehensive review journals; includes pilot estimate demonstrating AsI tractability; reframed framework as testing methodology)

References: 163 (one duplicate removed: Harvey 2022 = Le Gall 2022; Round 13 citation cleanup: removed Adikesavan 2021 [uncited], added Wang et al. 2019, fixed Bürmann 2023 title, removed Jude et al. 2022 [not in reference list], added Cayley et al. 2000)

Revision: Round 13 Complete (all 15 initial concerns + 14 additional concerns addressed; reframed as framework for testing causal asymmetry; pilot estimate added; Min system recategorized as ambiguous; CCGC presented as directional prediction only; Type A/B/C acknowledged as post-hoc descriptive)